

Models and Model Selection in Phylogenetic Paleobiology

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09.29.26

Agenda

- What are phylogenetic data?
 - What are the assumptions made when we collect phylogenetic characters?
- What models are available to us to work with morphological characters?
- Continuous traits

What are phylogenetic data

```
#NEXUS
```

```
BEGIN DATA;
```

```
  DIMENSIONS  NTAX=27 NCHAR=60;
```

```
  FORMAT DATATYPE = STANDARD GAP = - MISSING = ? SYMBOLS = " 0 1 2 3";
```

```
  MATRIX
```

```
Ctenocystis      011???000?0?????0000?000000000???????0020?00????????????000
```

```
Gyrocystis_platessa      1101110100(0 1)100000000-0000000000010012020110000001200001-1300
```

```
Gyrocystis_testudiformis 00011111000100000000-0000100000010012020101110001200001-1310
```

```
Gyrocystis_cruzae      1001?10?000100000000?000100000010012???0?00?0001200001-?310
```

```
Gyrocystis_badulesiensis 11011102000110000000-01101011000201120?0010000001200001-1310
```

```
Gyrocystis_erecta      110?????00110000000-?????????0201120?????00001200001-?3?0
```

```
Progyrocystis_disjuncta 010?????100100000000-0000000000001001000200000001200001-0300
```

```
Protocinctus_mansillaensis 010111??200010000000-0022200000000-120??2000?0?122000011?310
```

```
Elliptocinctus_barrandei (0 1)(0 1)0111010(1 2)(0 1 2)000000000022201022200201120?
```

```
01100000122100011?210
```

```
Elliptocinctus_vizcainoi 0100?0??0100?1000000-0220102??00201120?001000001221?001??210
```

```
Sucocystis_theronensis 130111011(0 1)(0 1 2)010000000022201022200100121??0100000122000011?310
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Sucocystis_bretoni      13011??111(0 1)010000000002201022200100120??0100000122000111?310
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```
Lignanicystis_barriosensis 13110112011000010000022201022201101120112000000122100000?210
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```
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```
Sucocystis_acrofera     230?????010010000000-02201022200100120??01000001222100110110
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```
Undatacinctus_quadricornuta 100?????010010000202-0222222220000-120??0100000122100001?211
```

```
Undatacinctus_melendezi 130?????010010001111-0220102220000-120??0100?0012210001??211
```

```
Asturicystis_jaekeli    121111?2000010000000-0000000000000-100??2000001222210110100
```

```
Sotocinctus_ubaghsi     100?01??0(0 1)0010000000-0000000000000-0000020000012222100110200
```

What are phylogenetic data

'Brachymyrmex_sp'

-----TTATGGTTCATTAGATTGTACC--CACATTTACTTG
GATAACTGTGGTAATTCTAGAGCTAATACATGCAAA--ACAGAGTCCCGACCAGTGATGG-AAGGGACGCTTTTATTAGATCAAAACCAATCGCCA
CCGTTTA-TTTTGGT-GACTCTGAATAACTTTGTGCTGATCGCACGGT-C--T-TAGCACCGGCG-ACGCATCTTTCAAATGTCTGCCTTATCAACT
GTCGATGGTAGGTTCTGCGCCTACCATGGTTGTAACGGG-TAACGGGRAATCAGGGTTCGA--TTCCGGAGAGGGAGCCTGAGAAACGGC-TACCA
CATCCAAGGAAGGCAGCAGGCGCGCAAA-TTACCCACTCCCGGCACGGGGAGGTAGTGACGAAAAATAACGATA-CGGGACTCATCCGAGGCCCCCGT
AATCGGAATGAGTACACTTTAAATCCTTTAACGAGGATCCATTGGAGGGCAAGTCTGGTGCCAGCAGCCGCGGTAATTCCAGCTCCAATAGCGTATA
TTAAAGTTGTTGCGGTAAAAAGCTCG-TAGTTGAATCTGTGTC--GCACACCGTCGGTT--CACCGCT--CG--CGGT-GTTTA-ACTGGCGTGAT
GTGGGACGTCCTACCGG---TGGGCTTAGCCTC--G-----

What are phylogenetic data

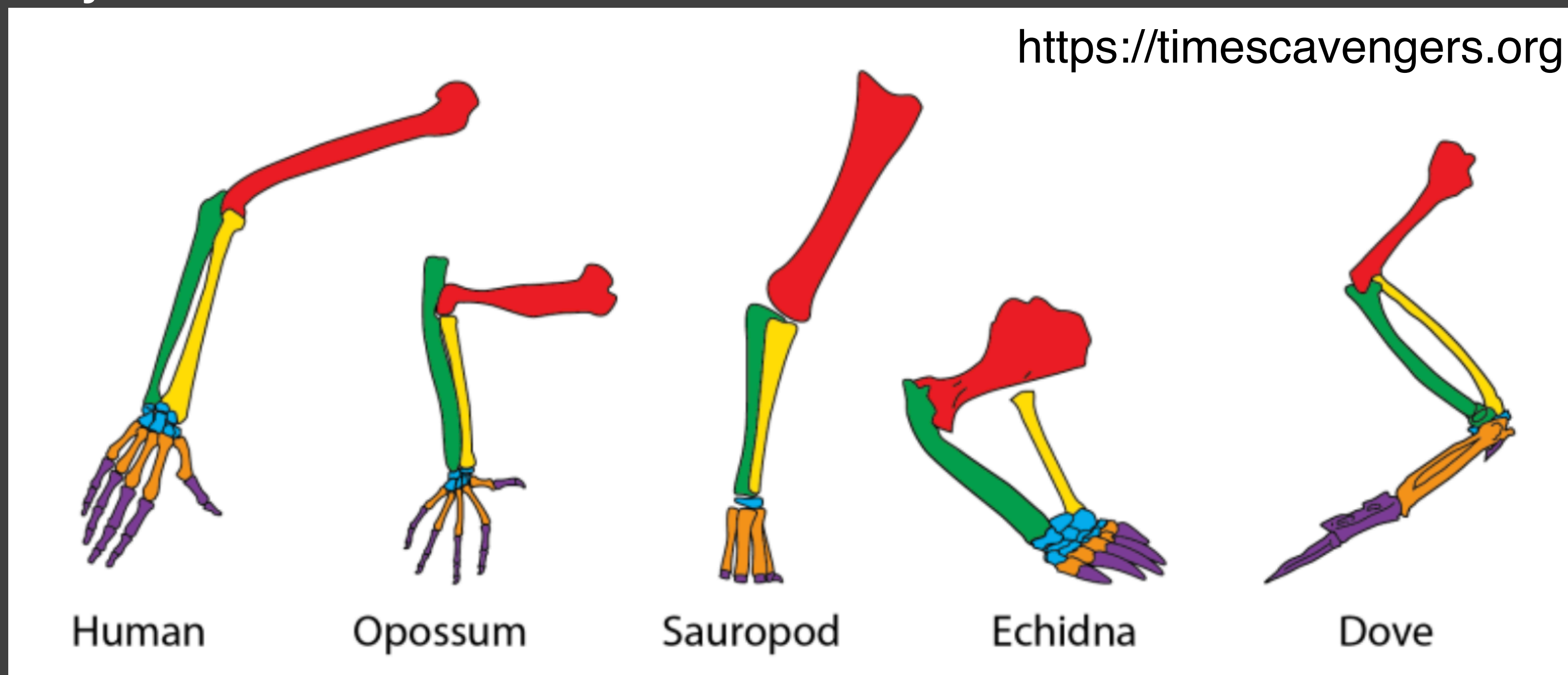
- Data are **representational**
 - Representational: A rendering of the data in a way that makes analysis and sharing possible
 - We don't stick the rock into BEAST

What are phylogenetic data

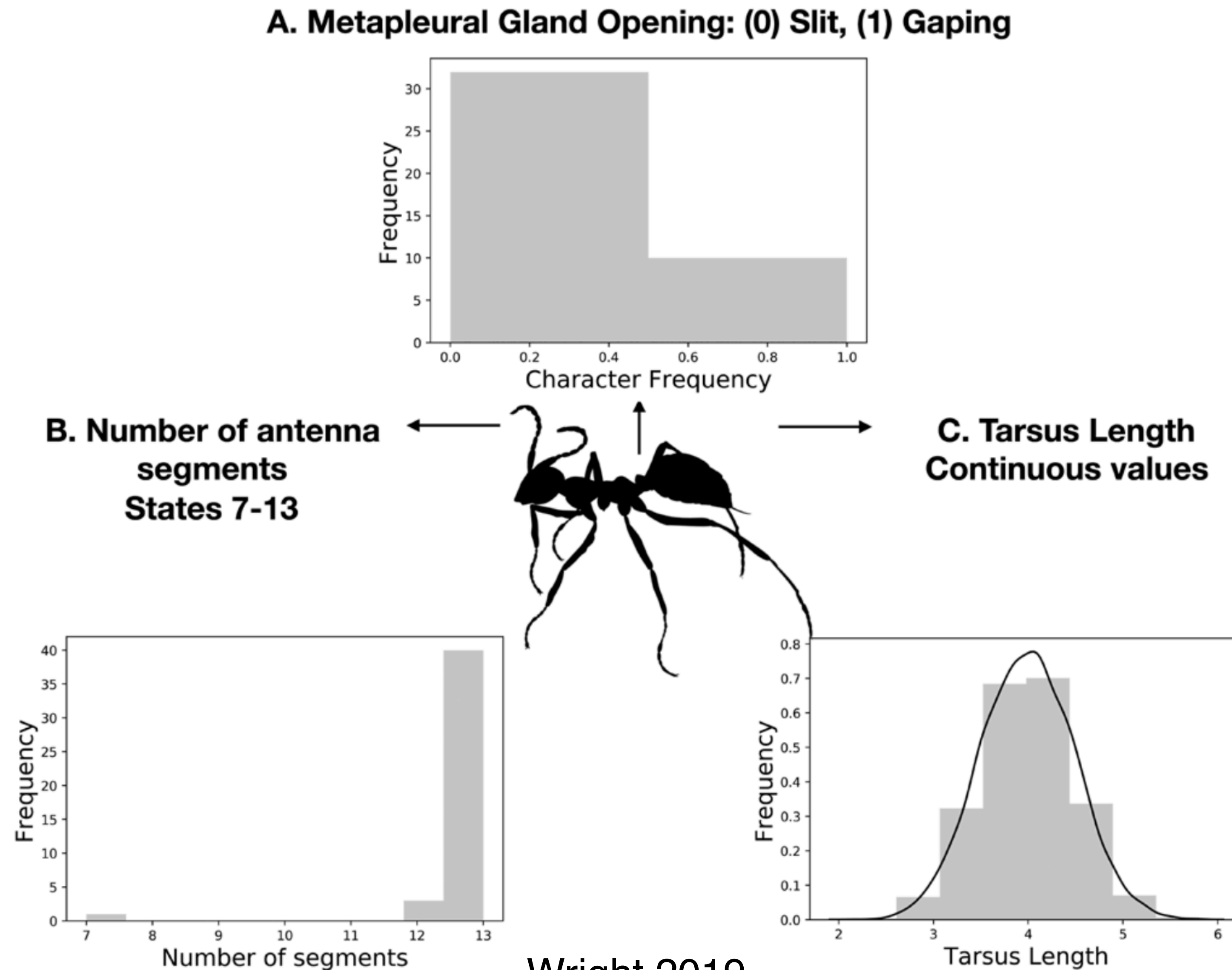
- Data are **representational**
 - Representational: A rendering of the data in a way that makes analysis and sharing possible
 - We don't stick the rock into BEAST
- The representations we make are informed by theory, implicitly or explicitly

What are phylogenetic data

- Phylogenetic data need to be **homologous**
 - Our aim: to estimate evolutionary histories
 - If characters are not homologous, then they don't share an evolutionary history to model



What are phylogenetic data



What are phylogenetic data

- How do we collect phylogenetic data?



Image via Wikimedia Commons

- Examine samples of your clade of interest, assess which characters across your sample are homologous
 - May do this based on modern analogs, developmental information, structural and placement similarity
 - If you're lucky, you may have multiple samples per O.T.U., which allows you to verify that a character actually exists and isn't stochastic variation

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Image via Wikimedia Commons

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 - Decide how many states are present for that character
 - How many unique ways are there to be this character?

What are phylogenetic data

- How do we collect phylogenetic data?

All of this is influenced by theory:

- You looked in this clade because you thought there was an interesting evolutionary question
- You collected these traits because you thought they contained evolutionary information
- Examine samples of your clade of interest, assess which characters across your sample are homologous
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What are phylogenetic data

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All of this is influenced by theory:

- You looked in this clade because you thought there was an interesting evolutionary question
- You collected these traits because you thought they contained evolutionary information
- Under parsimony, we don't collect invariants and autapomorphies



Our analytic model impacts our data collection



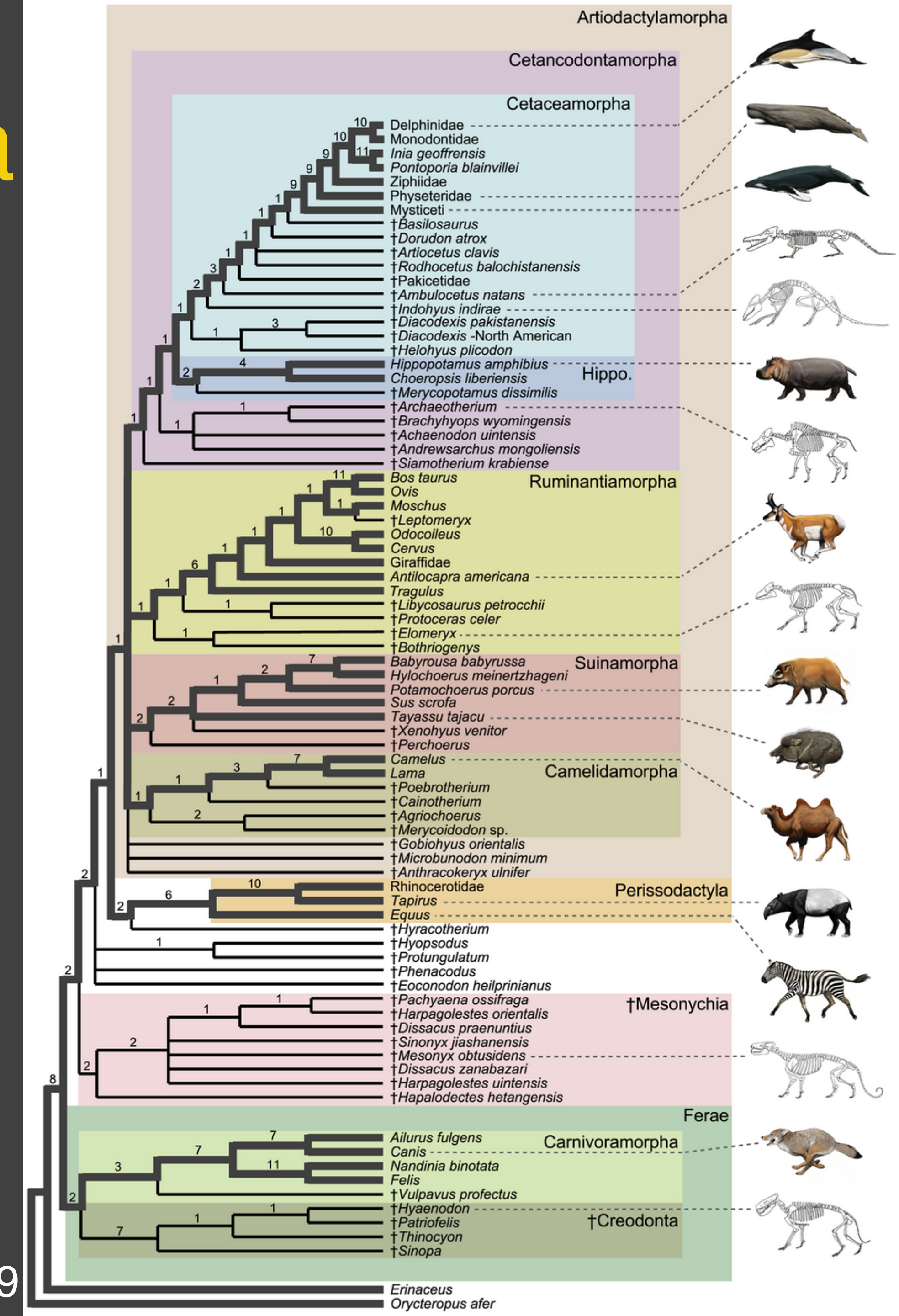
What are phylogenetic data

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All of this is influenced by theory:

- You looked in this clade because you thought there was an interesting evolutionary question
- You collected these traits because you thought they contained evolutionary information
- What characters are present is determined by your sample

Spaulding et al 2009

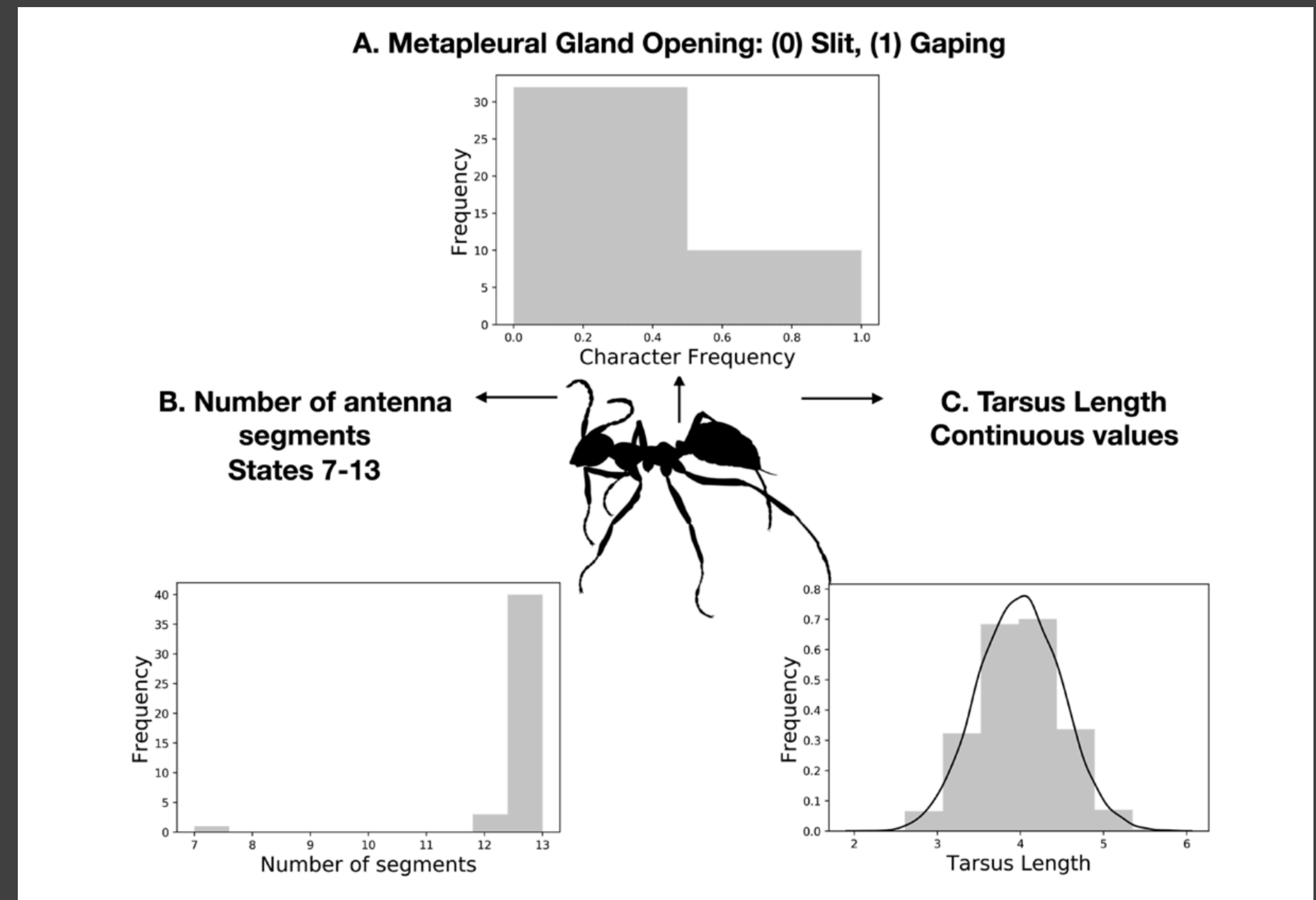


What are phylogenetic data

- How do we collect phylogenetic data?

All of this is influenced by theory:

- You looked in this clade because you thought there was an interesting evolutionary question
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- What characters are present is determined by your sample
- How we write down characters is model-informed
 - If a character is ordered, that is a model of evolution

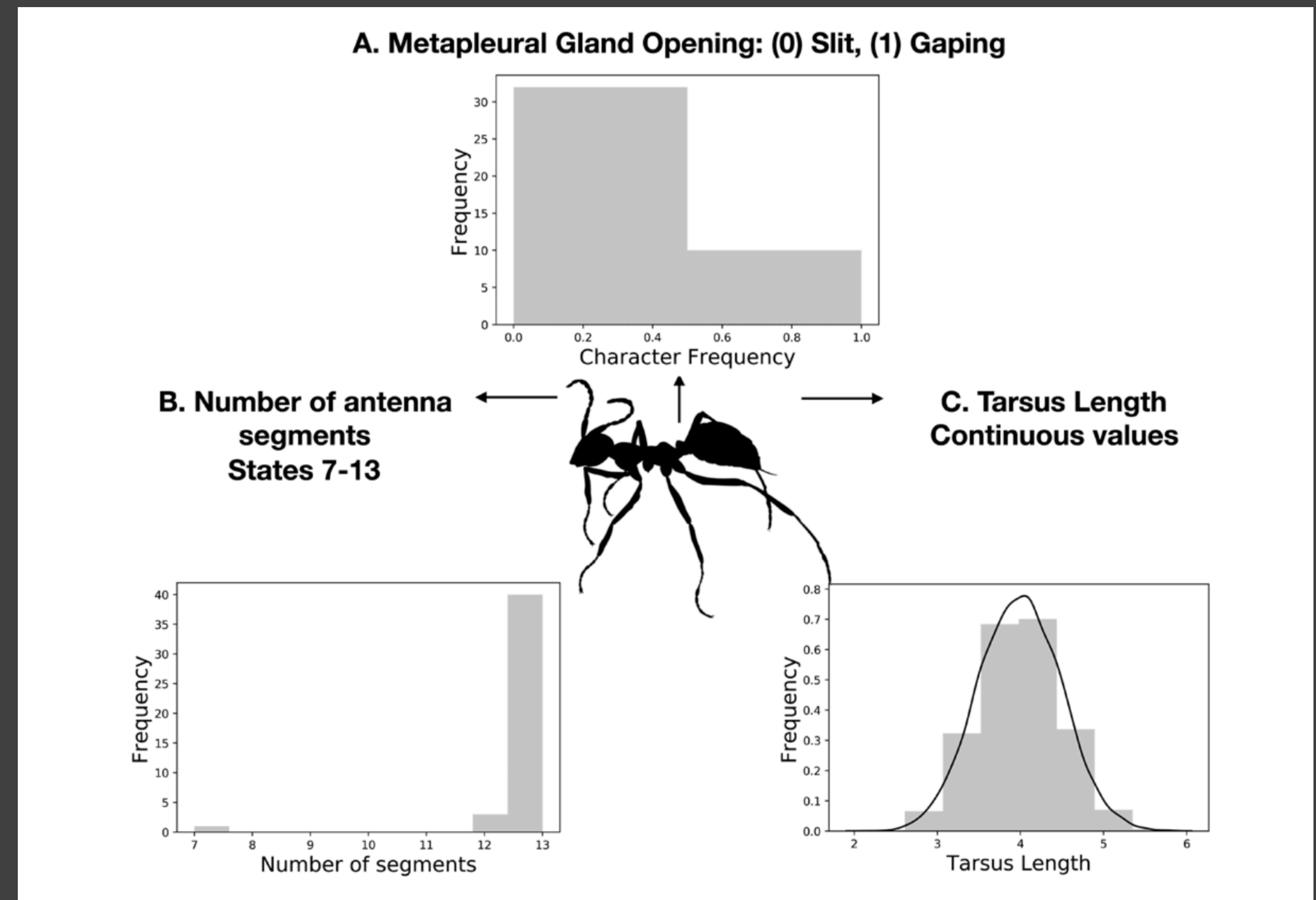


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What are phylogenetic data

- Hansen (1958) and Popper (1959) described data as being theory-laden

What are phylogenetic data

- Hansen (1958) and Popper (1959) described data as being theory-laden
- All phylogenetic data are representations, and how we make those representations is deeply embedded with a mental model of the evolutionary process

What are phylogenetic data

- Suppe (1966) described the idea of the data model, the idea that raw data is seldom useful to us as scientists
- Bokulich and Parker (2021) described an idea of fidelity to purpose that can be used to describe a model of your data

What are phylogenetic data

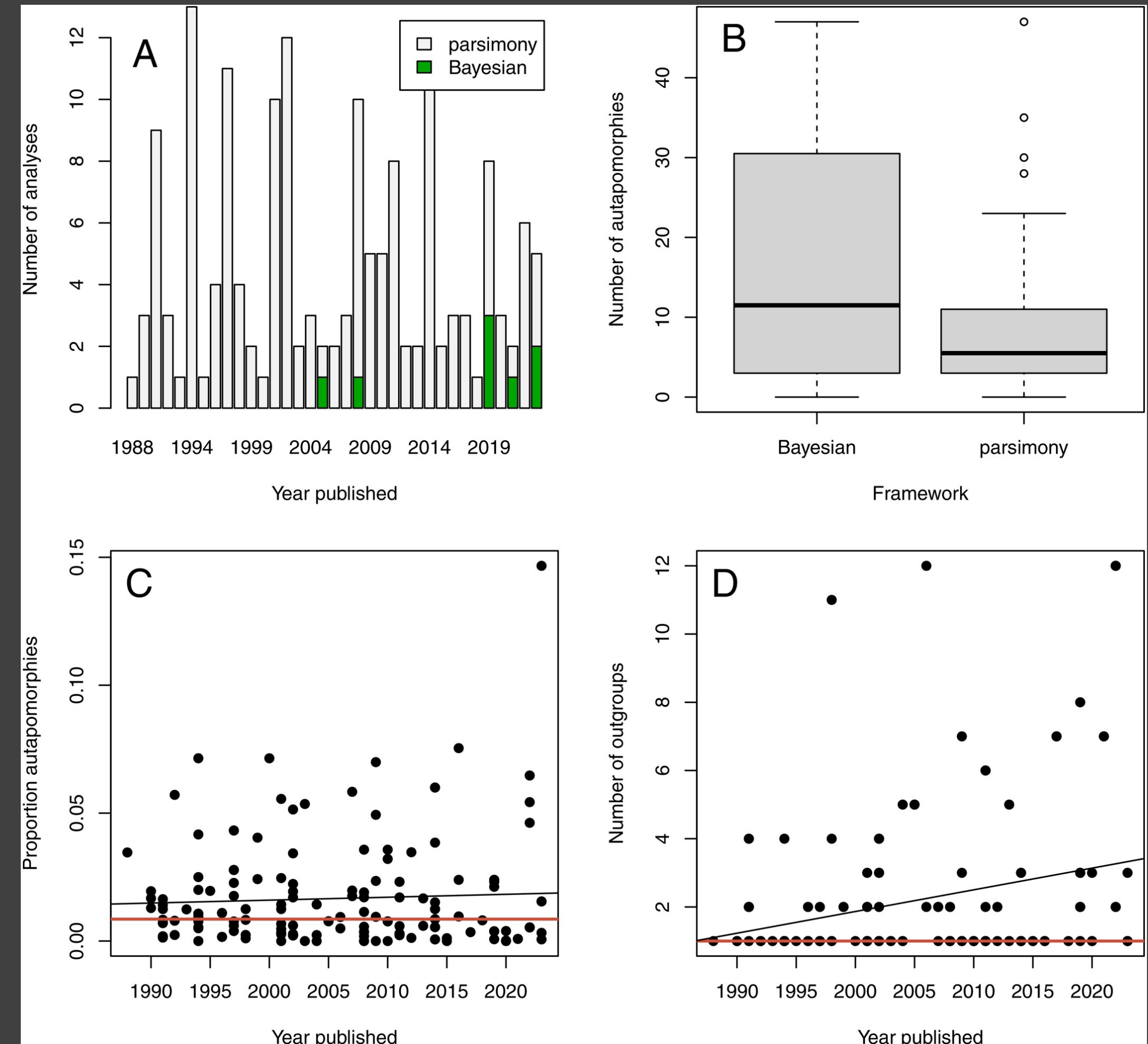
- In the case of phylogenetic data, most character matrices were assembled for use with parsimony
 - No autapomorphies
 - Bayesians can use autapomorphies
 - Therefore, a data matrix with autapomorphies is poor fit for purpose for parsimony, but good fit for purpose with Bayesian methods

What are phylogenetic data

- Indeed, Hopkins et al (2025) have demonstrated that as Bayesian methods become more common, more autapmorphies are being collected



Data collection impacted by model of evolution



Hopkins et al 2025

What are phylogenetic data

OK, so what?

- When you pick up a phylogenetic character matrix to reuse or add to:
- Why was this collected?
- What were the assumptions of the person collecting it about evolution?
- What were the goals of their analysis?
- What methods did they intend to use?

What are phylogenetic data

Is this a morphological problem?

No.

- When we use molecular data, we need to align those data to assess homology.
- Assume a model of insertions, deletions, and mutations at the genetic level

What are phylogenetic data

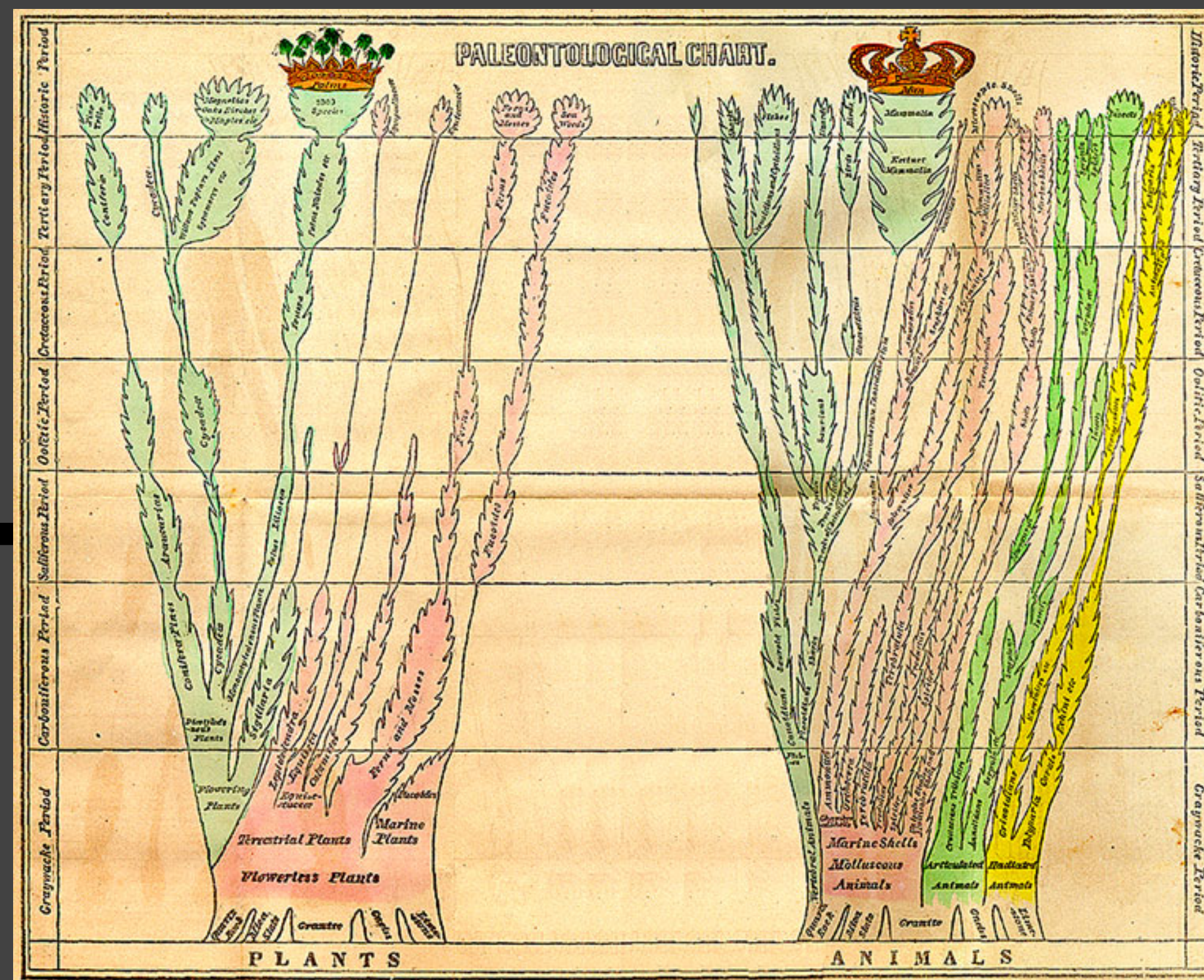
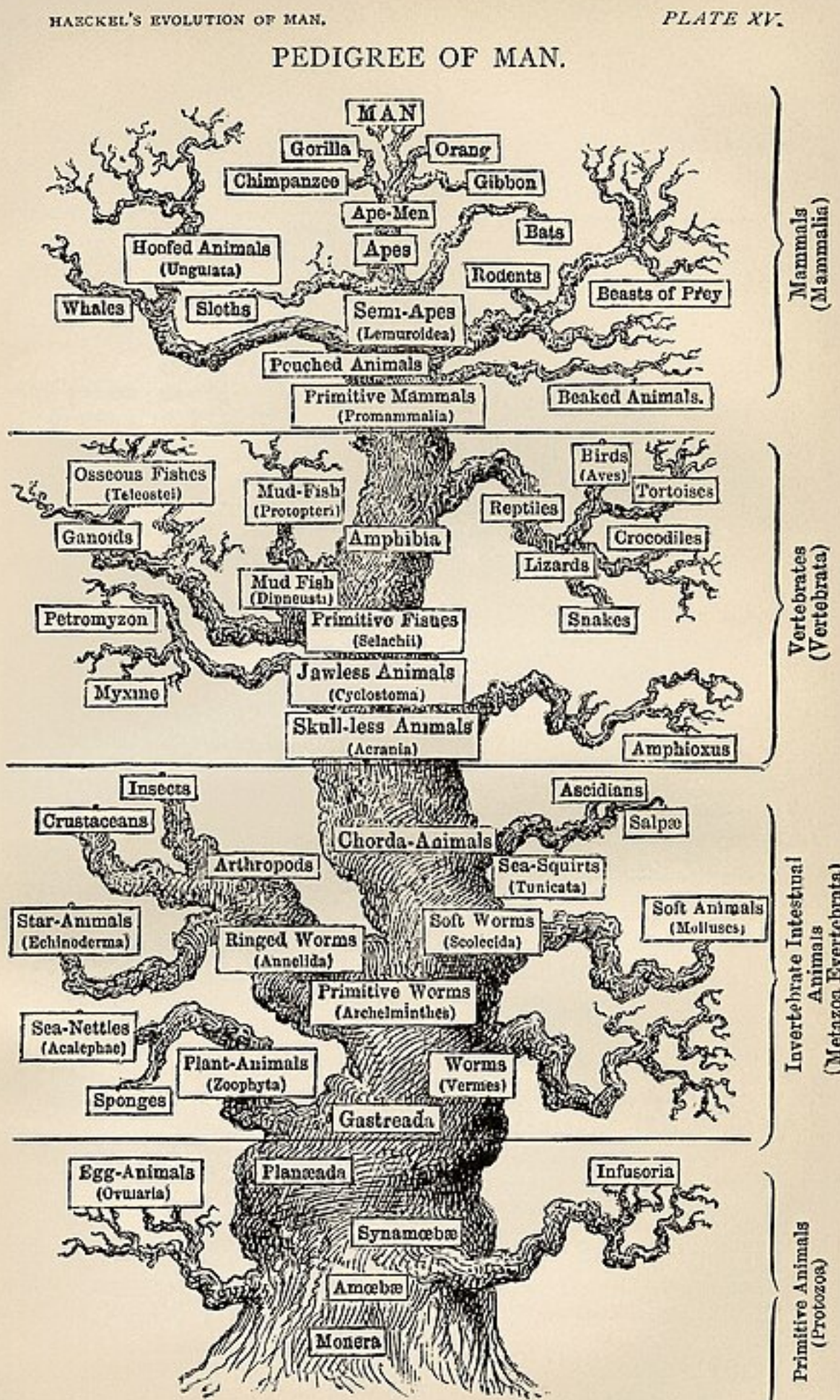
Is this a morphological problem?

No.

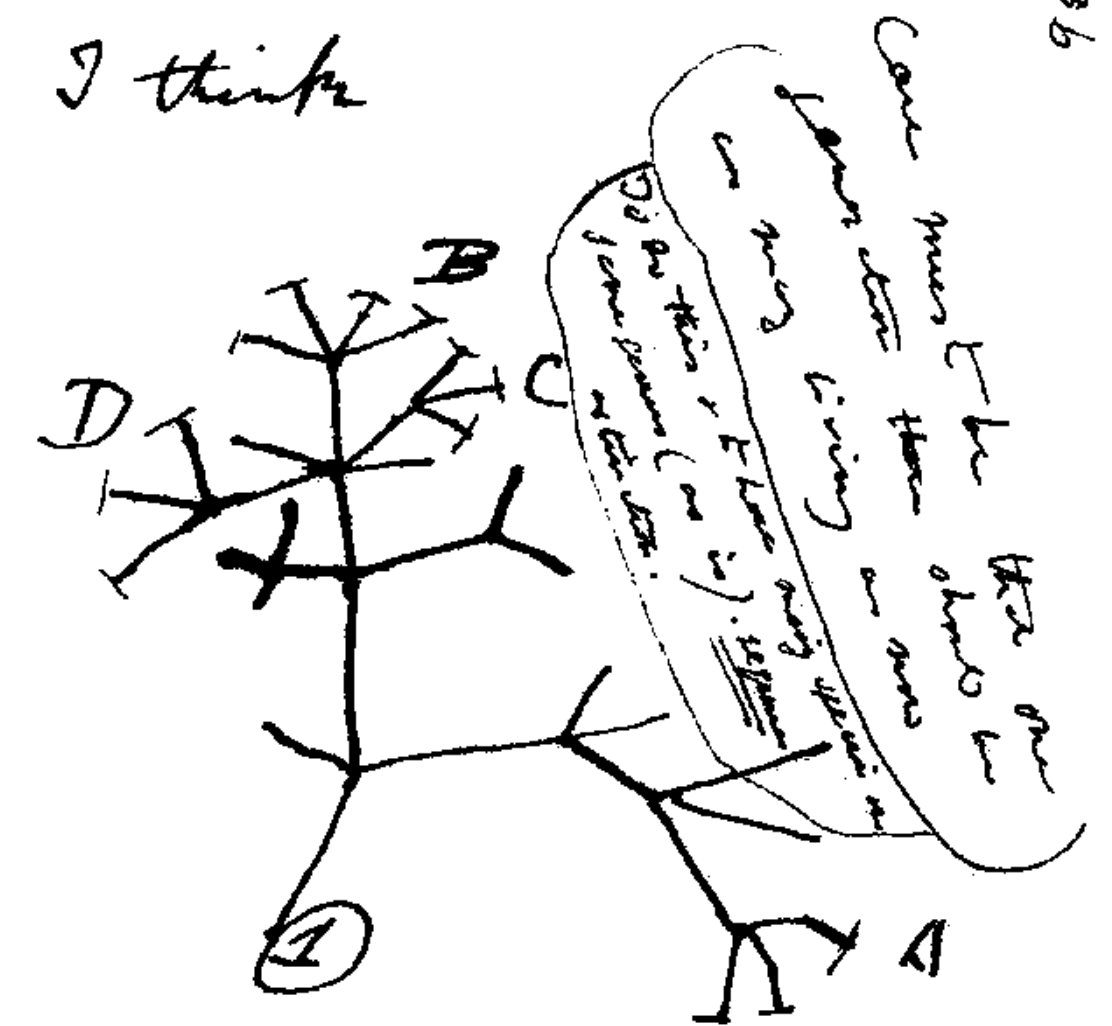
- When we use molecular data, we need to align those data to assess homology.
- Assume a model of insertions, deletions, and mutations at the genetic level
- **Done by machine but not more objective**
 - Dependent on your assumptions of molecular evolution
 - A different person could specify a different method and get a totally different dataset from the raw machine read data
 - Molecular data every bit as theory laden as morphological data

Pause: Questions

Mid-1800s

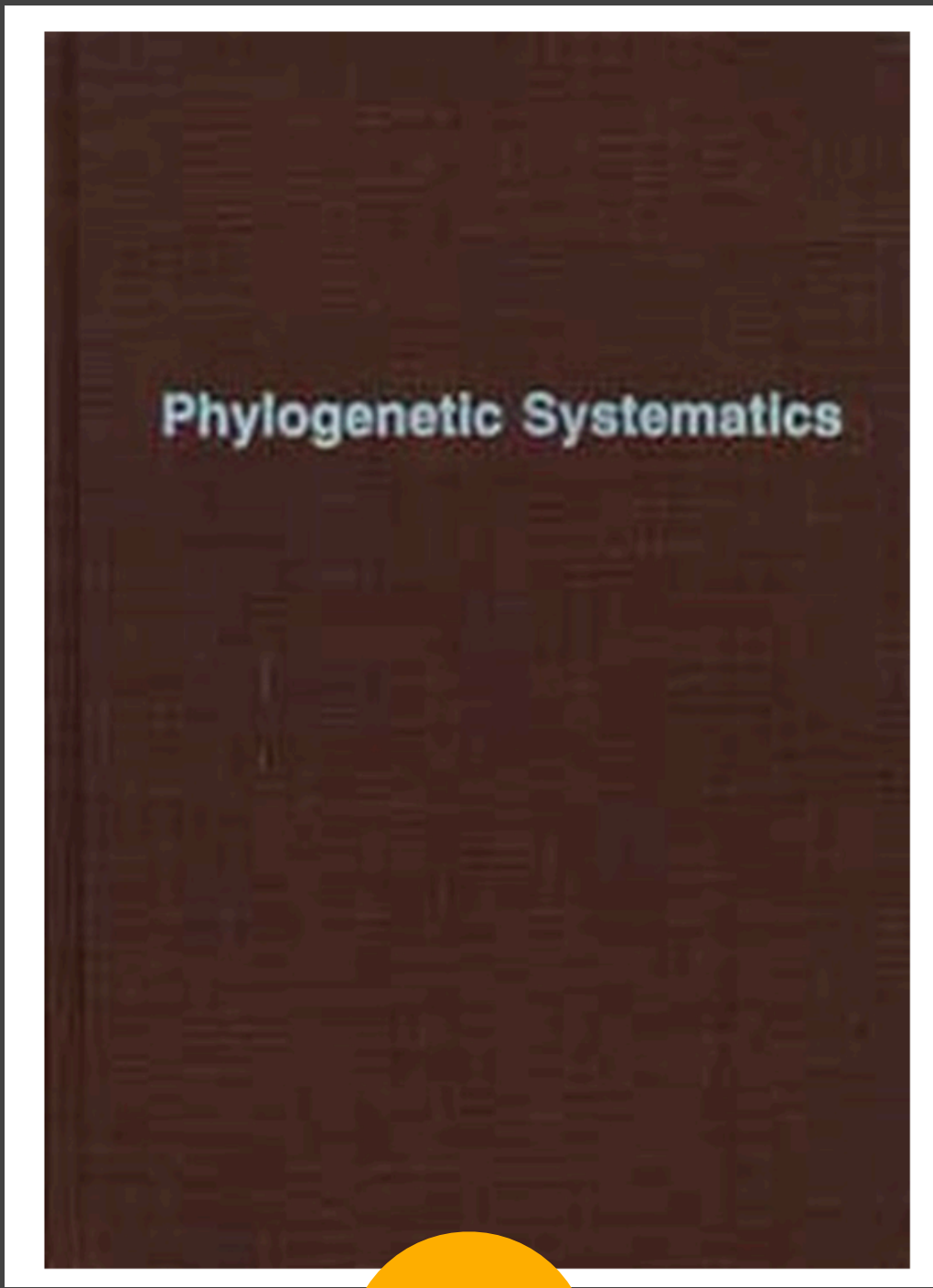
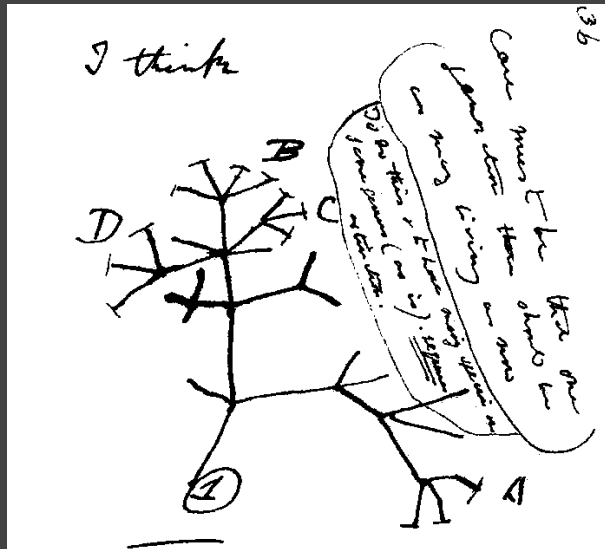


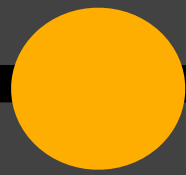
I think



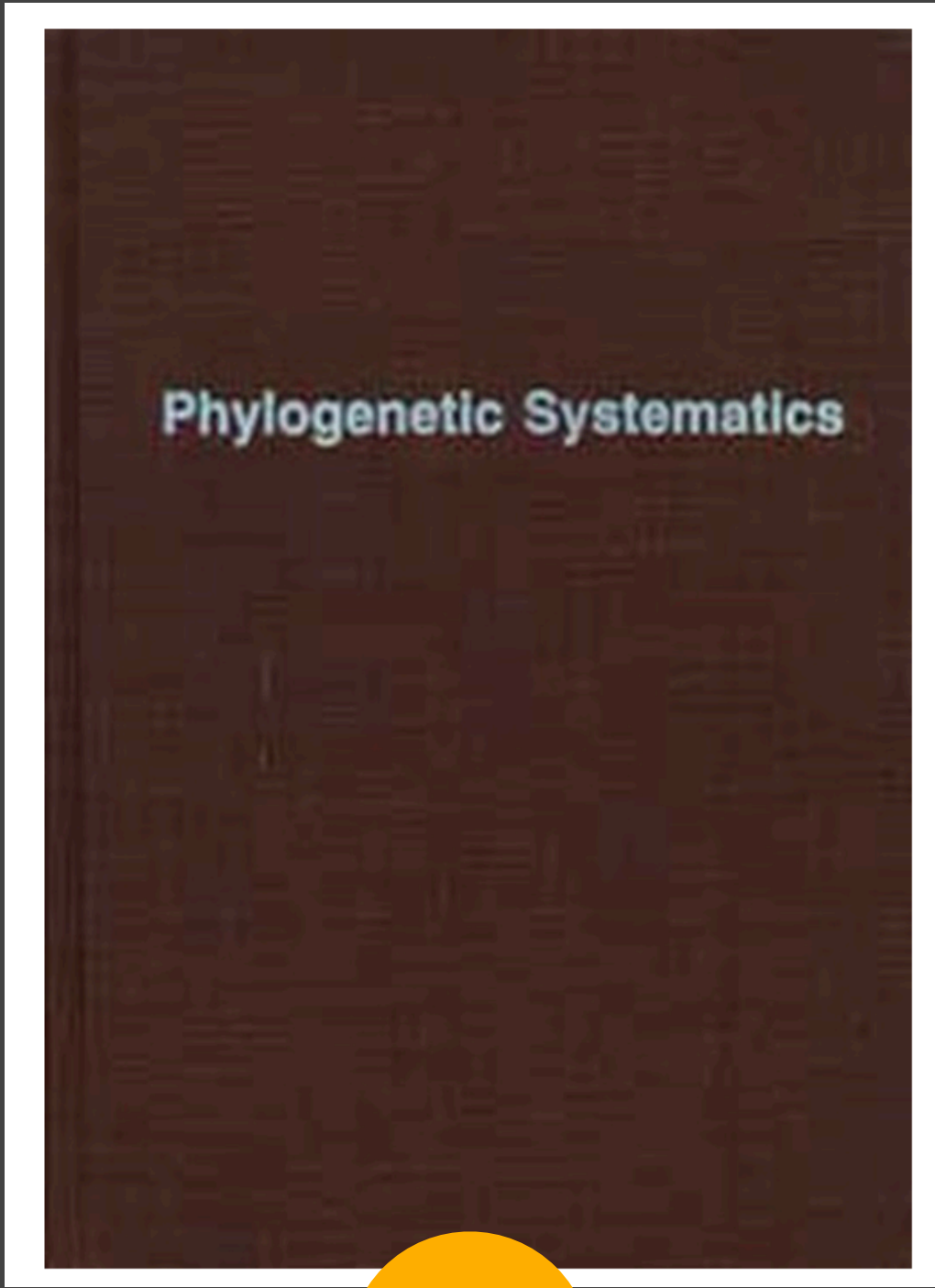
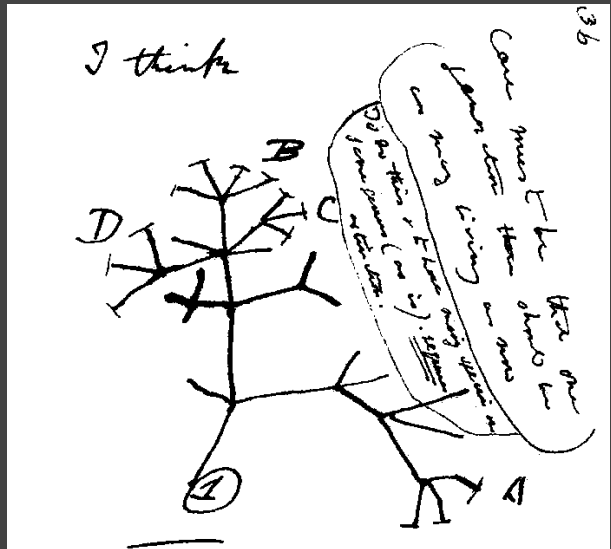
Then between A & B. various
 sort of relation. C & B. The
 first gradation, B & D
 rather greater distinction
 Then genus would be
 formed. - binary relation

Mid-1900s





Mid-1900s



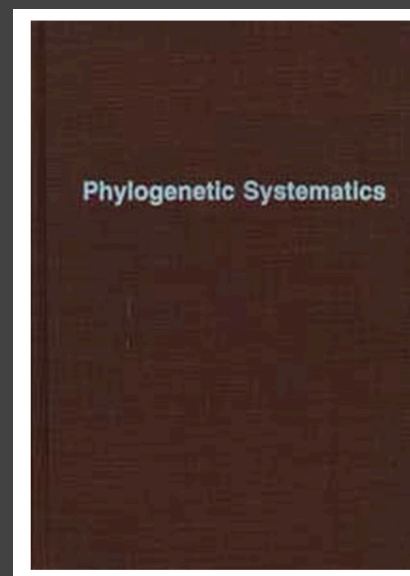
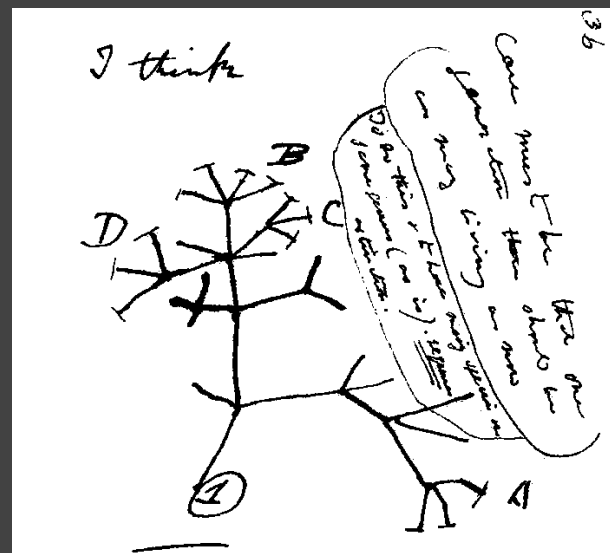
Parsimony: Application of Occam's Razor

1960s

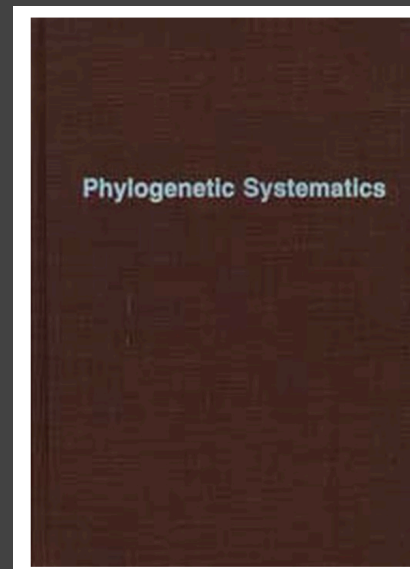
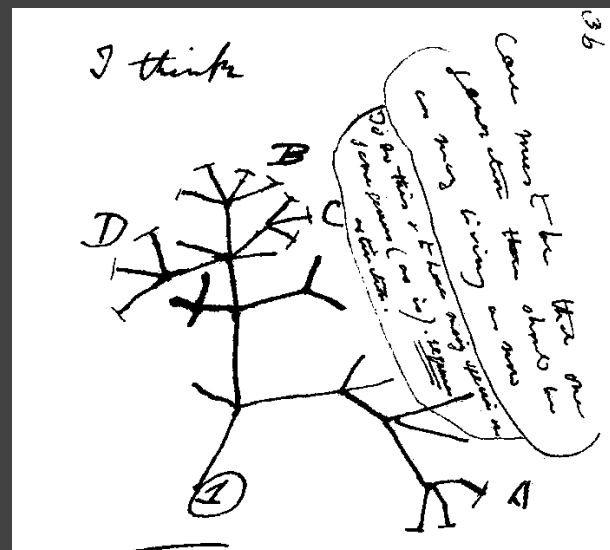
Sokal and Sneath: Numerical Taxonomy

First sequence models pop up

Era defined by data availability and compute availability



1960s



Margaret Dayhoff develops
first computational
parsimony algorithms, alignments,
and pioneers data sharing

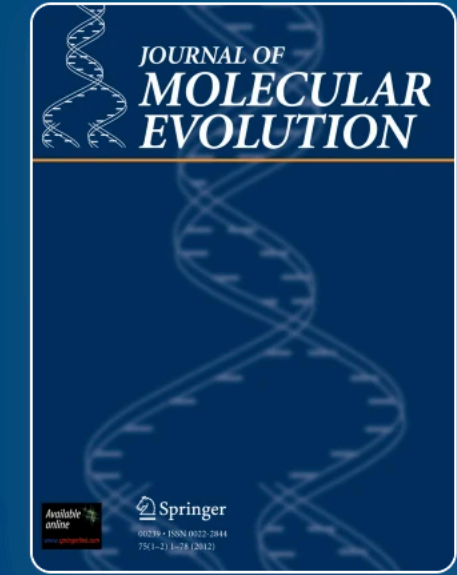
1981

[Home](#) > [Journal of Molecular Evolution](#) > [Article](#)

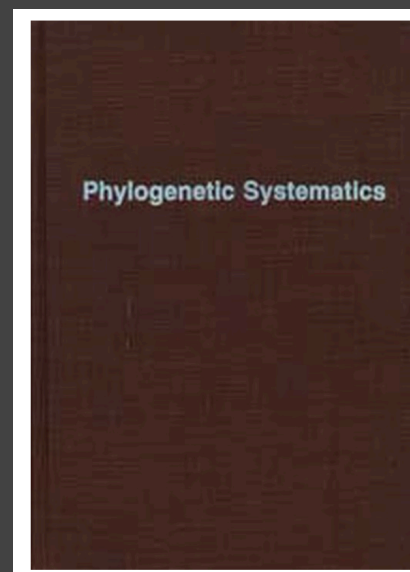
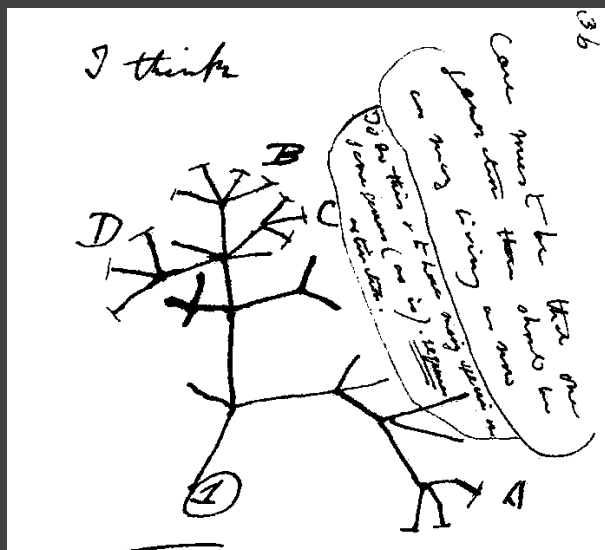
Evolutionary trees from DNA sequences: A maximum likelihood approach

Published: November 1981

Volume 17, pages 368–376, (1981) [Cite this article](#)

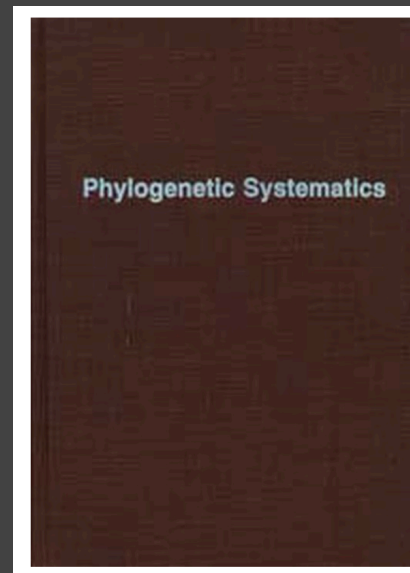
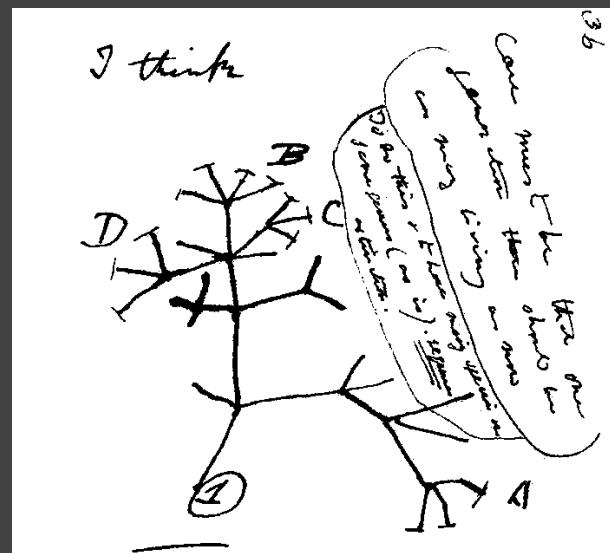


[Journal of Molecular Evolution](#)

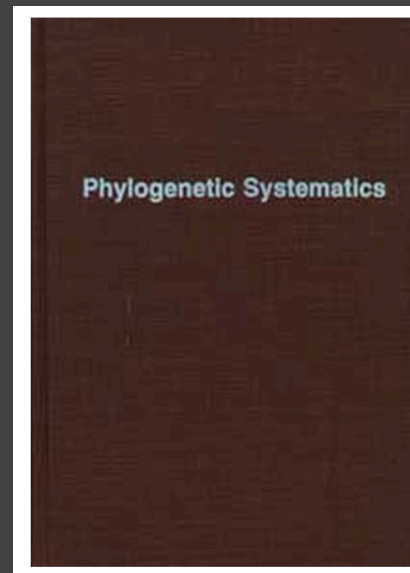
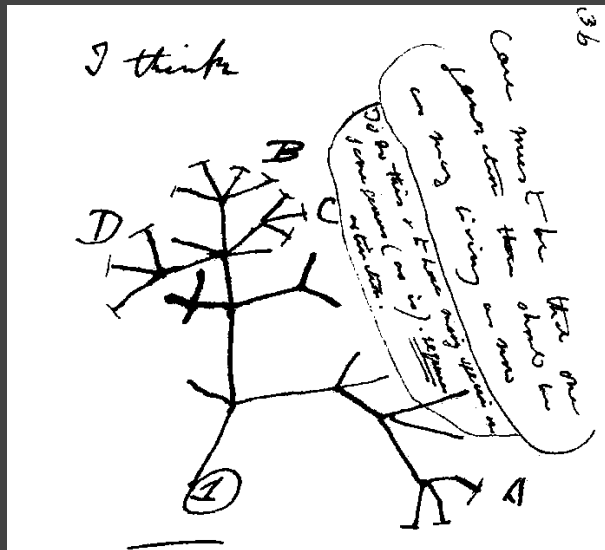


Felsenstein 1981

1981



1982



GenBank Overview

What is GenBank?

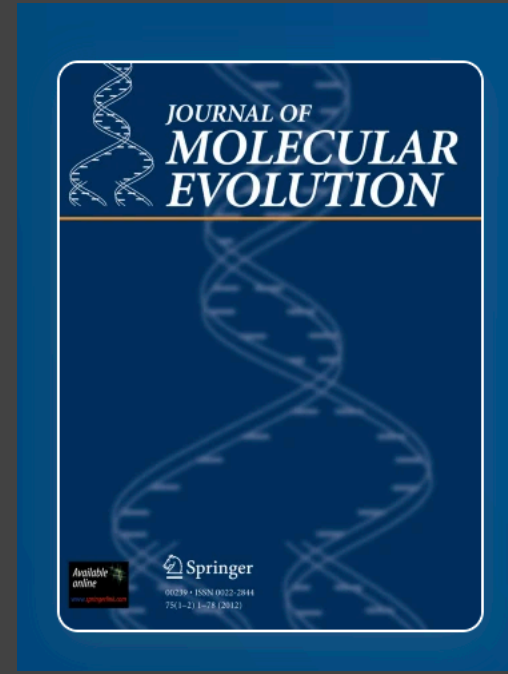
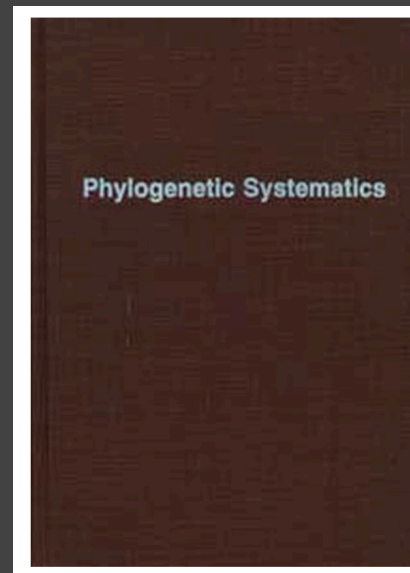
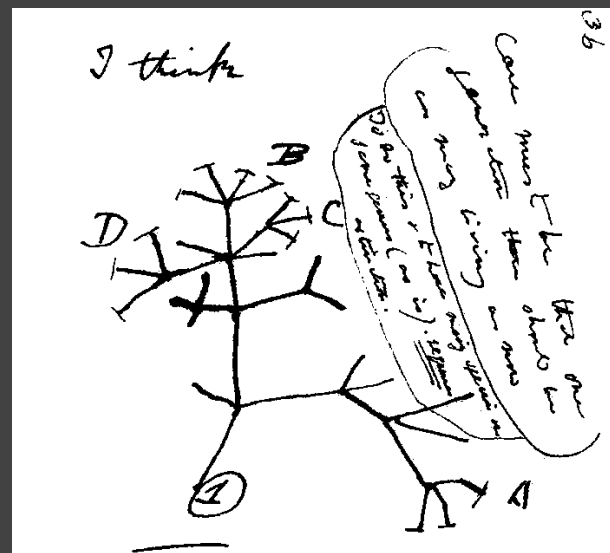
GenBank[®] is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences ([Nucleic Acids Research, 2013 Jan;41\(D1\):D36-42](#)). GenBank is part of the [International Nucleotide Sequence Database Collaboration](#), which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI. These three organizations exchange data on a daily basis.

A GenBank release occurs every two months and is available from the [ftp site](#). The [release notes](#) for the current version of GenBank provide detailed information about the release and notifications of upcoming changes to GenBank. Release notes for [previous GenBank releases](#) are also available. GenBank growth [statistics](#) for both the traditional GenBank divisions and the WGS division are available from each release.

An [annotated sample GenBank record](#) for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

1983

Polymerase Chain Reaction



GenBank

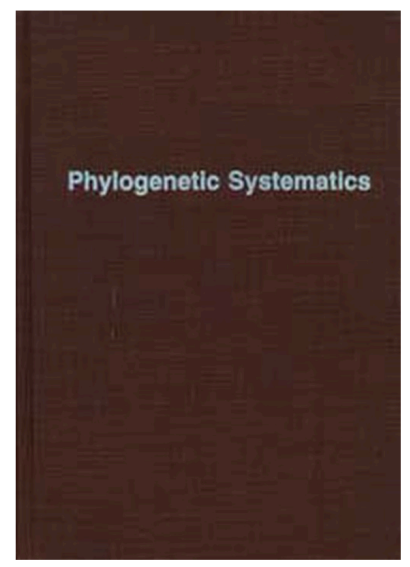
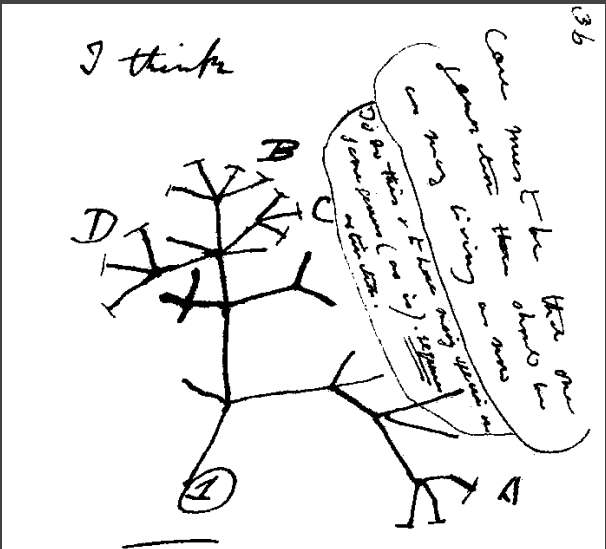


Late '80s-00s



MrBayes: Bayesian Inference of Phylogeny

Download MrBayes

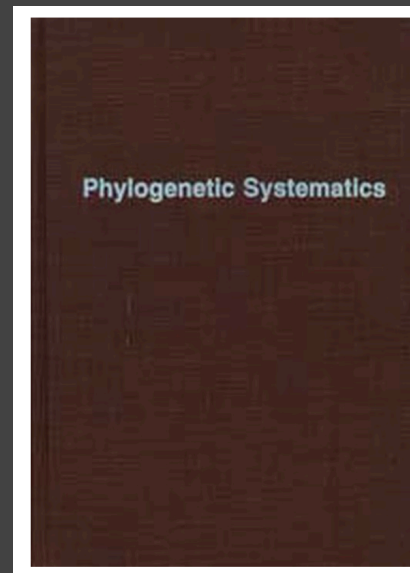
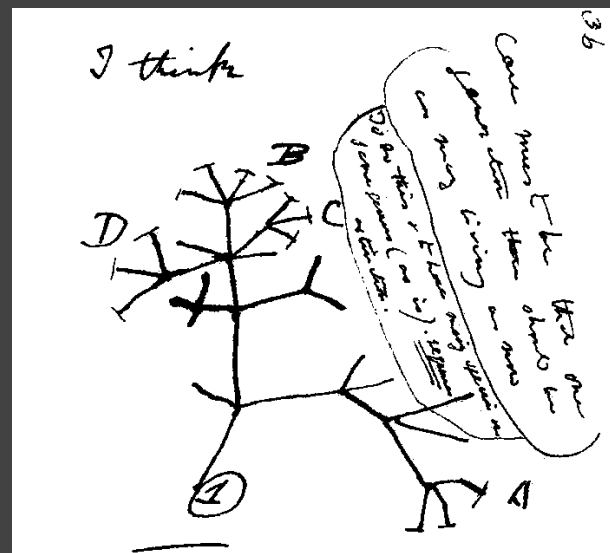


GenBank



2010s-now

High-throughput sequencing, data reduction
But also: data integration

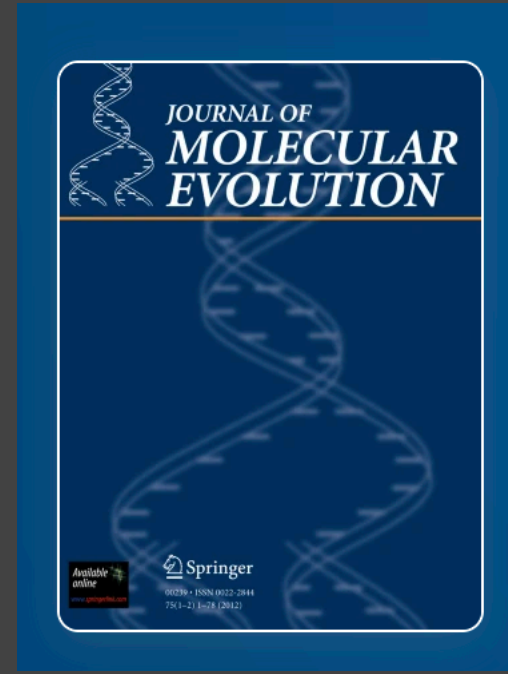
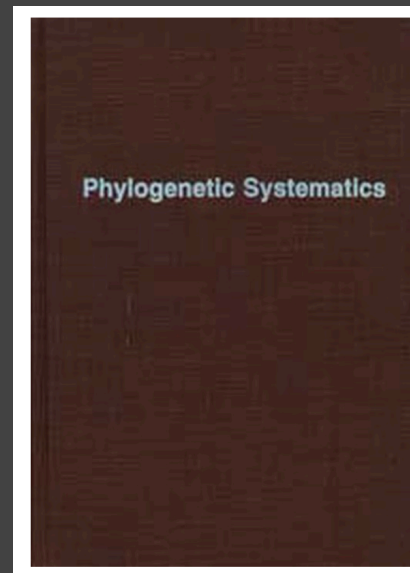
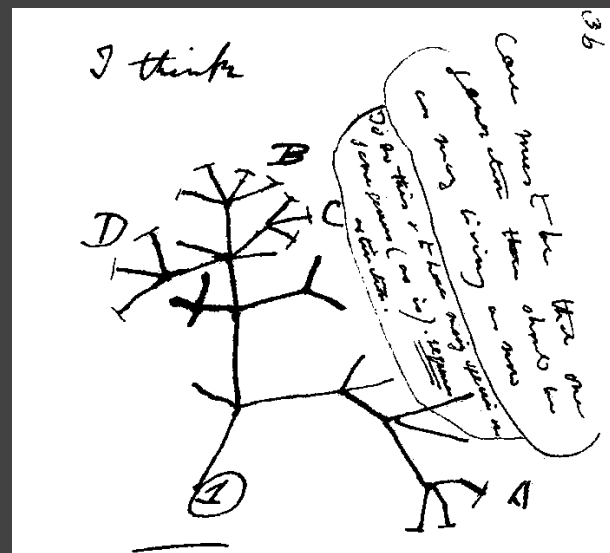


GenBank



2010s-now

High-throughput sequencing, data reduction
But also: data integration



GenBank



The molecular data revolution



Many modern phylogenetic methods are designed to take advantage of molecular data, not morphological data

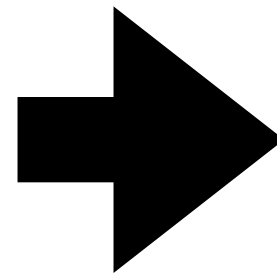
Q5E940_BOVIN	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_HUMAN	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_MOUSE	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_RAT	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_CHICK	-----MPREDRATWKSNYFMKIIQLLDDYPKCFVVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_RANSY	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--SALE	76
Q7ZUG3_BRARE	-----MPREDRATWKSNYFLKIIQLLDDYPKCFIVGADNVGSKQMOQIRMSLRGK-AVVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_ICTPU	-----MPREDRATWKSNYFLKIIQLLNDYPKCFIVGADNVGSKQMOQIRMSLRGK-AIVLMGKNTMMRKAIRGHLENN--PALE	76
RLA0_DROME	-----MVRENKAAWKAQYFIKVVFLFDEFKPKCFIVGADNVGSKQMOQIRMSLRGL-AVVLMGKNTMMRKAIRGHLENN--PQLE	76
RLA0_DICDI	-----MSGAG-SKRKKLFIEKATKLFTTYDKMIVAEADFVGSSQLQKIRKSIRGI-GAVLMGKNTMIRKVIRDLADSK--PELD	75
Q54LP0_DICDI	-----MSGAG-SKRKNVFIKATKLFTTYDKMIVAEADFVGSSQLQKIRKSIRGI-GAVLMGKNTMIRKVIRDLADSK--PELD	75
RLA0_PLAF8	-----MAKLSKQKKQMYIEKLSSLIQQYSKLILVHVDNVGNSQMASVRKSLRGK-ATILMGKNTIRIRTALKKNLQAV--PQIE	76
RLA0_SULAC	-----MIGLAVTTTKKIAKWKVDEVAELTEKLKTHKTIIIANIEGFPADKLHEIRKKLRGK-ADIKVTKNLNFNIALKNAG----YDTK	79
RLA0_SULTO	-----MRIMAVITQERKIAWKIEEVEKELEQKLREYIIANIEGFPADKLHDIRKKMRGM-AEIKVTKNTLFGIAAKNAG----LDVS	80
RLA0_SULSO	-----MKRIALALKQRKVASWKLEEVKELTELIKNSNTILIGNLEGFPADKLHEIRKKLRGK-ATIKVTKNTLFGIAAKNAG----IDIE	80
RLA0_AERPE	MSVVSILVGQMYKREKIPPEWKTLMLELEELFSKHRVVLADLTGTPTFVVQVRVKKLWKK-YPMMVAKKRIILAMKAAGLE---LDDN	86
RLA0_PYRAE	MMLAIGKRRYVTRQYPARKVKIVSEATELLQKYPYVFLFDLHGLSSRILHEYRYRLRRY-GVIKIIKPTLFFKIAFTKVYGG---IPAE	85
RLA0_METAC	-----MAEERHHTEHIPQWKKDEIENIKELIQSHKVFQGMVGIEGILATKMOKIRRDLDV-AVLKVSNTLTTERALNQLG----ETIP	78
RLA0_METMA	-----MAEERHHTEHIPQWKKDEIENIKELIQSHKVFQGMVRIEGLATKIQKIRRDLDV-AVLKVSNTLTTERALNQLG----ESIP	78
RLA0_ARCFU	-----MAAVRGS---PPEYKVRAVEEIKRMISSKPVVAIVSFRNVFAGOMOKIRREFRGK-AEIKVVKNTLLERALDALG----GDYL	75
RLA0_METKA	MAVKAKGQPPSGYEPKVAEWKRREVEKELKELMDEYENVGLVDLEGIPAPQLOEIRAKLRERDTIIRMSRNTLMRIALEEKIDER--PELE	88
RLA0_METTH	-----MAHVAEWKKKEVQELHDLIKGYEVVGIANLADIPAROLOKMRQTLRDS-ALIRMSKKTLLISLAEKAGREL--ENV	74
RLA0_METTL	-----MITAESEHKIAPWKIEEVNKLKELLKNGQIVALVDMMEVPAROLOEIRDKIR-GTMTLKMSRNTLIERAIKEVAEETGNPEFA	82
RLA0_METVA	-----MIDAKSEHKIAPWKIEEVNALKELLKSANVIALIDMMEVPAROLOEIRDKIR-DQMTLKMSRNTLIKRAVEEVAEETGNPEFA	82
RLA0_METJA	-----METKVKAHVAPWKIEEVKTLKGLIKSKPVVAIVDMMDVPAPQLOEIRDKIR-DKVKLMSRNTLIIRALKEAAEELNPNKLA	81
RLA0_PYRAB	-----MAHVAEWKKKEVEELANLIKSYPVIALVDVSSMPAYPLSQMRRLIRENGGLLRVSRNTLIELAIKKAAQELGKPELE	77
RLA0_PYRHO	-----MAHVAEWKKKEVEELAKLIKSYPVIALVDVSSMPAYPLSQMRRLIRENGGLLRVSRNTLIELAIKKAAKELGKPELE	77
RLA0_PYRFU	-----MAHVAEWKKKEVEELANLIKSYPVIALVDVSSMPAYPLSQMRRLIRENNGLLRVSRNTLIELAIKKVAQELGKPELE	77
RLA0_PYRKO	-----MAHVAEWKKKEVEELANLIKSYPVIALVDVAGVPAYPLSKMRDKLE-GKALLRVSRNTLIELAIKRAAQELGQPELE	76
RLA0_HALMA	MSAESERKTETIPEWKQEEVDAIVFMIESYESVGVVNIAGIPSRQLODMRRDLHGT-AELRVSRNTLLERALDDVD----DGLE	79
RLA0_HALVO	MSSEVRQTEVIPQWKREEVDLVDFIESYESVGVVGAGIPSRQLOSMRRELHGS-AAVRMSRNTLVNRALDEVN----DGFE	79
RLA0_HALSA	MSAEEQRTTEEVPEWKRQEVAEVLDDLETYDSVGVVNVGTGIPSKQLODMRRGLHGQ-AALRMSRNTLLVRALEEAG----DGLD	79
RLA0_THEAC	-----MKEVSQKKKELYNEITORIKASRSVAIVDTAGIRTRQIODIRGKNRGK-INLKVIKKTLLFKALENLGD----EKLS	72
RLA0_THEV	-----MRKINPKKKEIVSELAQDITKSKAIVDIKGVTRQMODIRAKNRDK-VKIKVVKKTLLFKALDSIND----EKLT	72
RLA0_PICTO	-----MTEPAQWKIDFVKNLNEINSRKVAAIVSIKGLRNNFQKIRNSIRDK-ARIKVSARLLRLAIENIGK---NNIV	72
ruler	1.....10.....20.....30.....40.....50.....60.....70.....80.....90	



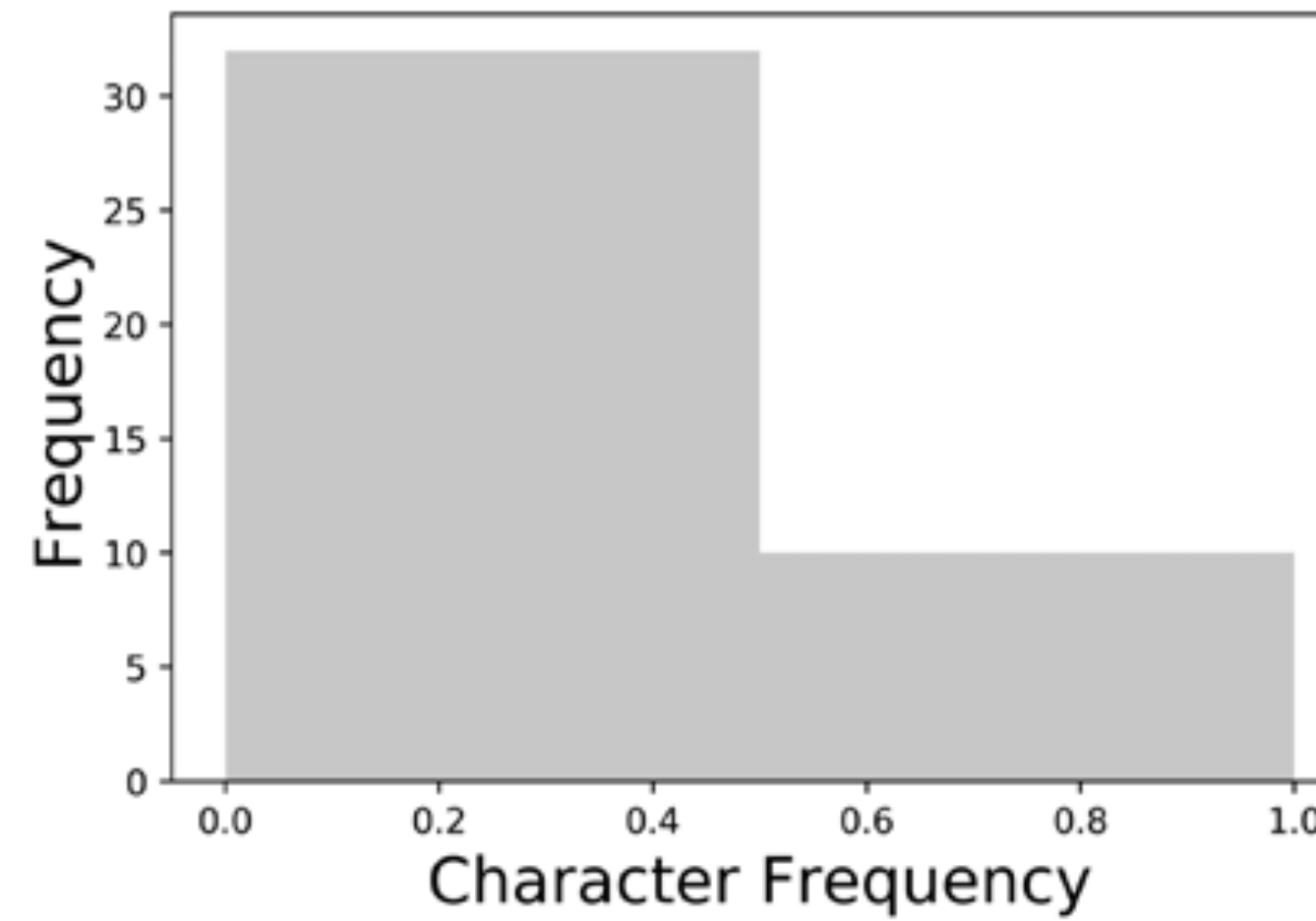
- Sequence data sets large
- Take the whole sequence
- Characters have more clearly-defined properties

- Smaller
- Selection bias
- Morphology has to be interpreted

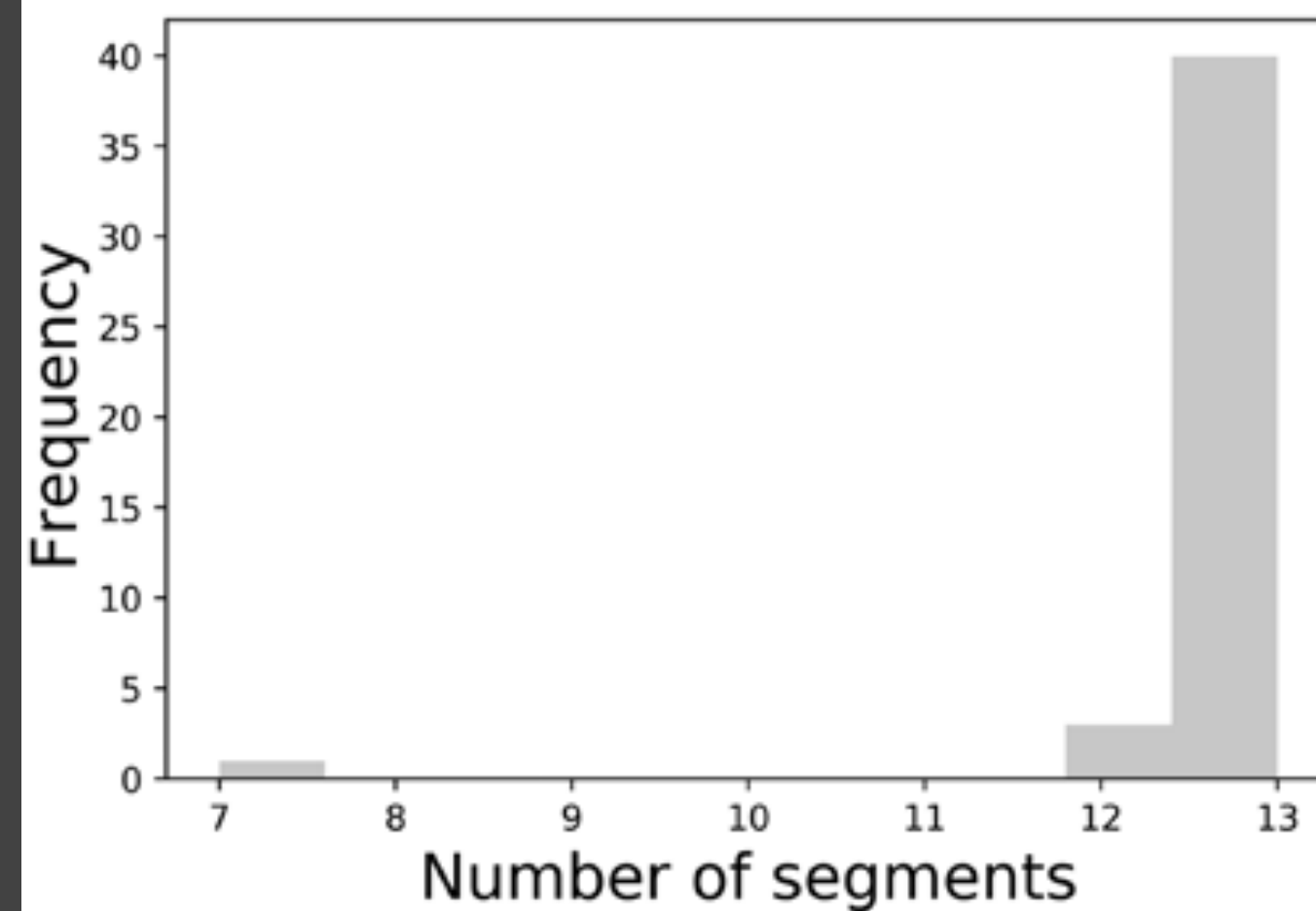
Discrete Data:
Vast majority of
the
collected paleo
phyla data



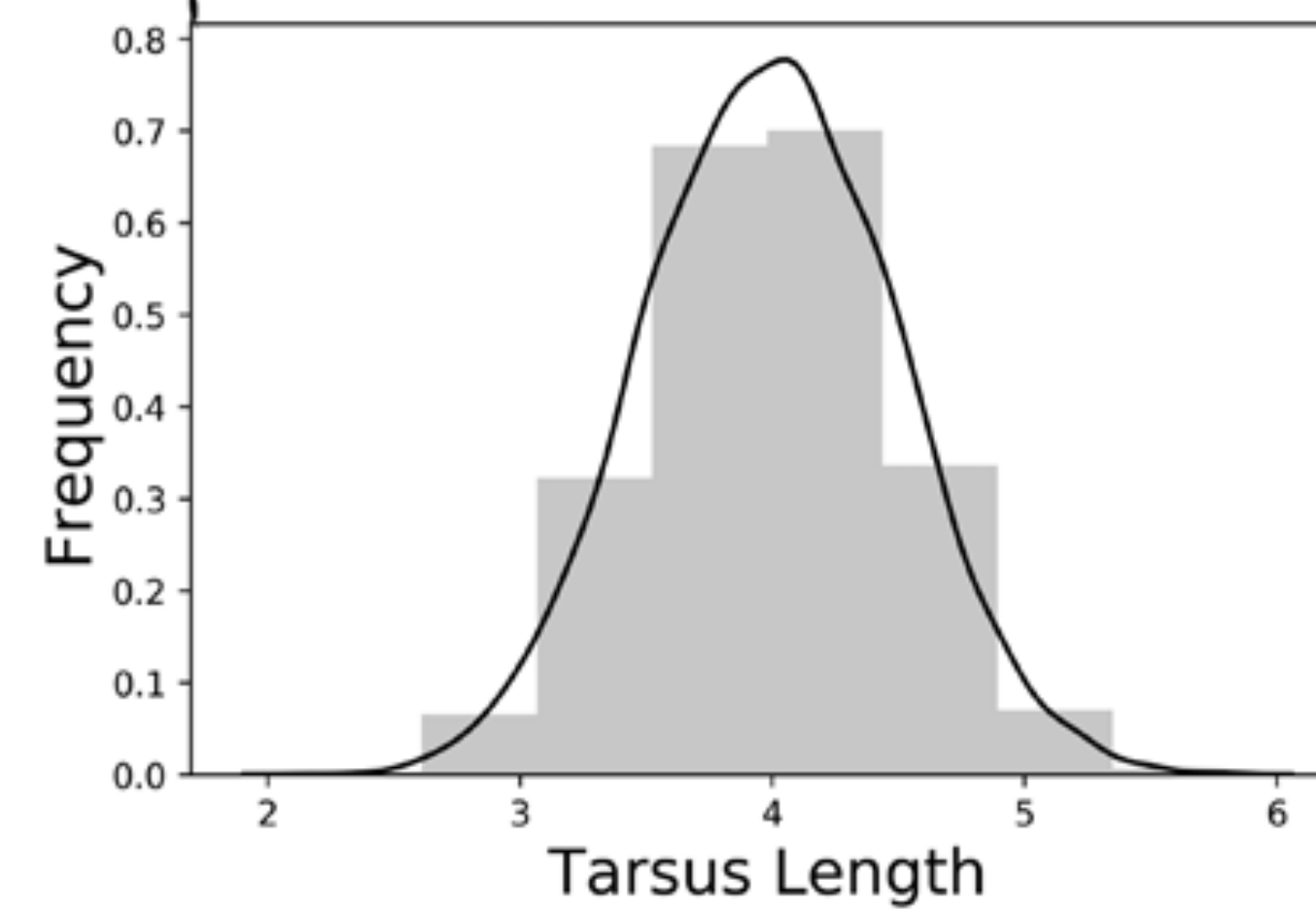
A. Metapleural Gland Opening: (0) Slit, (1) Gaping



B. Number of antenna segments States 7-13



C. Tarsus Length Continuous values



Taxon	Character 1	Character 2	Character 3
Taxon 1	0	1	1
Taxon 2	0	1	0
Taxon 3	1	0	0
Taxon 4	1	0	1

Character Likelihood (L_i) = $\text{Pr}(\text{ data } | \text{ model })$

Total Likelihood (L) = $L_1 + L_2 + L_3$

Bayes' theorem

$$P(\text{parameters} \mid \text{data, model}) = \frac{P(\text{data} \mid \text{parameters, model}) P(\text{parameters} \mid \text{model})}{P(\text{data} \mid \text{model})}$$

likelihood

priors

posterior

marginal probability of the data

Figure 1. Schematic representation of Bayes’ theorem and the Hastings ratio, used in Bayesian inference.

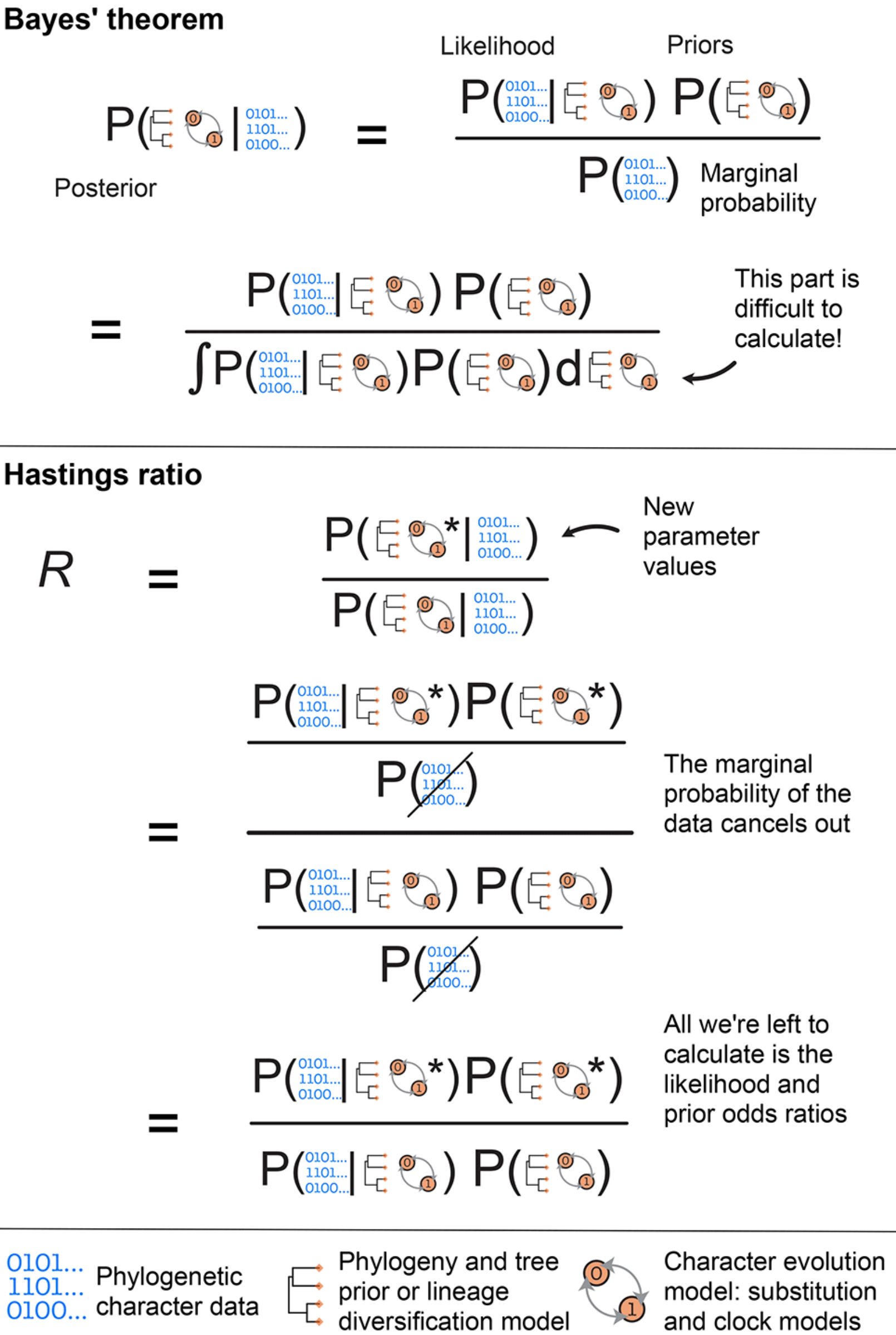
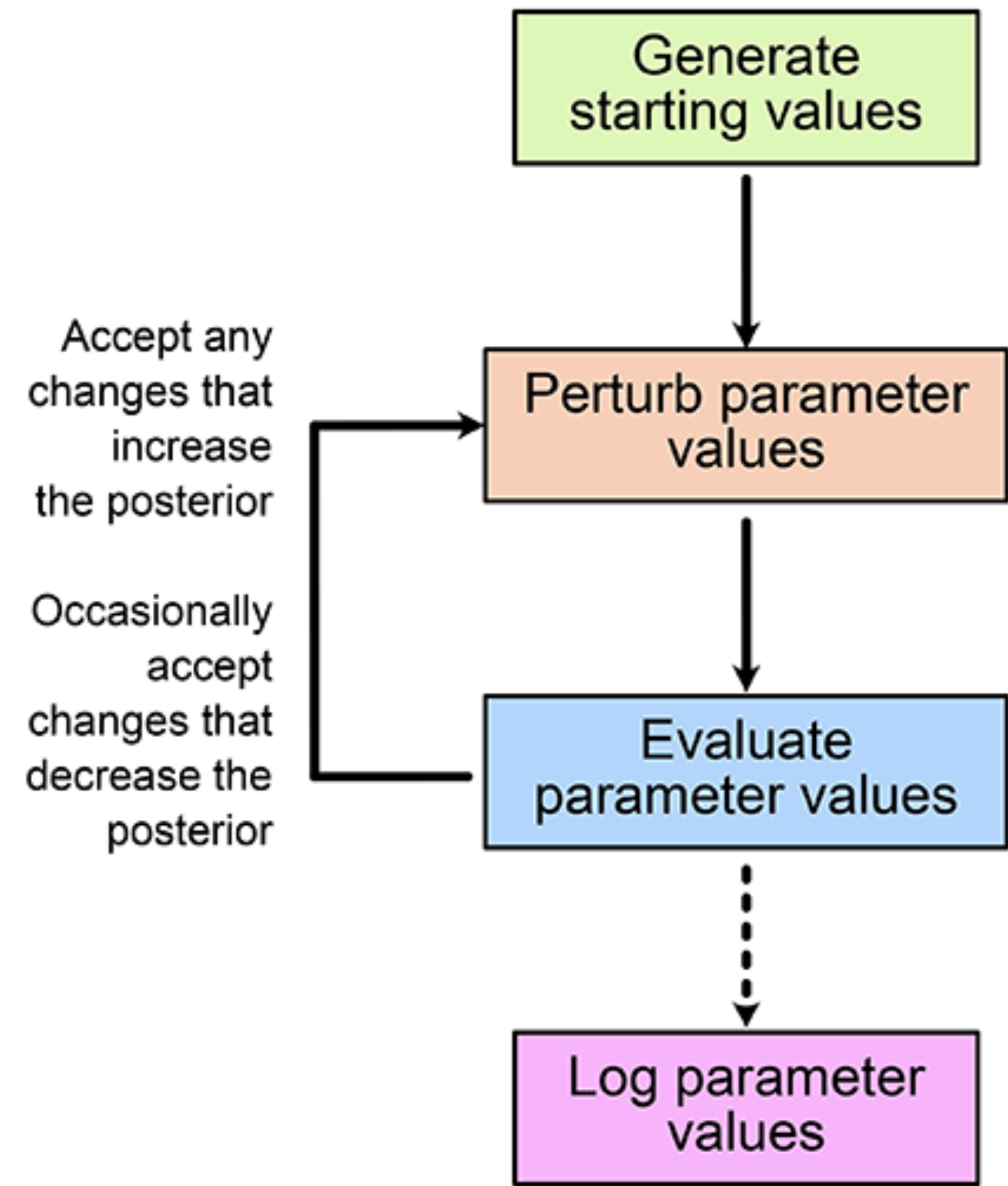


Figure 2. Flowchart and pseudocode showing the main steps in the Metropolis-Hastings algorithm.

The Metropolis-Hastings algorithm

Flowchart



Pseudocode

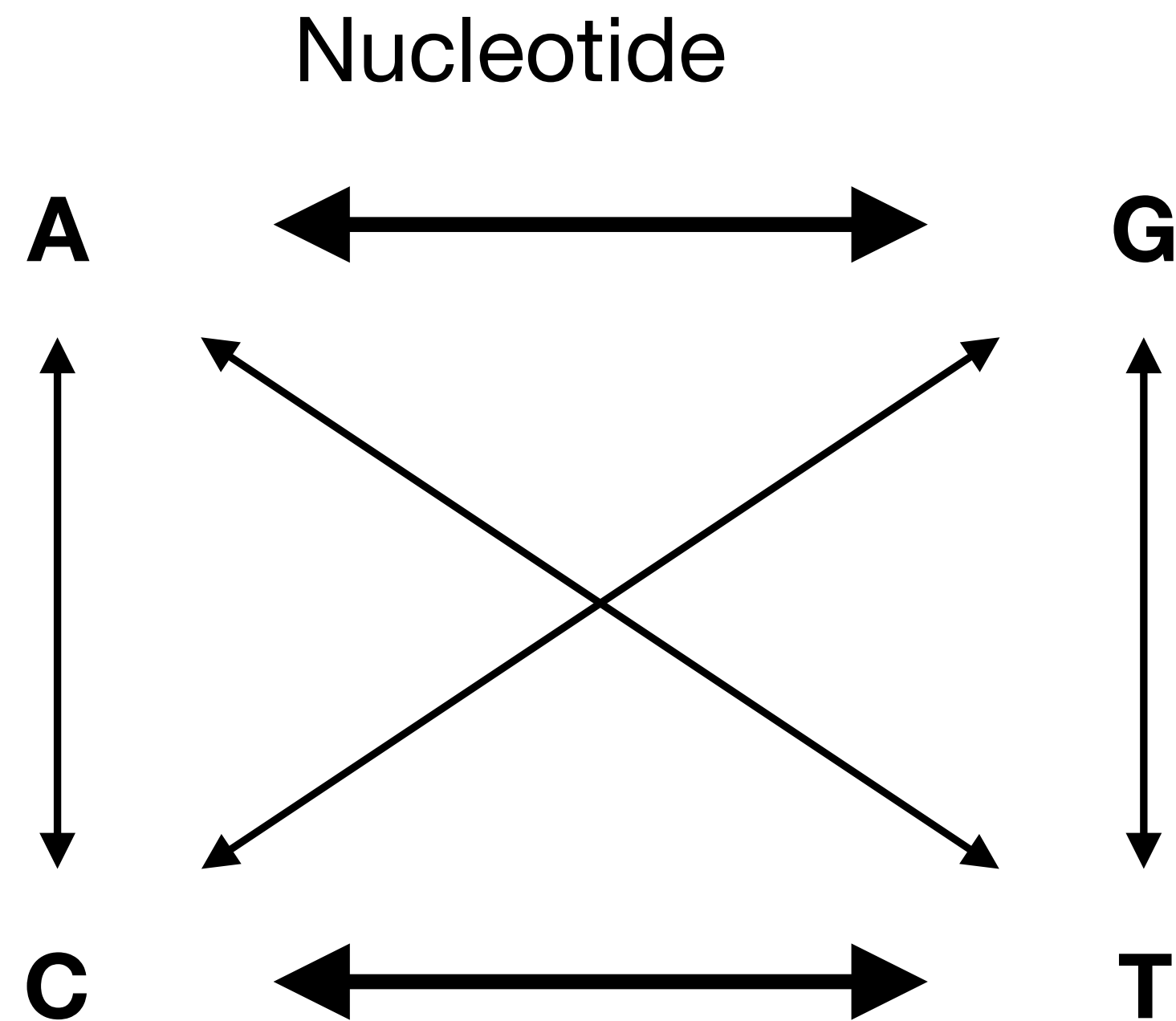
```
initialize starting values;

for i in mcmc steps
do
    propose new parameter values;
    calculate the Hastings ratio R;

    if( R > 1 )
        accept the new values;
    else
        accept the new values with Pr = R;

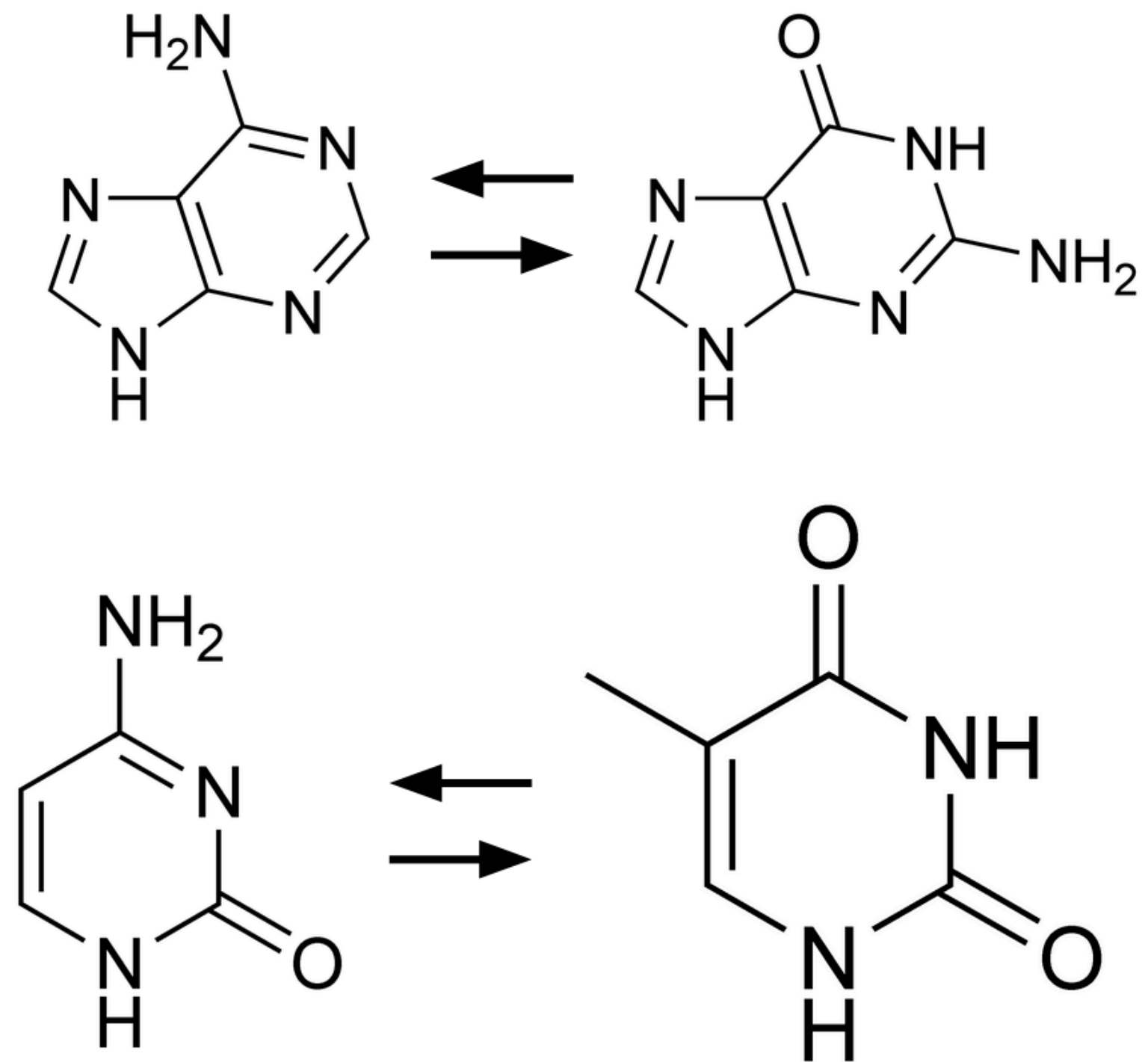
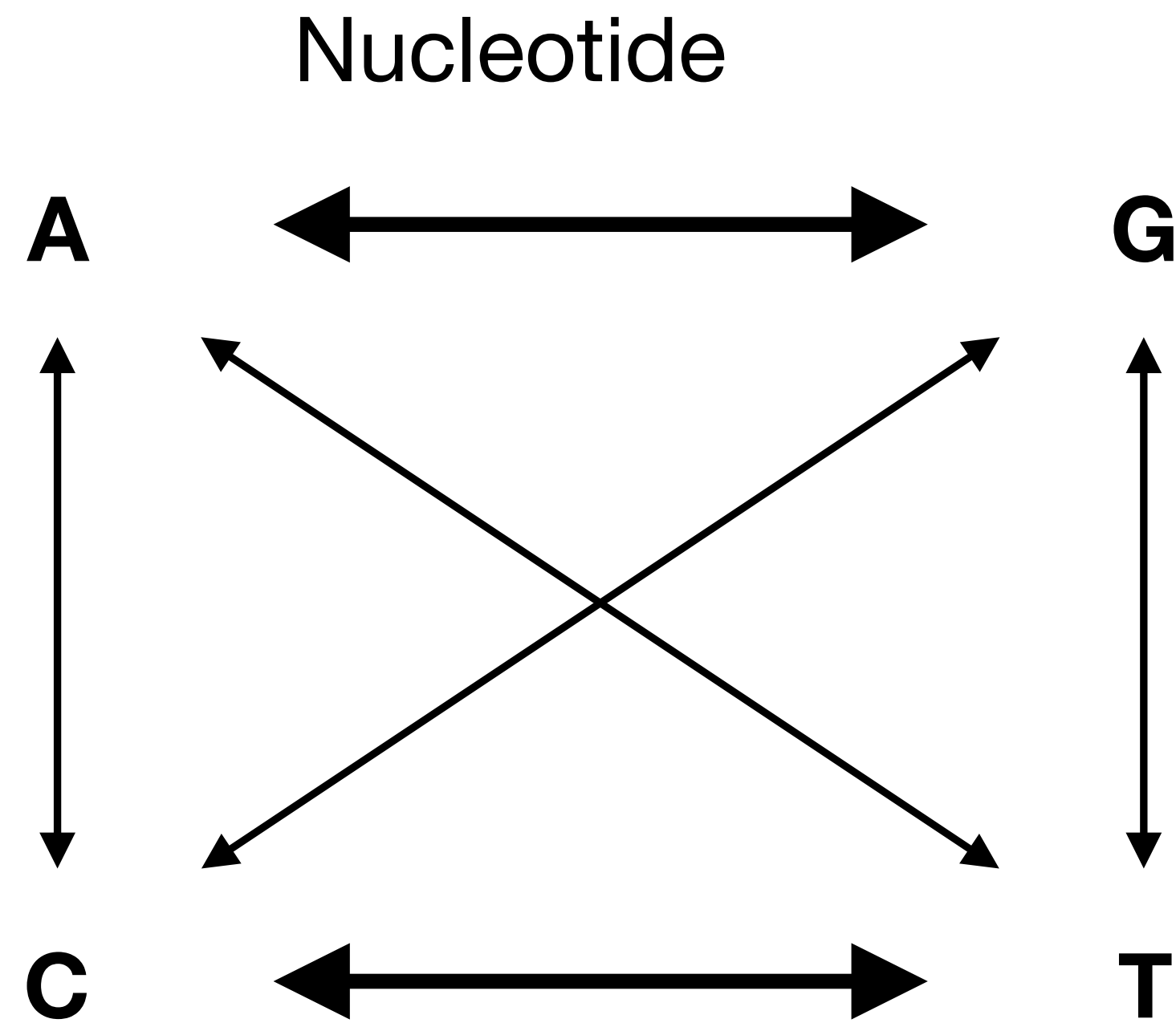
    store the values with frequency j;
done
```

Character Change Model

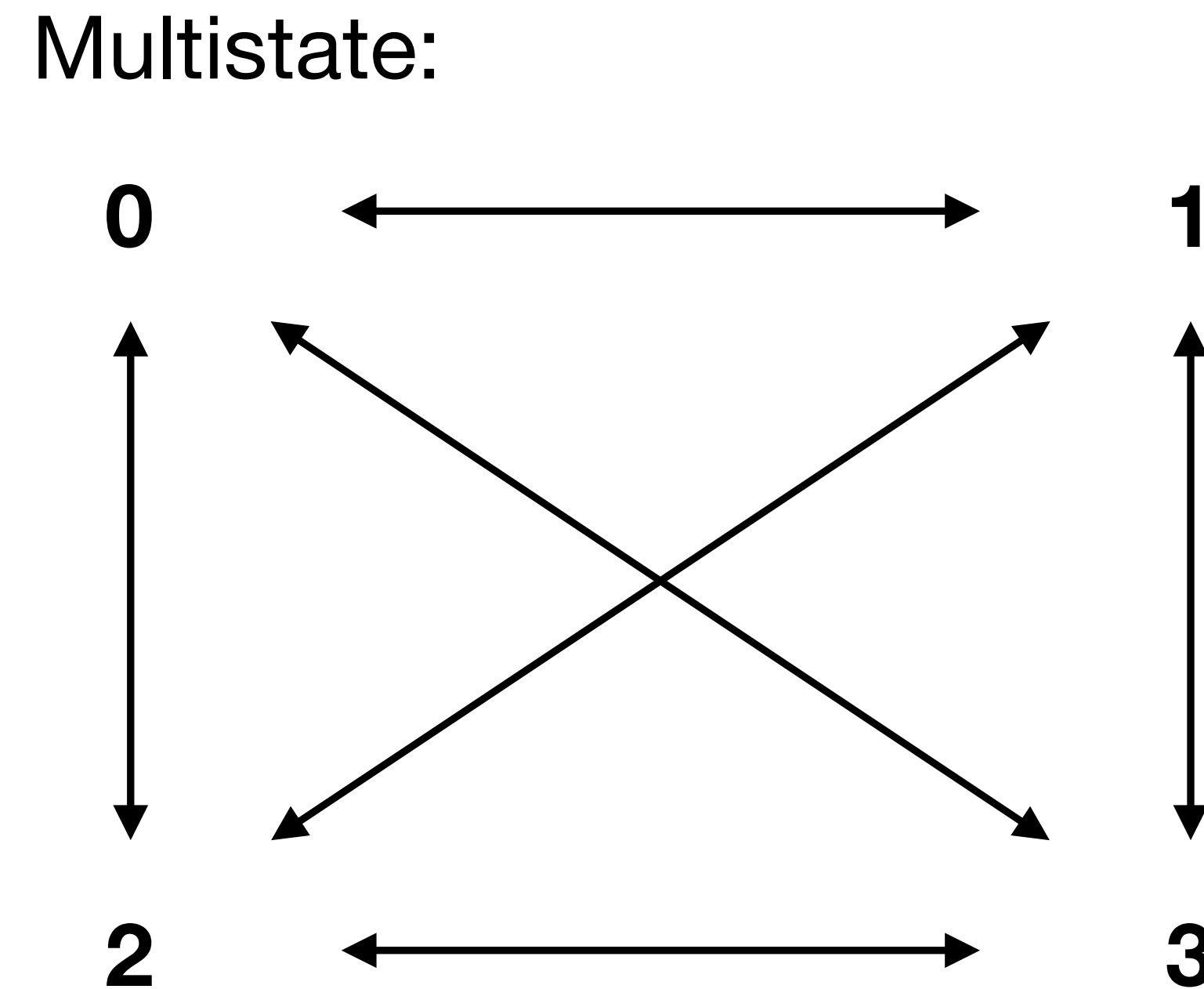
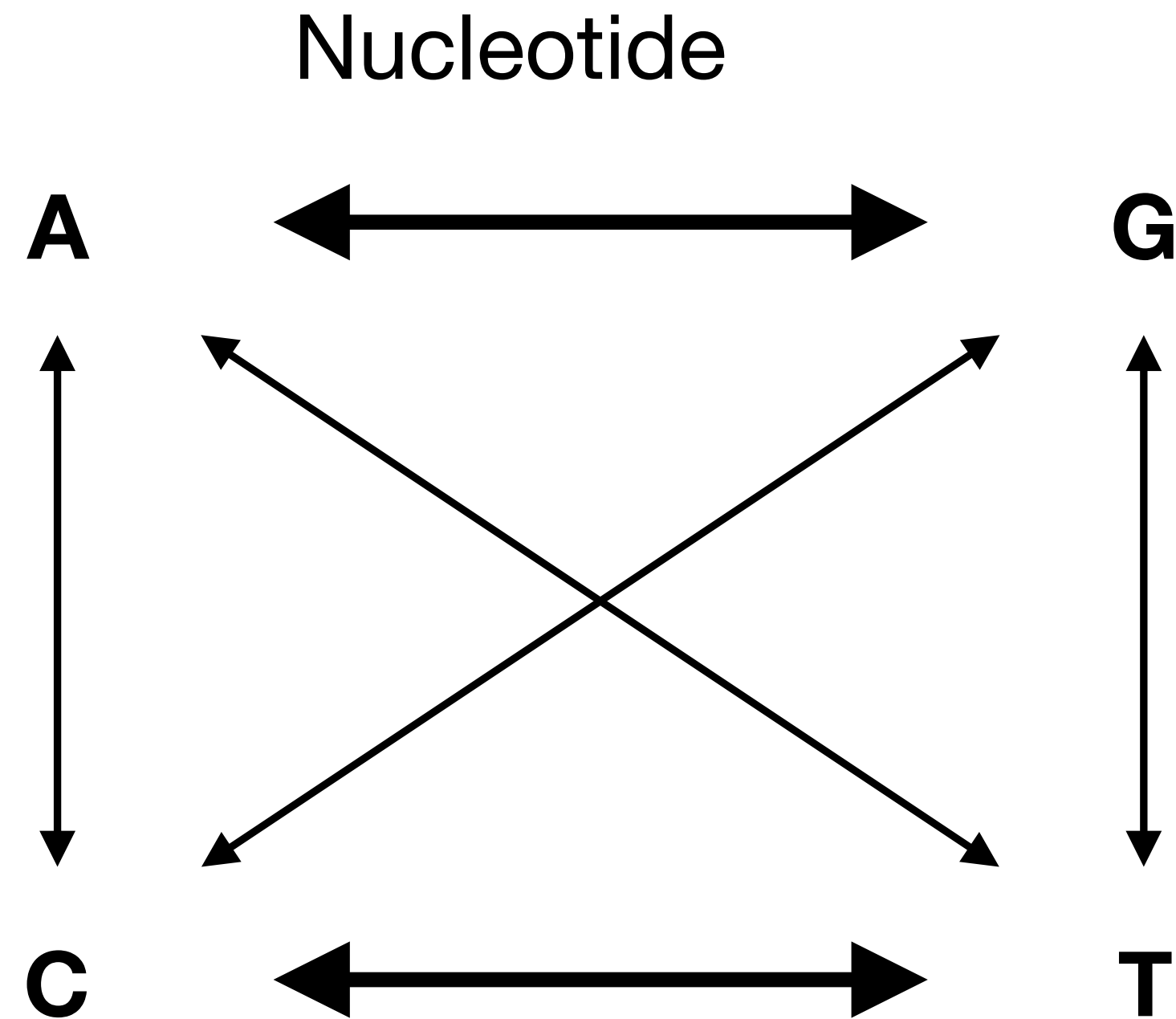


GTR model (Tavaré 1986)
K2P, F81 (Kimura 1979, Felsenstein 1981)

Character Change Model

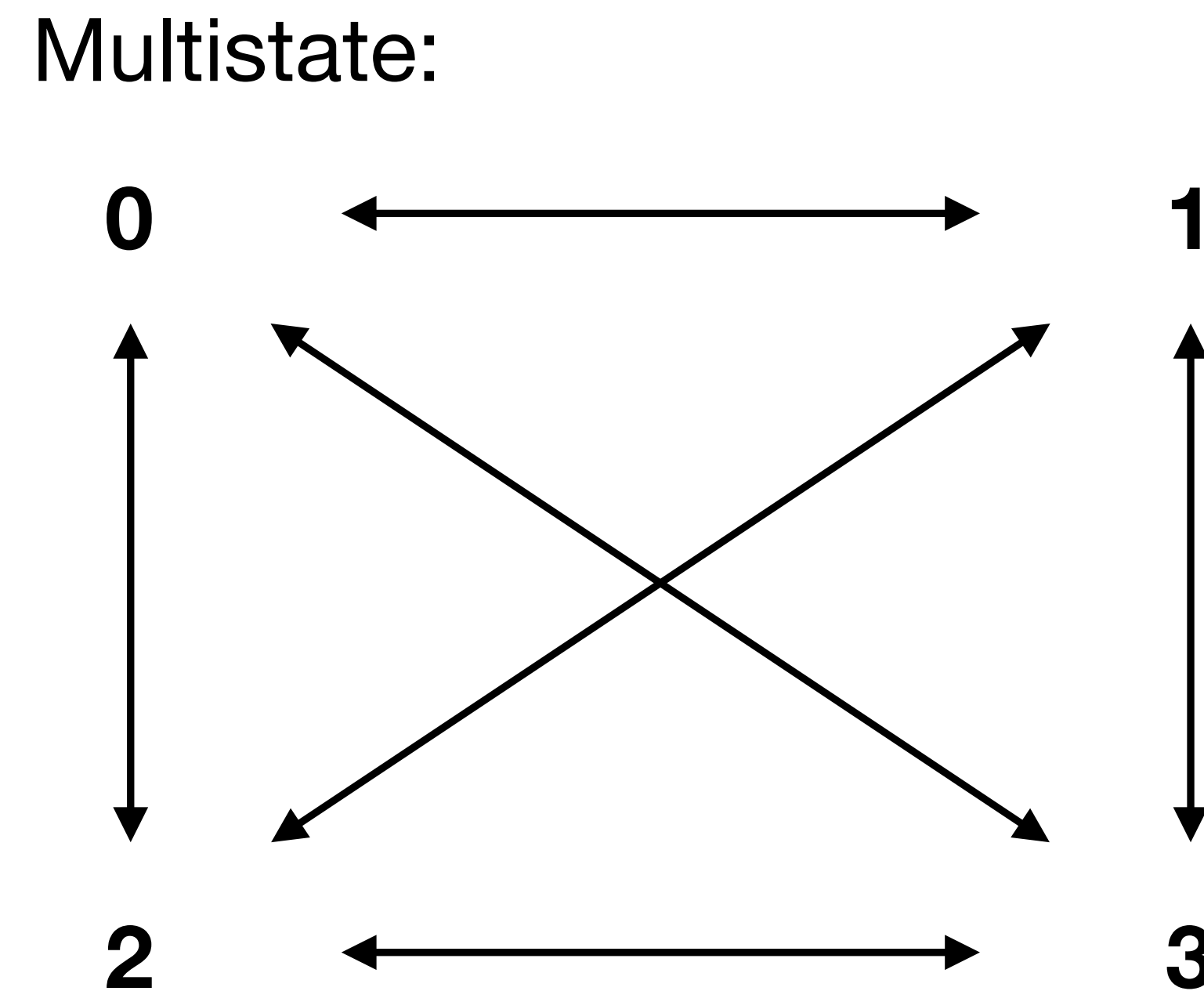
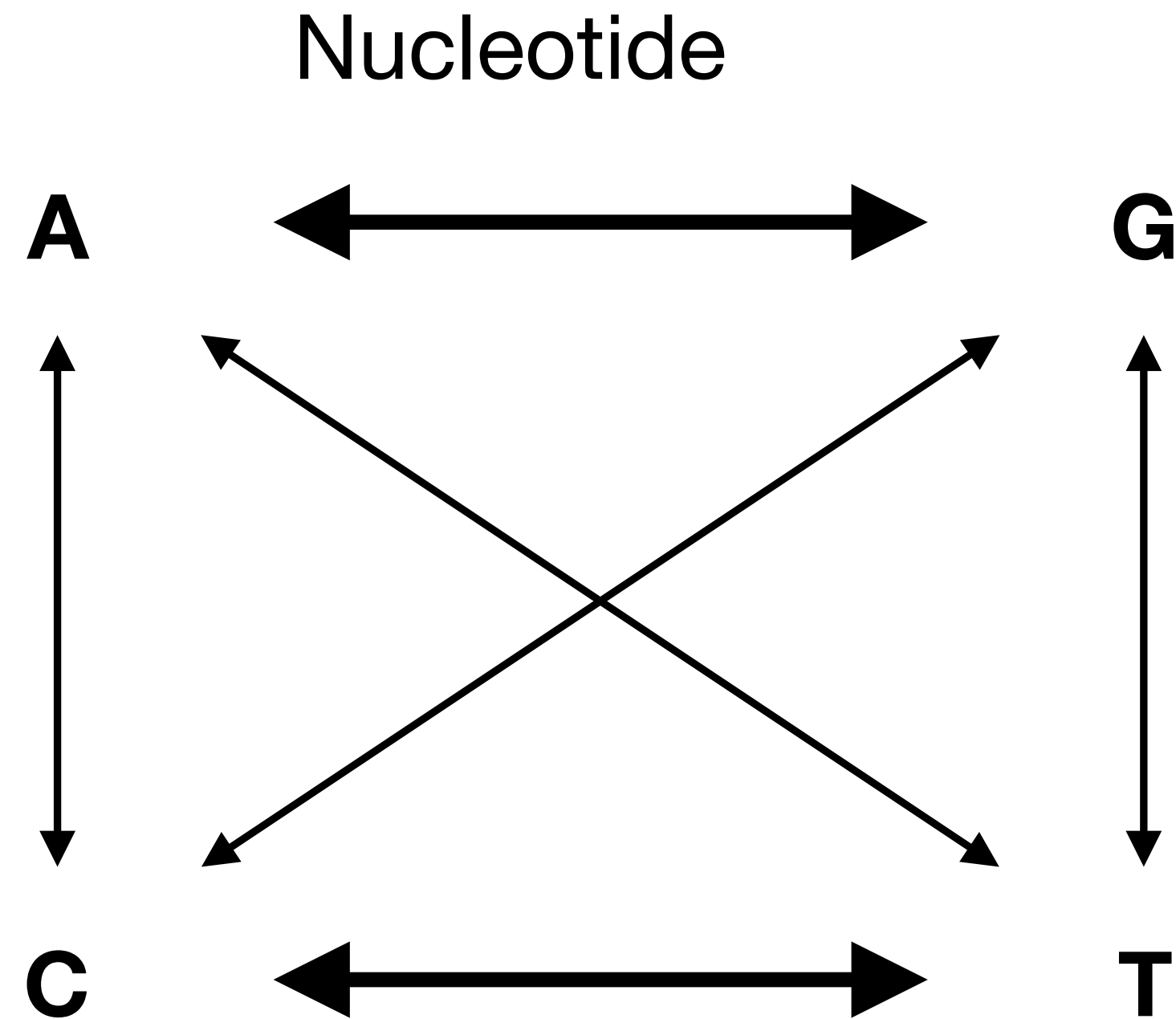


Character Change Model



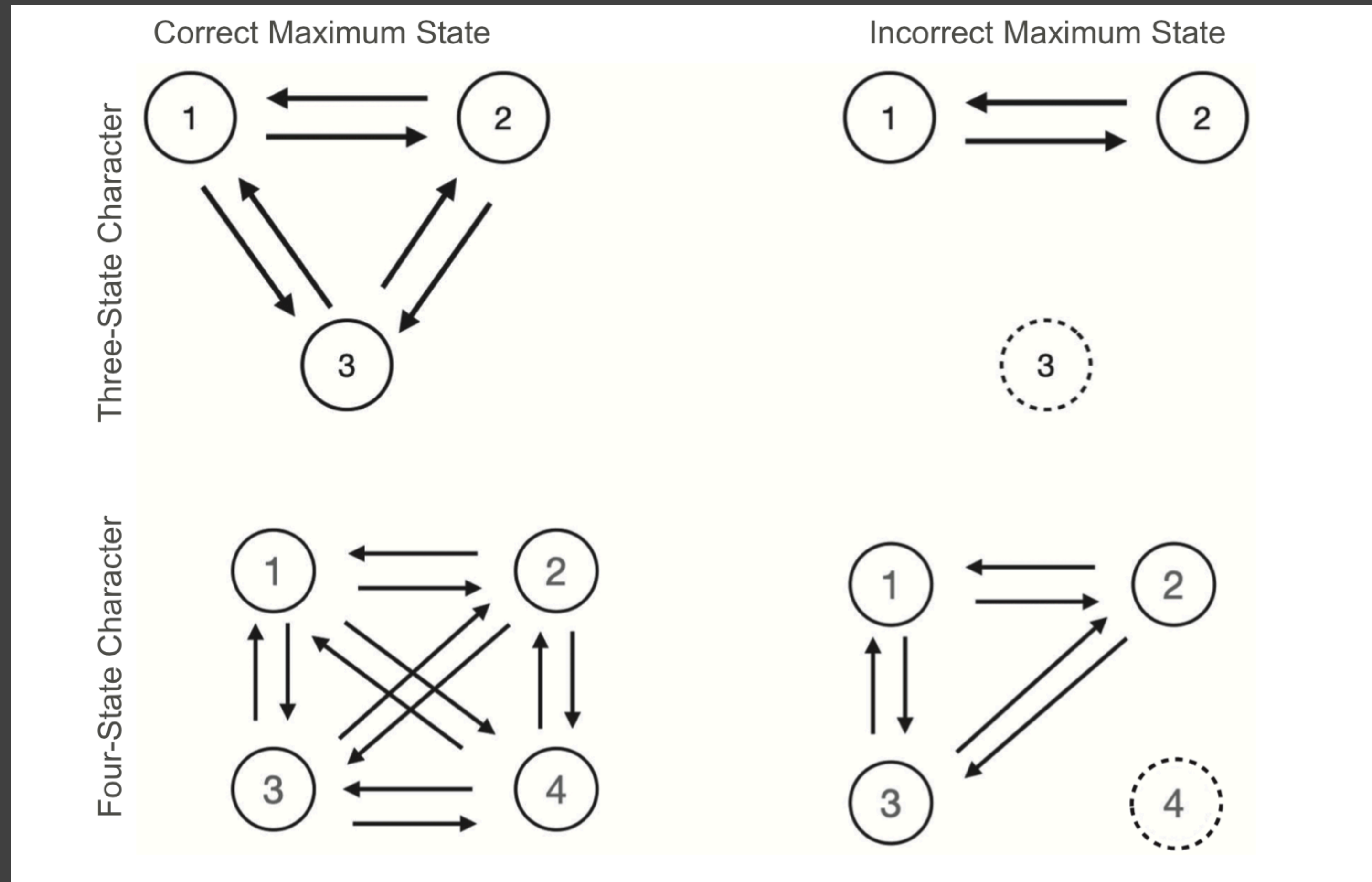
Mk model of Lewis 2001
Based on Jukes-Cantor 1969
Works well enough most of the time
(Wright and Hillis 2014)

Character Change Model

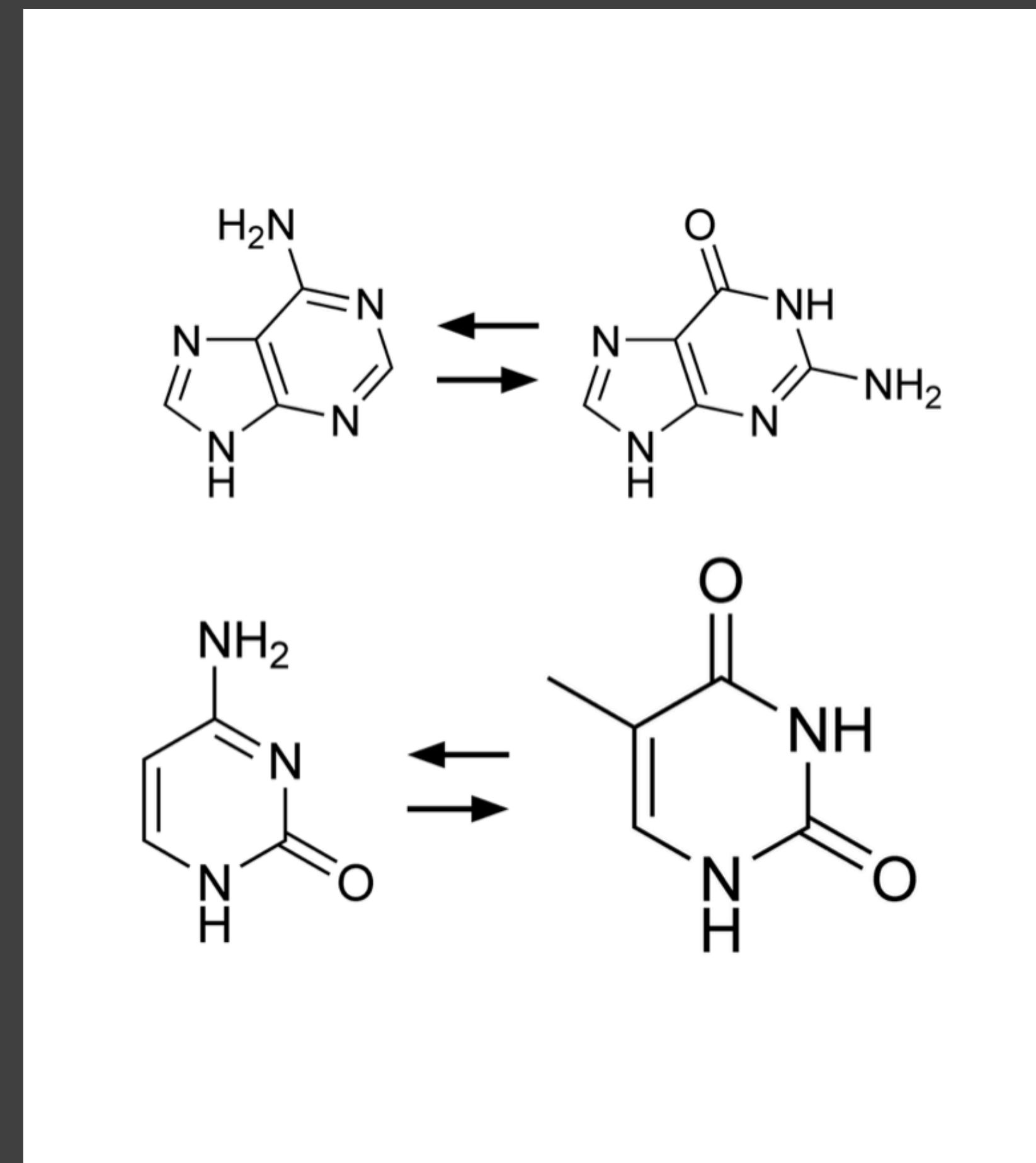
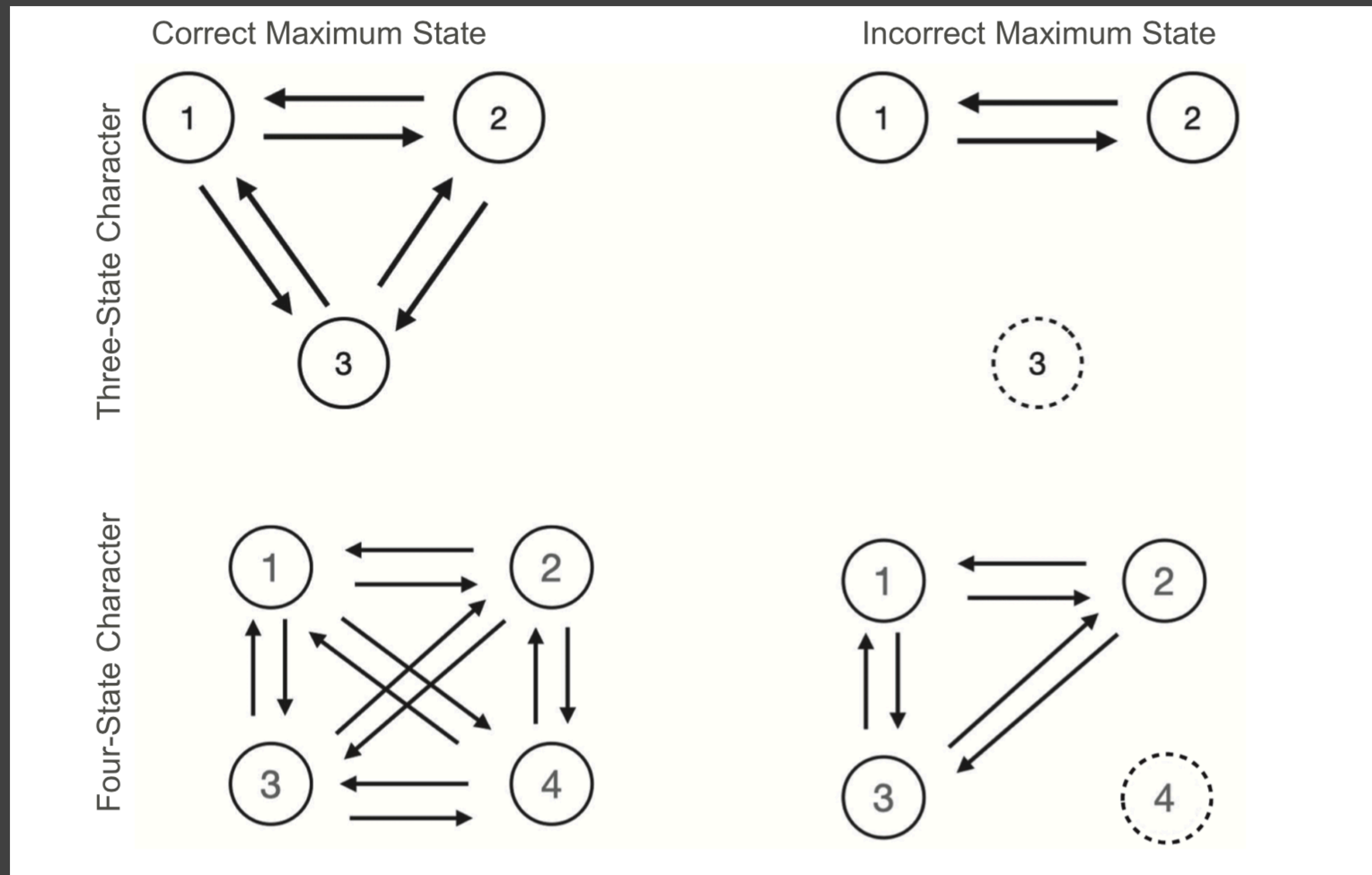


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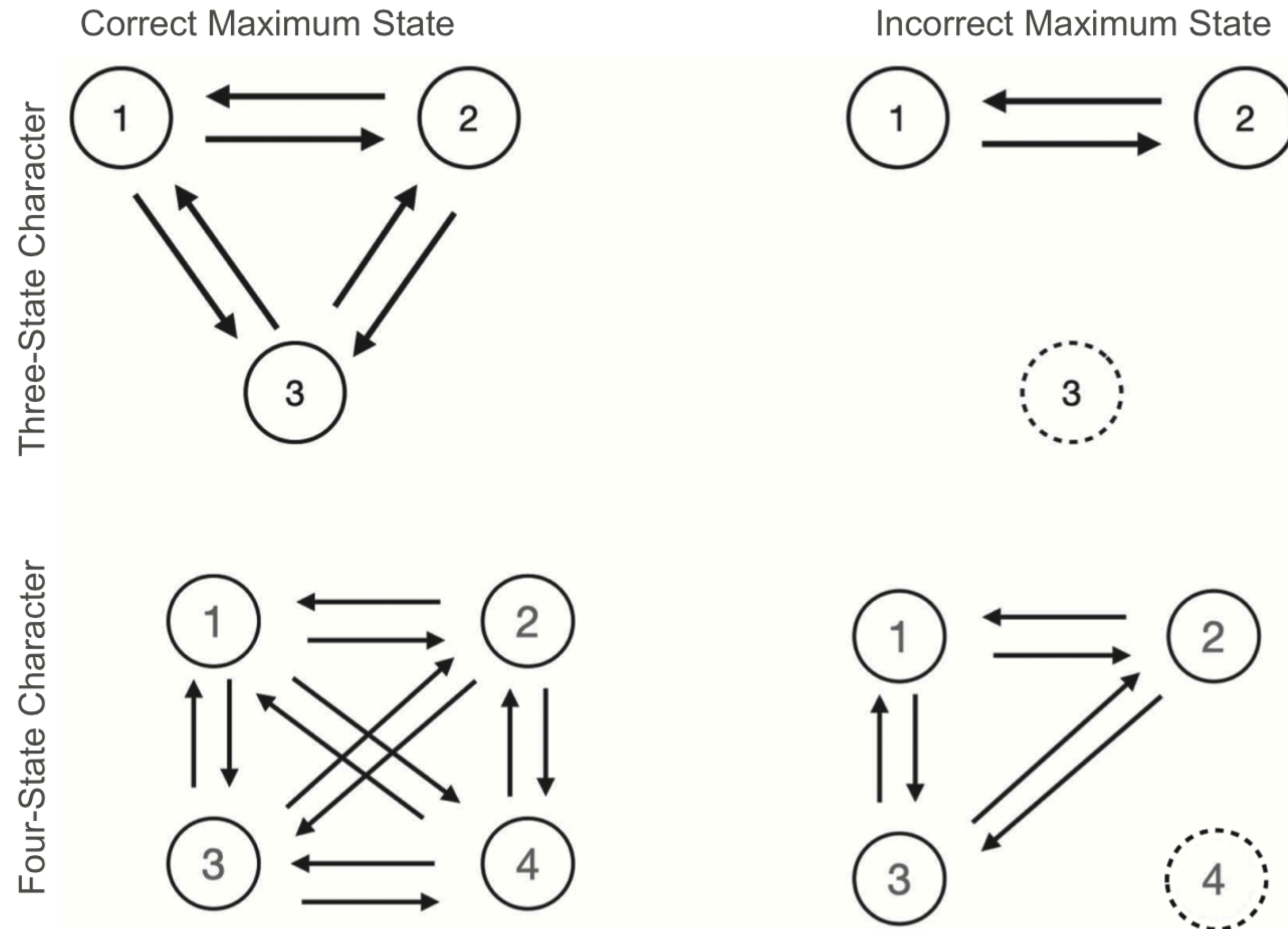
How do we model morphological data



How do we model morphological data



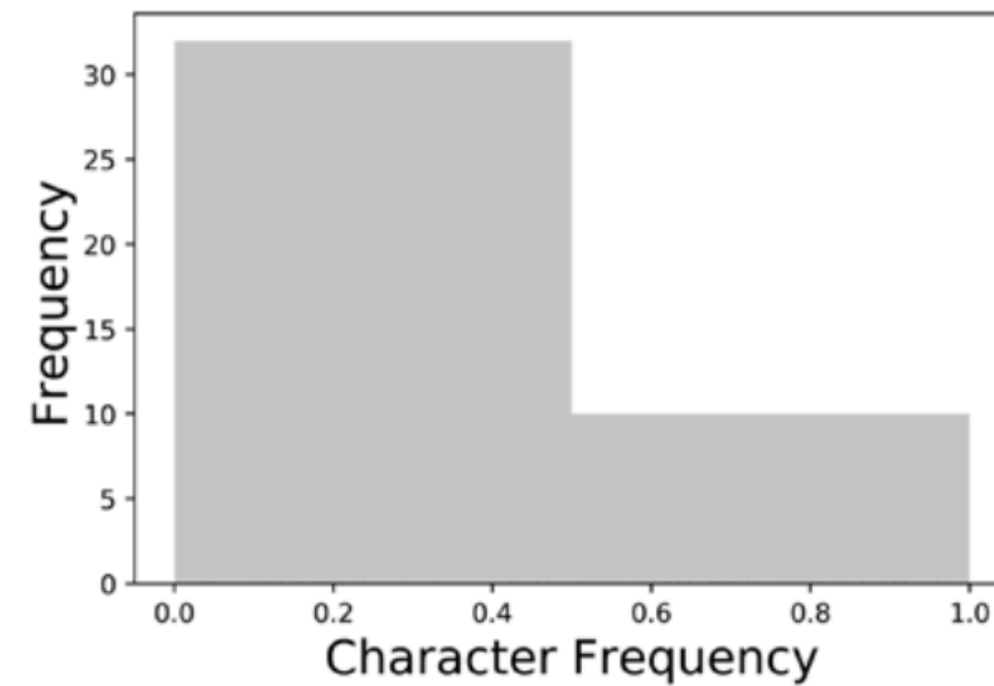
How do we model morphological data



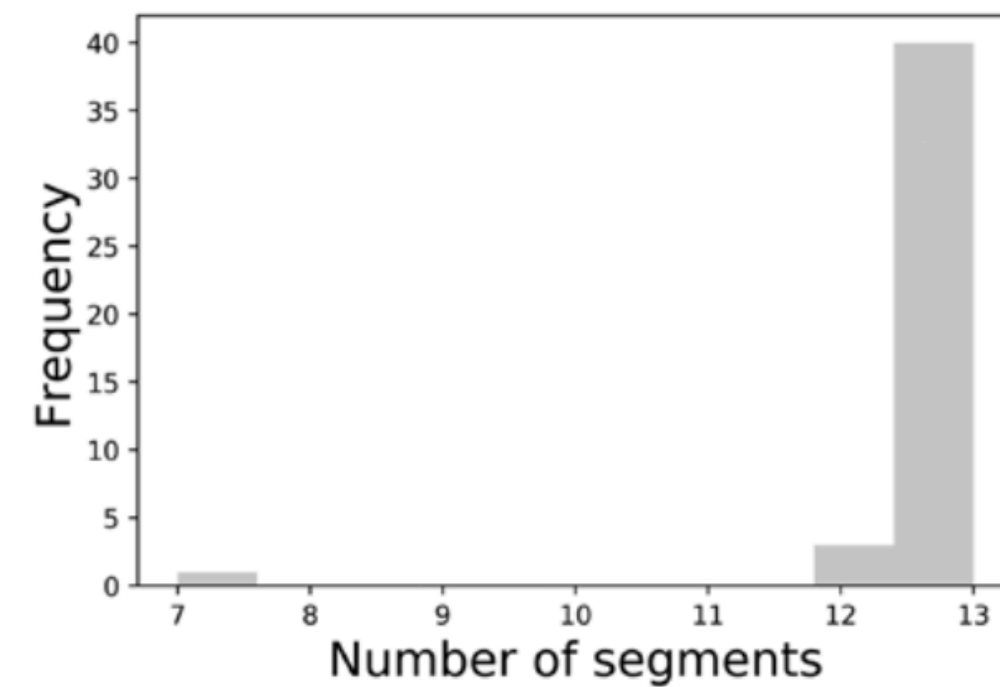
- On the fly assessment of state spaces
 - Khakurel et al 2024
 - Mulvey et al 2025

How do we model morphological data

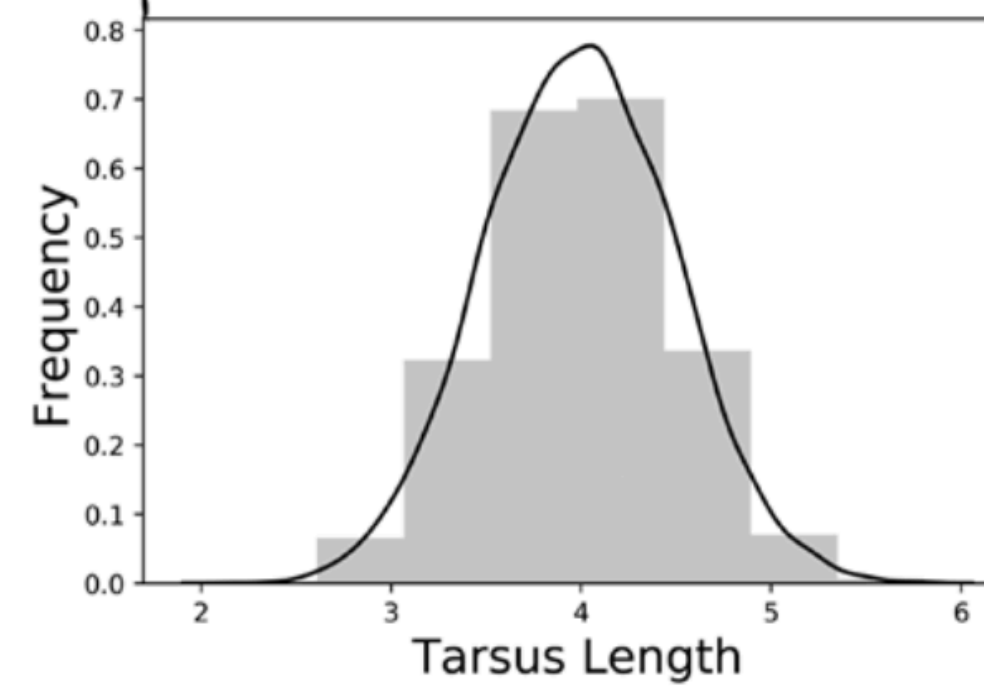
A. Metapleural Gland Opening: (0) Slit, (1) Gaping



**B. Number of antenna segments
States 7-13**



**C. Tarsus Length
Continuous values**



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a $Q = \begin{pmatrix} -\mu_0 & \mu_1 \\ \mu_0 & -\mu_1 \end{pmatrix},$ $0 \longleftrightarrow 1$

b $Q = \begin{pmatrix} -\mu_0 & \mu_1 & \mu_2 & \mu_3 \\ \mu_0 & -\mu_1 & \mu_2 & \mu_3 \\ \mu_0 & \mu_1 & -\mu_2 & \mu_3 \\ \mu_0 & \mu_1 & \mu_2 & -\mu_3 \end{pmatrix},$

c $Q = \begin{pmatrix} -\mu_0\pi_0 & \mu_1\pi_1 \\ \mu_0\pi_0 & -\mu_1\pi_1 \end{pmatrix},$ $0 \rightleftharpoons 1$

d

	0	1
0	-	3
1	1	-

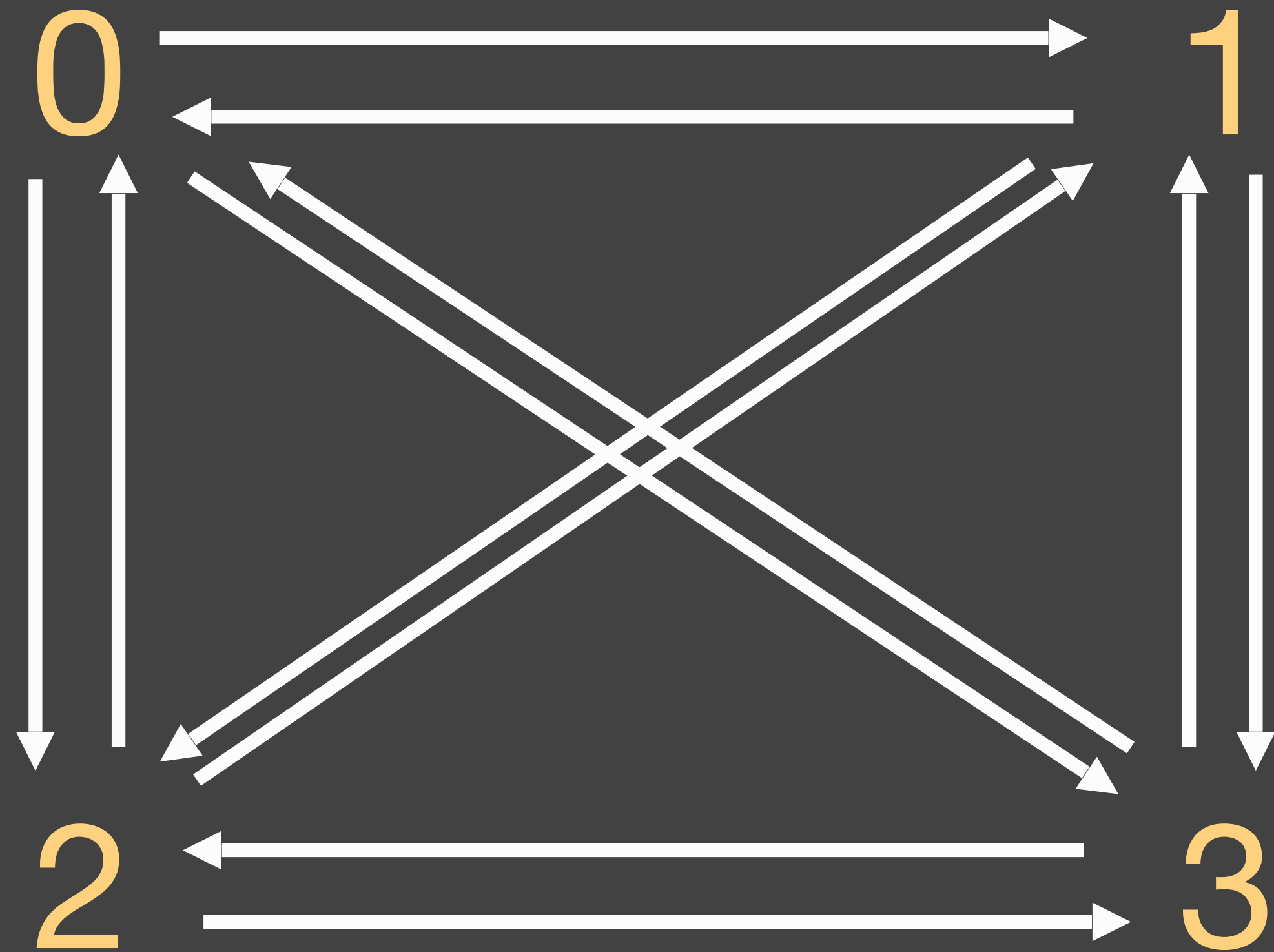
 $0 \rightleftharpoons 1$

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 - Khakurel et al 2024
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- No common meaning to characters

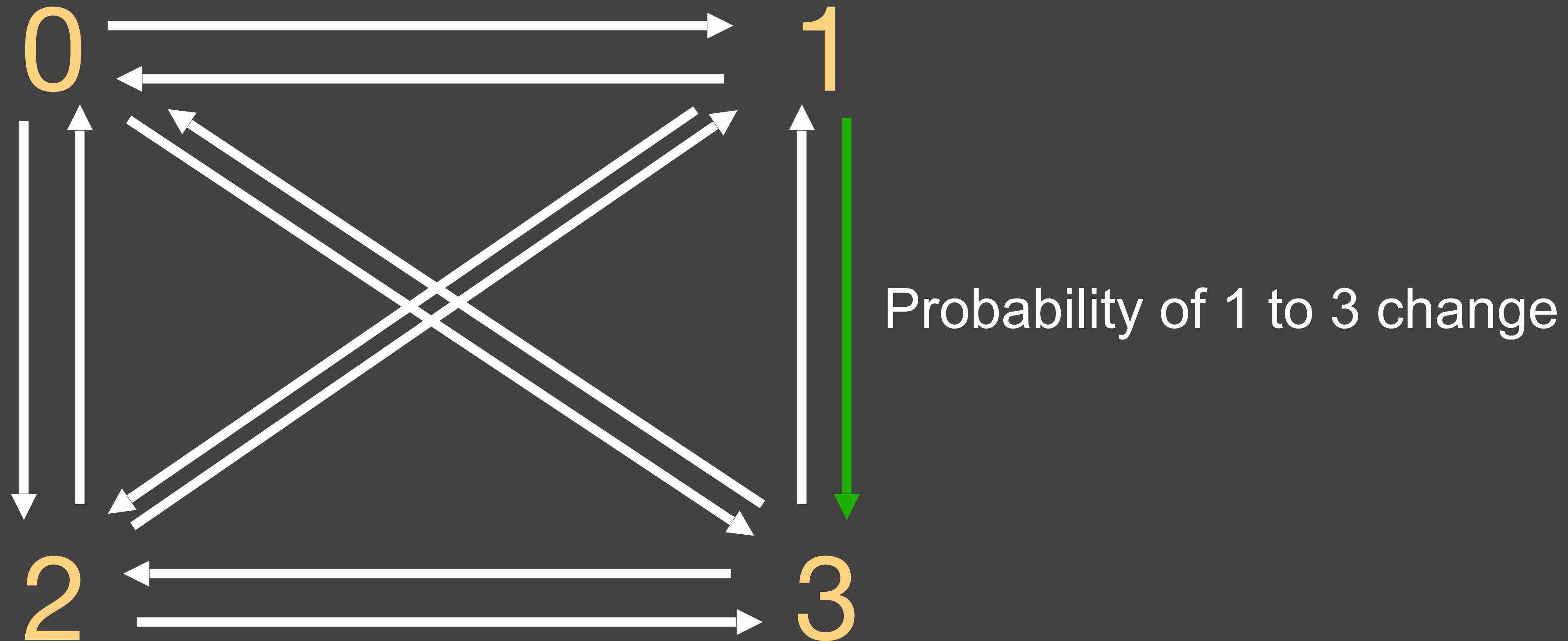
Relaxing this assumption

- In Bayesian estimation, we can put priors on the parameters in our analyses
- Transition probabilities are the product of exchangeabilities and frequencies

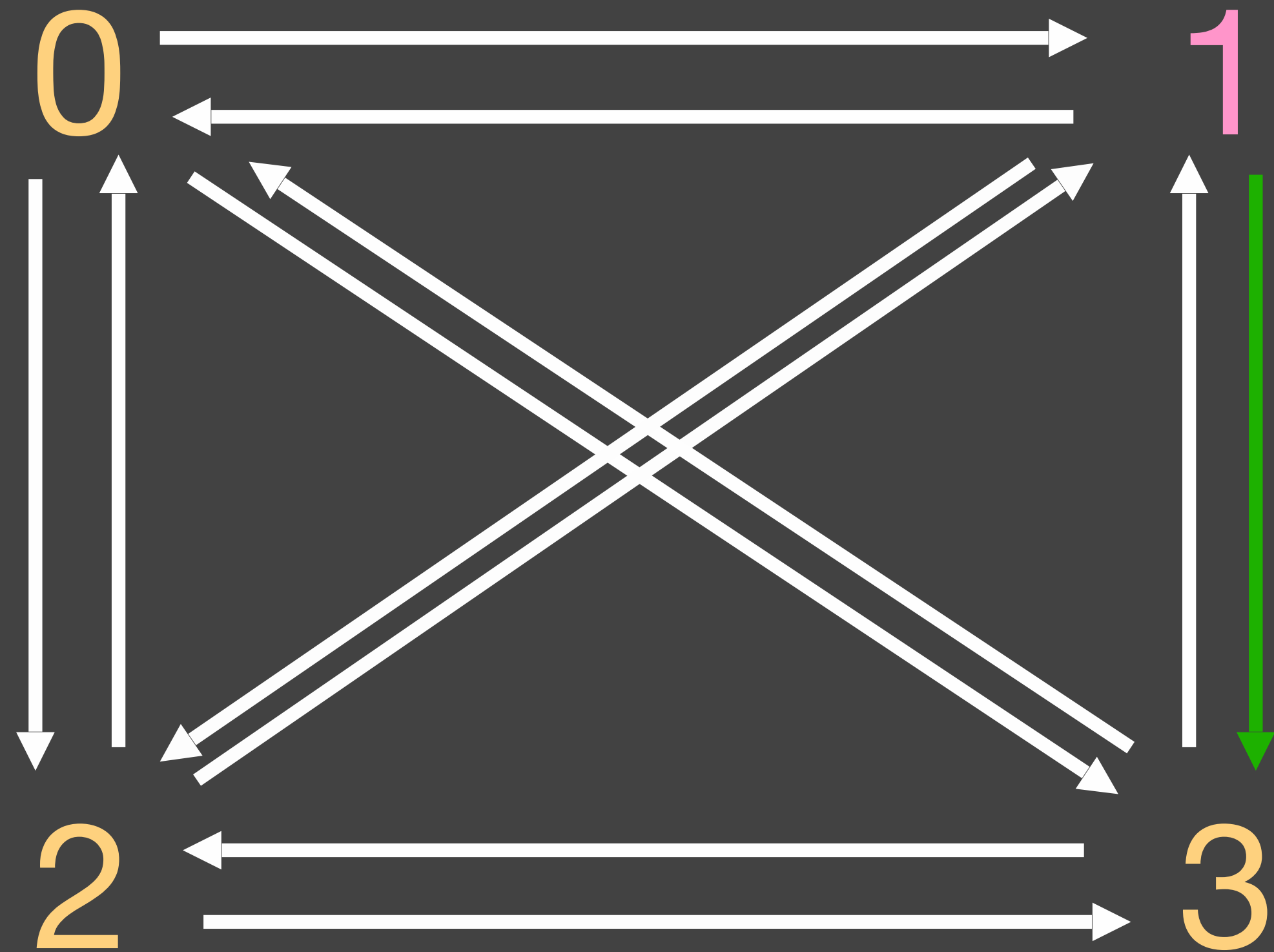
Relaxing this assumption



Relaxing this assumption

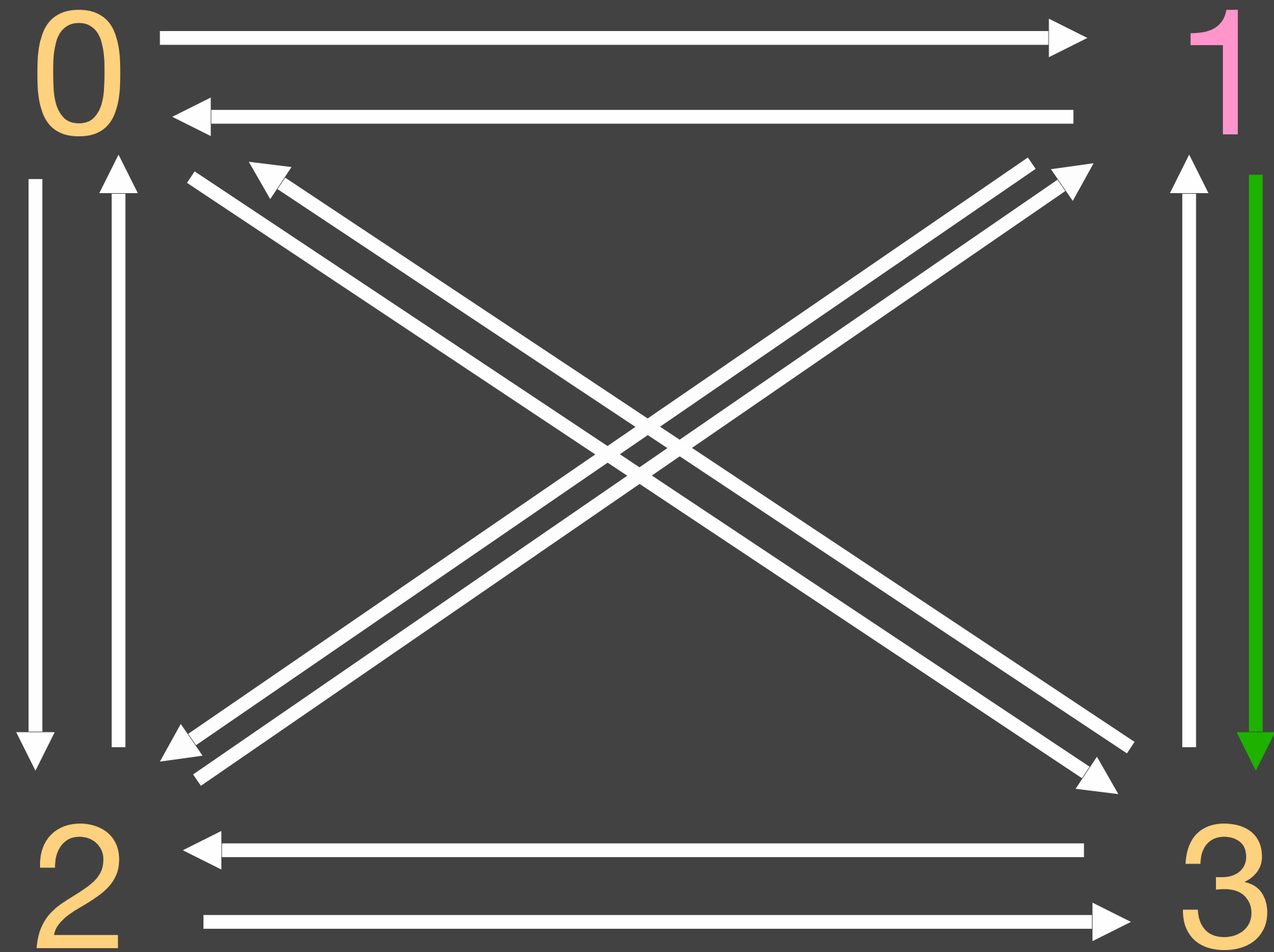


Relaxing this assumption



Probability of 1 to 3 change (exchangeability)
Equilibrium frequency of 1

Relaxing this assumption

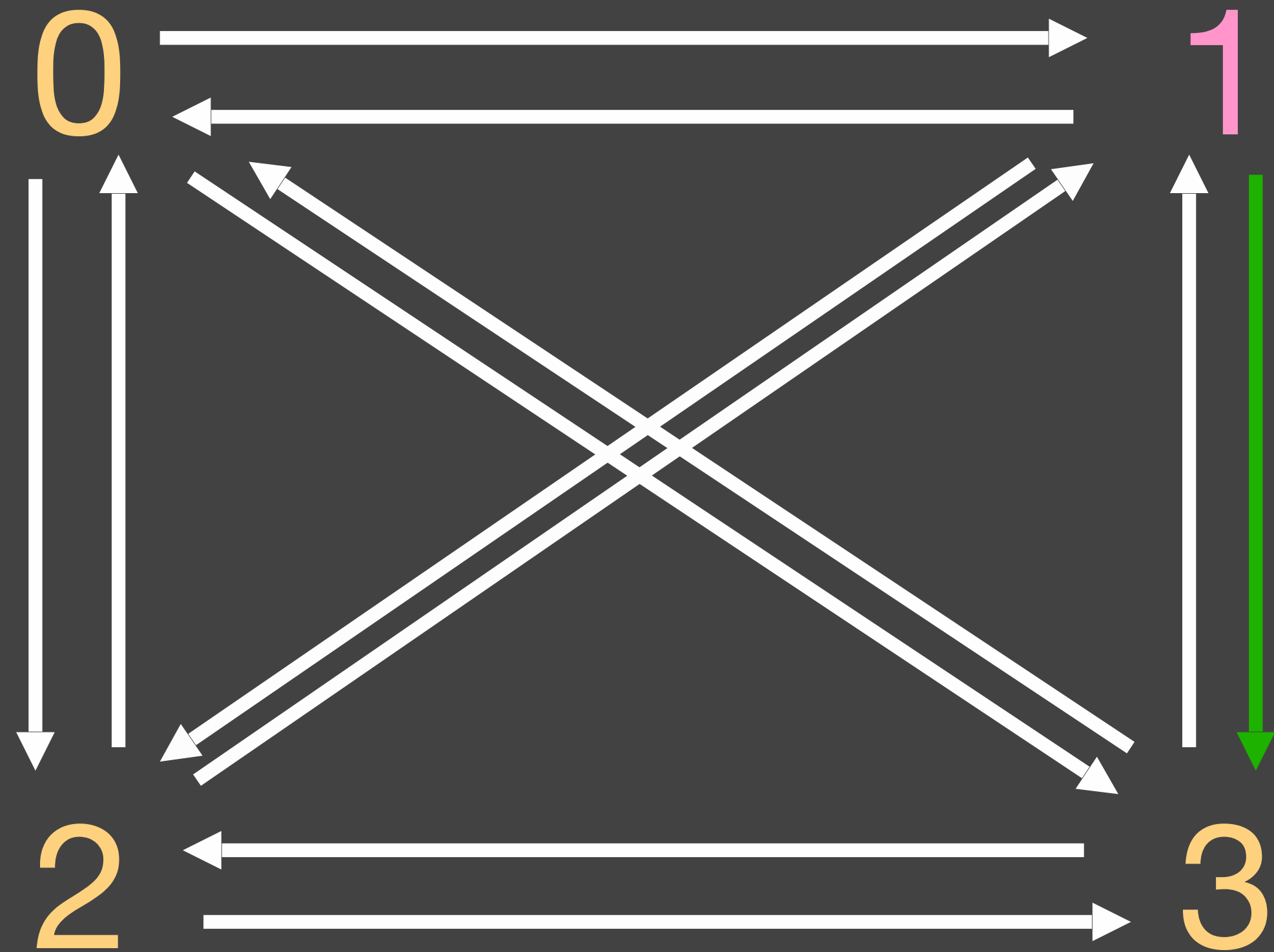


Probability of 1 to 3 change (exchangeability)

Equilibrium frequency of 1

$$.75 * .0 = 0$$

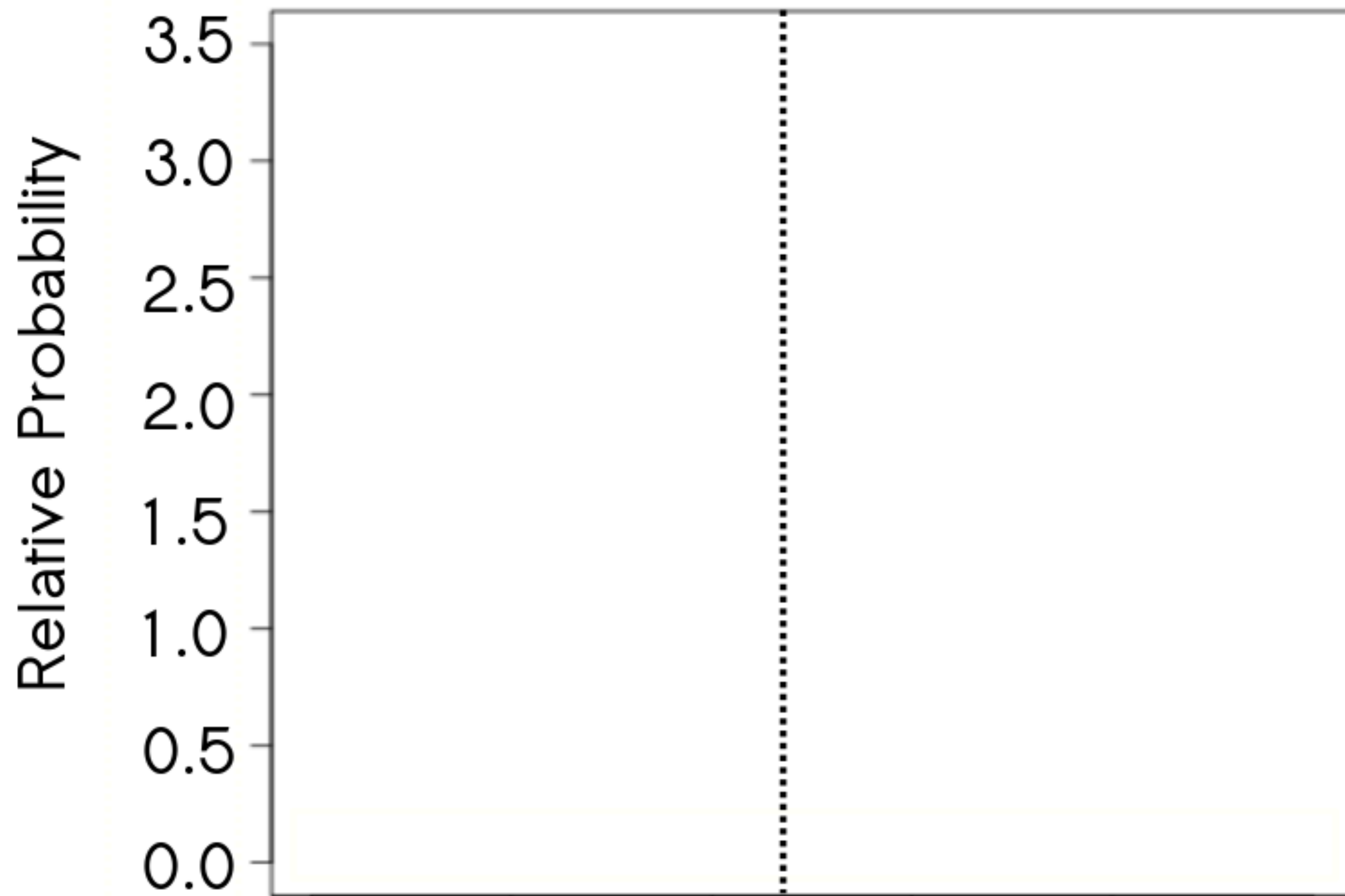
Relaxing this assumption



Probability of 1 to 3 change (exchangeability)

Equilibrium frequency of 1

$$.75 * .25 = 0.1875$$



Parameters

$$\alpha = \infty$$

0 to 1 transition rate

1 to 0 transition rate

0

50

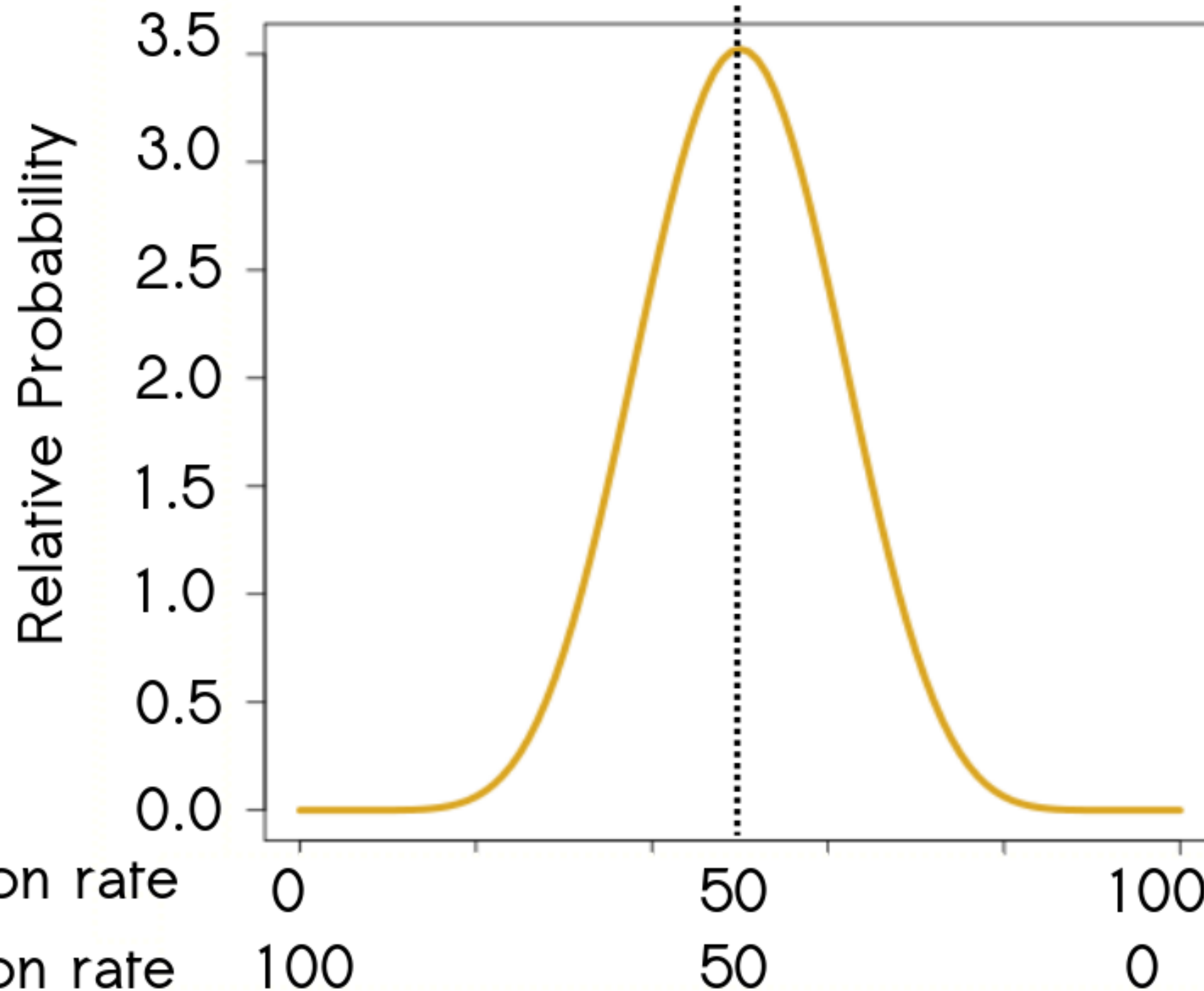
100

100

50

0

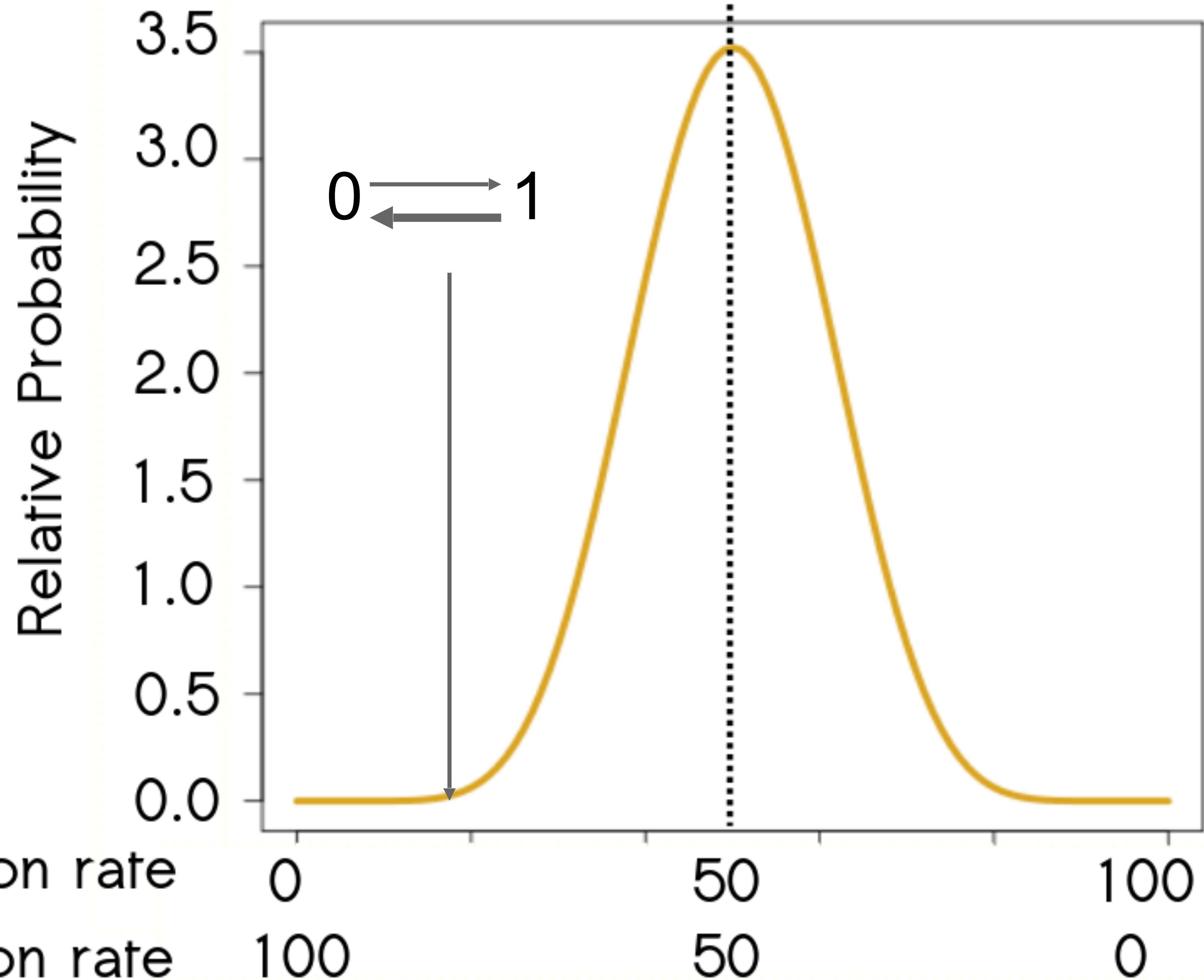
0 to 1 transition rate
1 to 0 transition rate



Parameters

$$\alpha = \infty$$

$$\alpha = 10$$

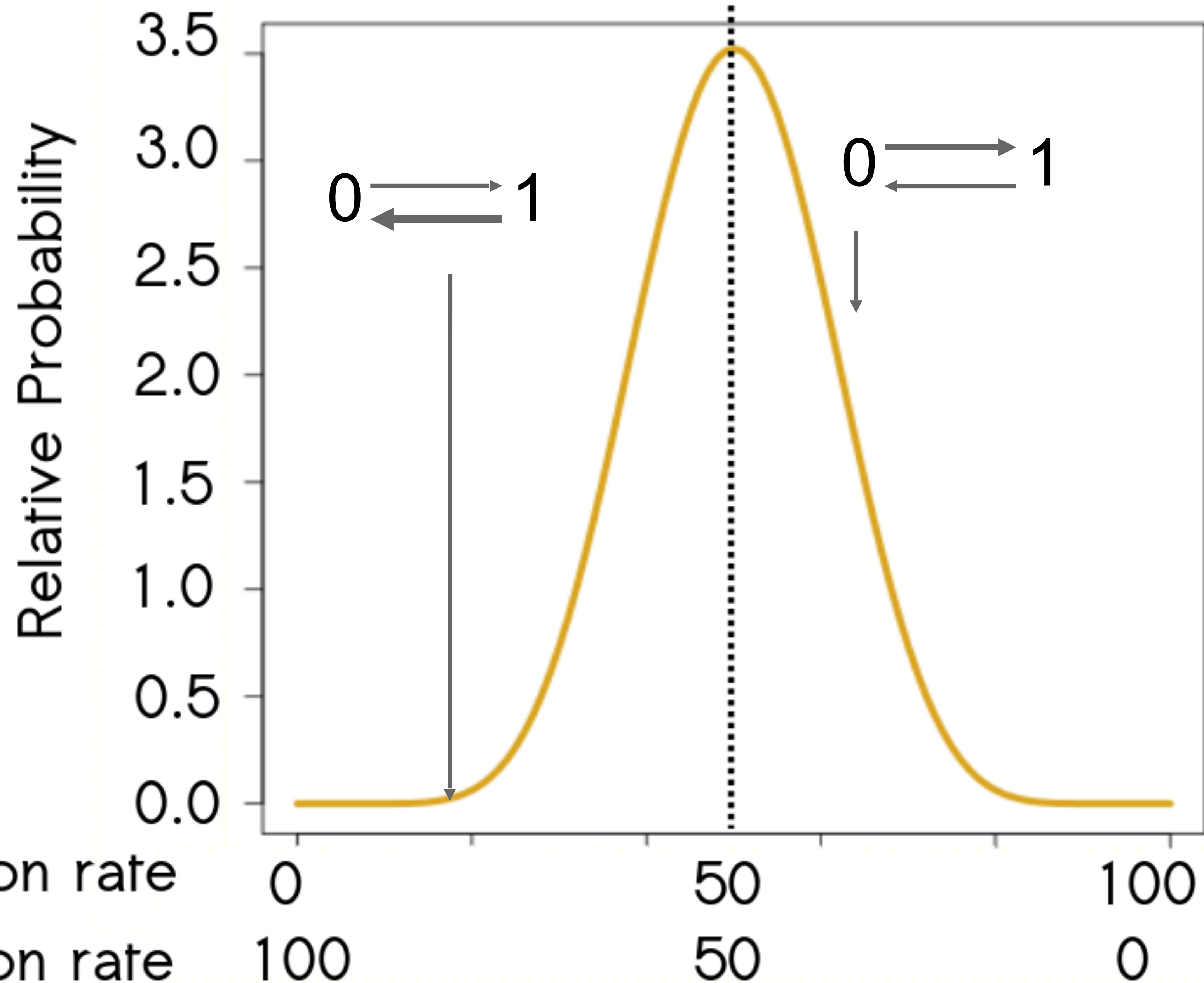


Parameters

$$\alpha = \infty$$

$$\alpha = 10$$

0 to 1 transition rate
1 to 0 transition rate

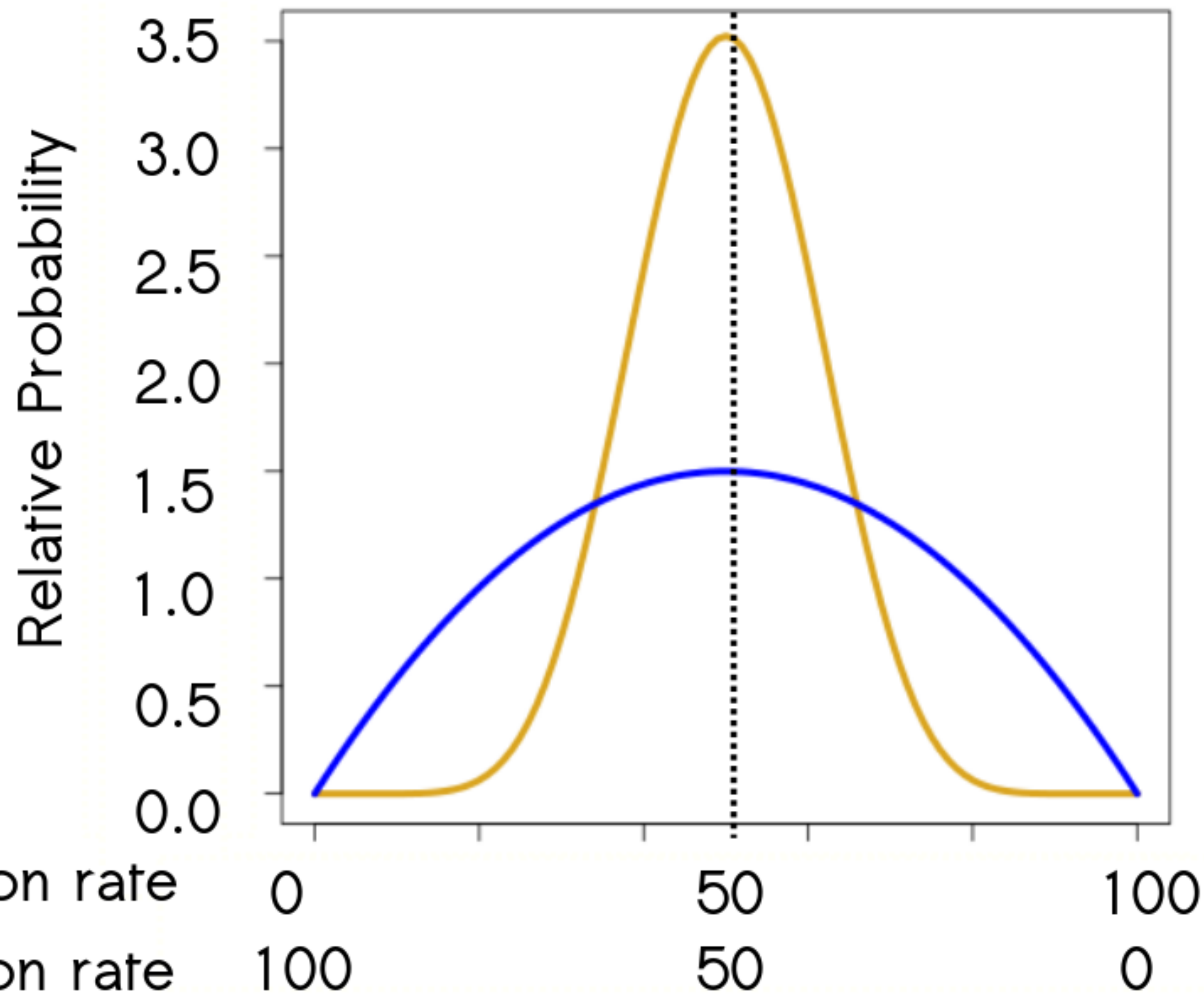


Parameters

$$\alpha = \infty$$

$$\alpha = 10$$

0 to 1 transition rate
1 to 0 transition rate



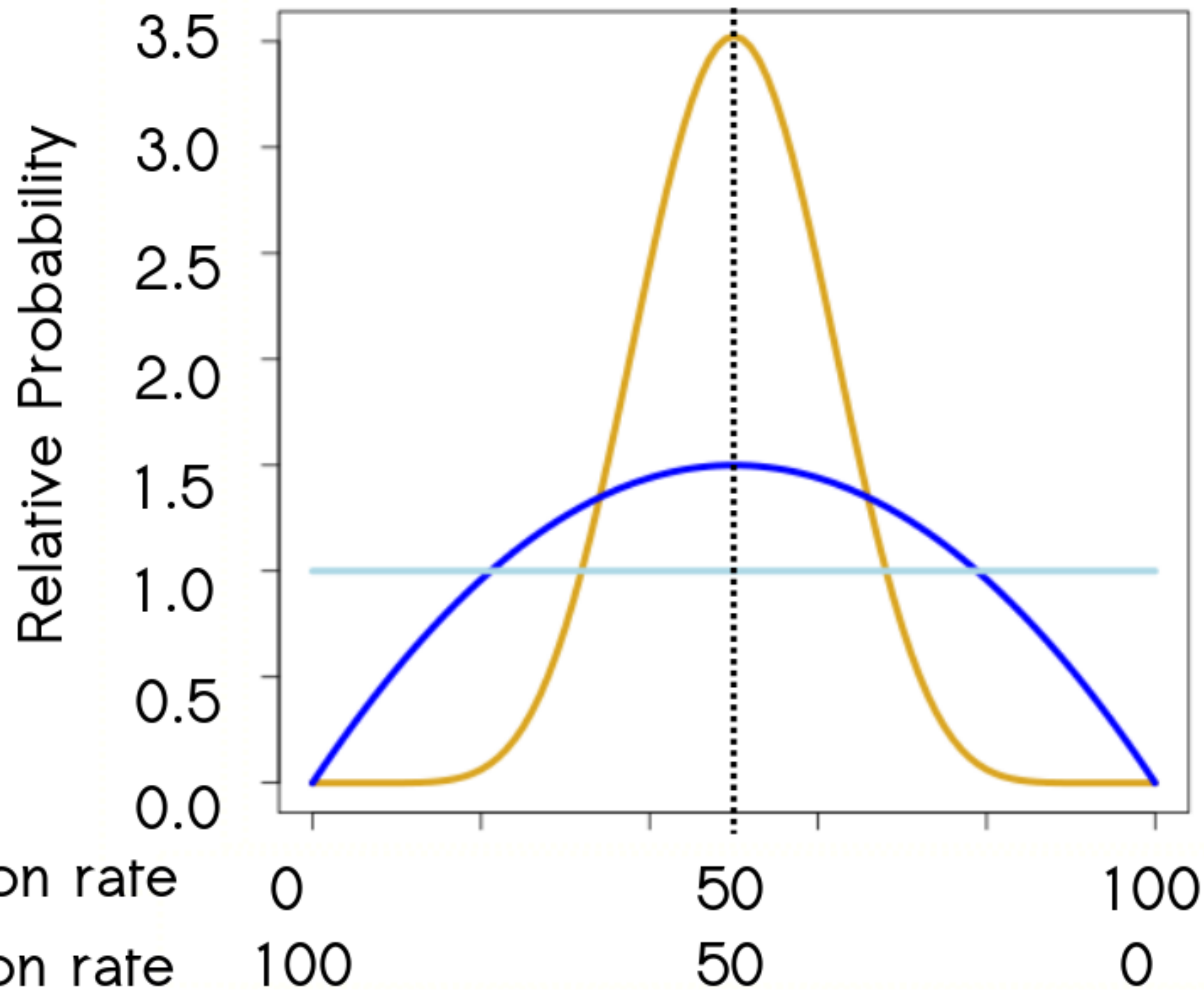
Parameters

$\alpha = \infty$

$\alpha = 10$

$\alpha = 2$

0 to 1 transition rate
1 to 0 transition rate



Parameters

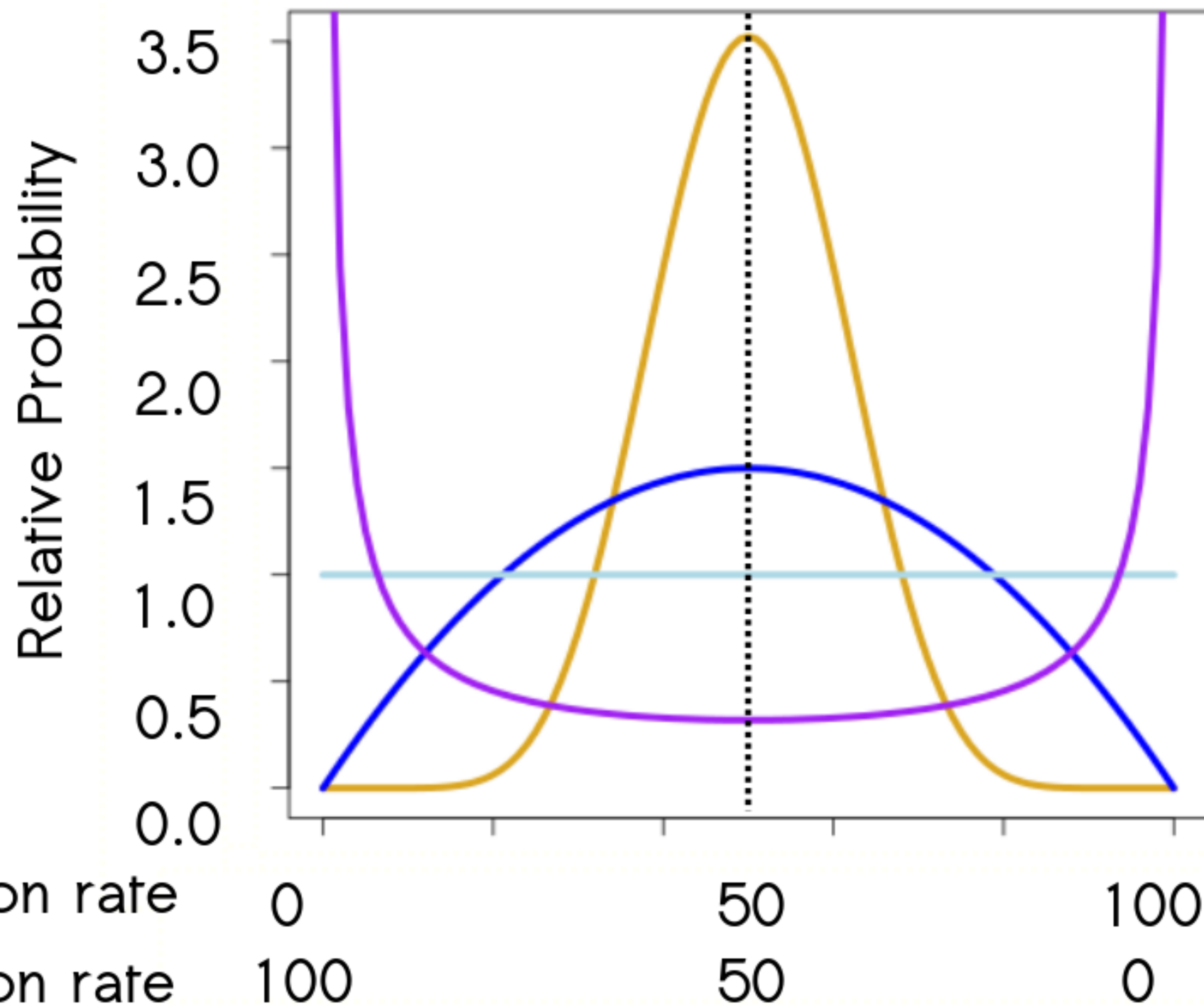
$\alpha = \infty$

$\alpha = 10$

$\alpha = 2$

$\alpha = 1$

0 to 1 transition rate
1 to 0 transition rate



Parameters

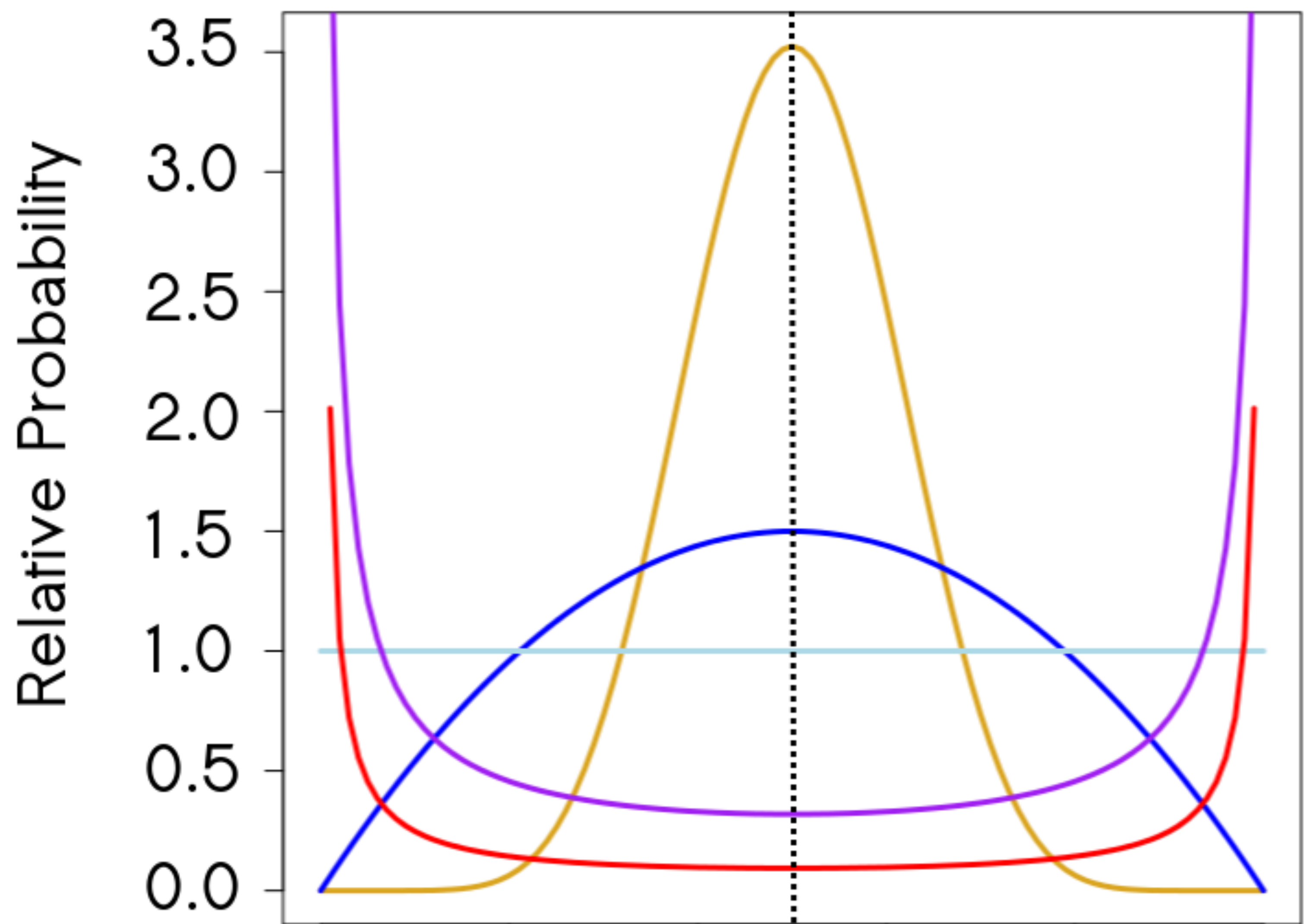
$\alpha = \infty$

$\alpha = 10$

$\alpha = 2$

$\alpha = 1$

$\alpha = 0.2$



Parameters

$\alpha = \infty$

$\alpha = 10$

$\alpha = 2$

$\alpha = 1$

$\alpha = 0.2$

$\alpha = 0.05$

0 to 1 transition rate

1 to 0 transition rate

0

50

100

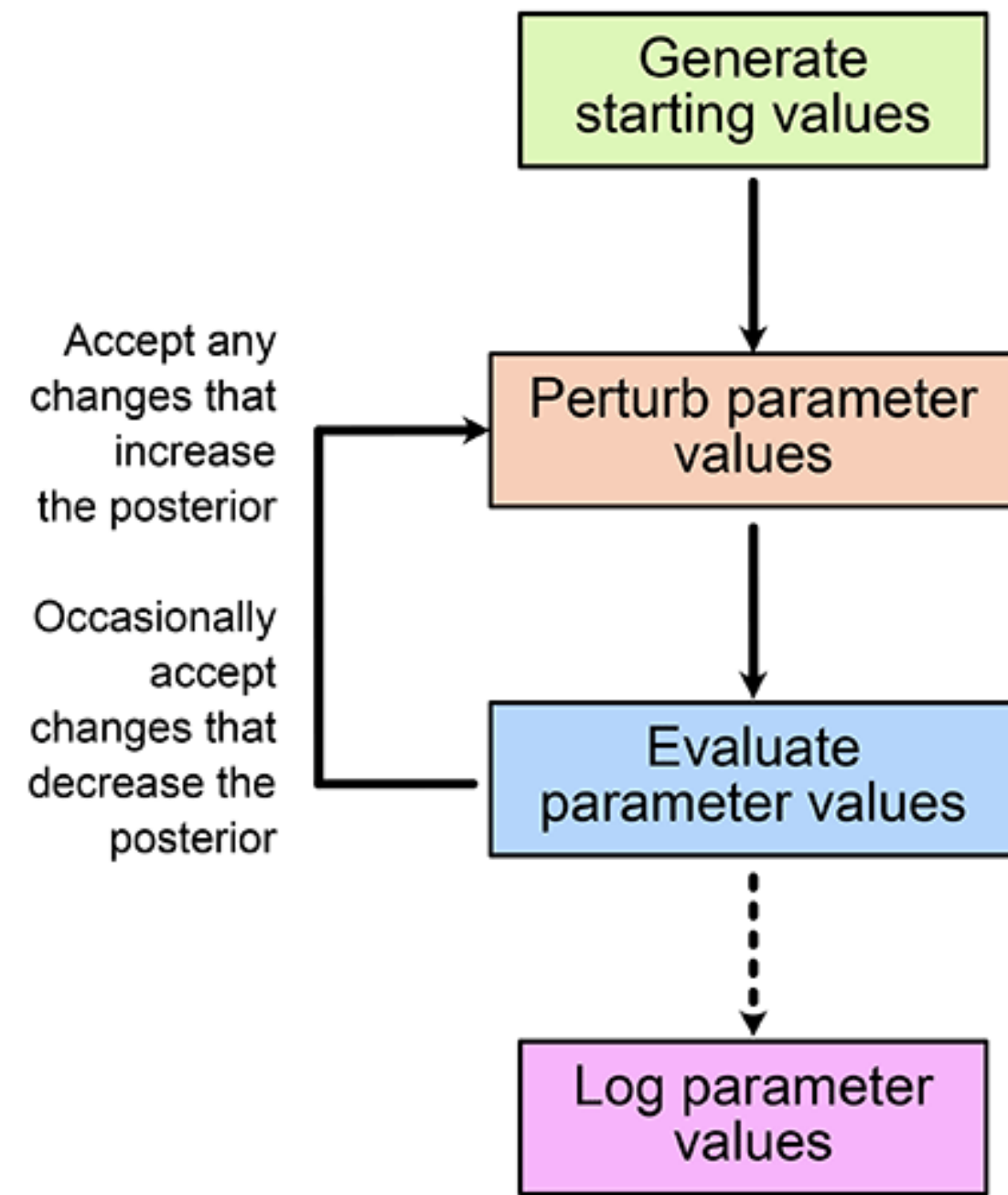
100

50

0

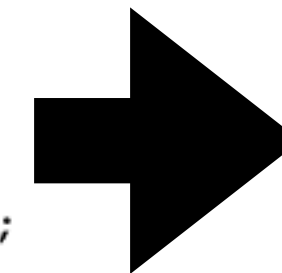
The Metropolis-Hastings algorithm

Flowchart



Pseudocode

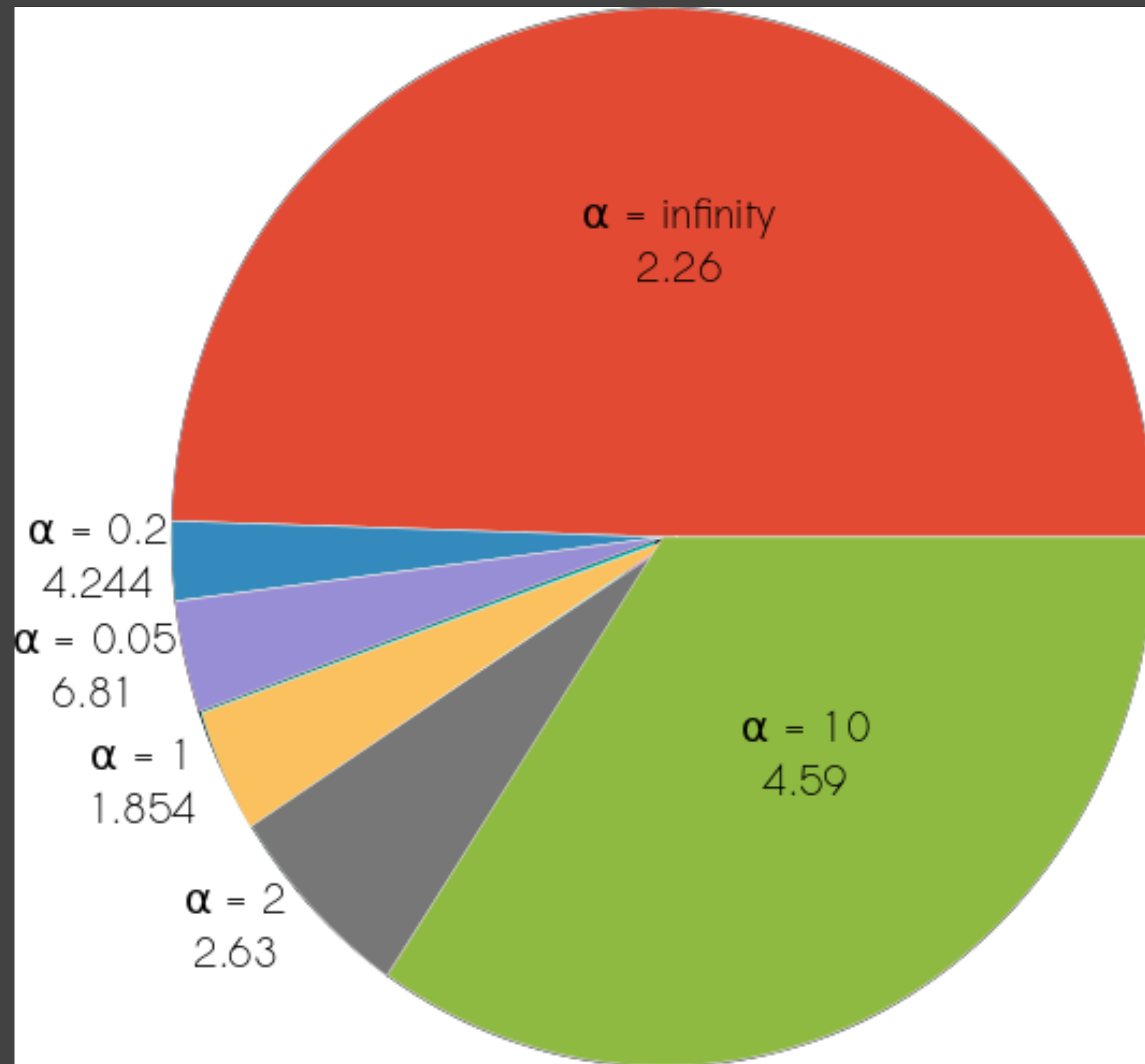
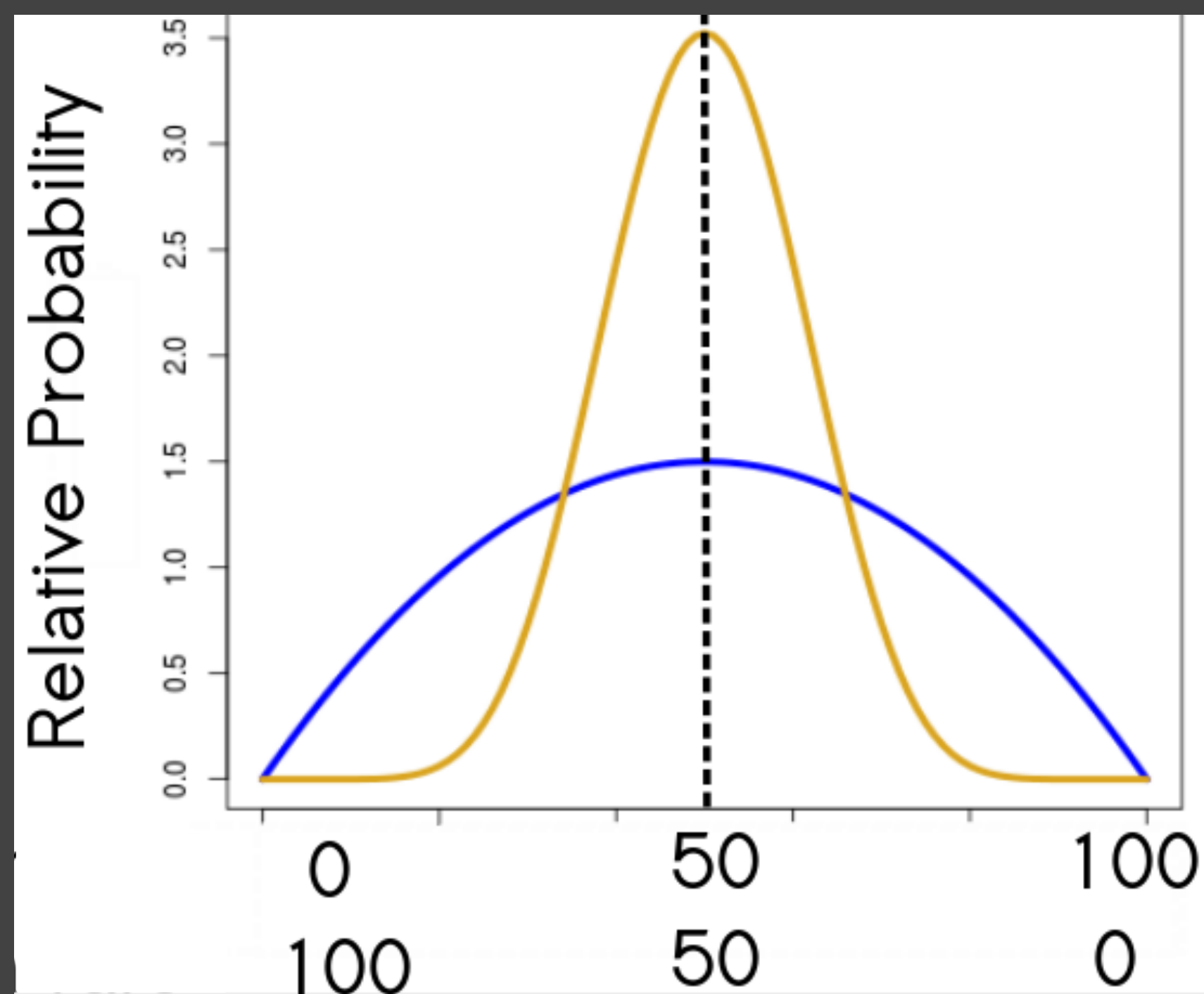
```
initialize starting values;  
  
for i in mcmc steps  
do  
    propose new parameter values;  
    calculate the Hastings ratio R;  
  
    if( R > 1 )  
        accept the new values;  
    else  
        accept the new values with Pr = R;  
  
    store the values with frequency j;  
done
```



Bias towards good values

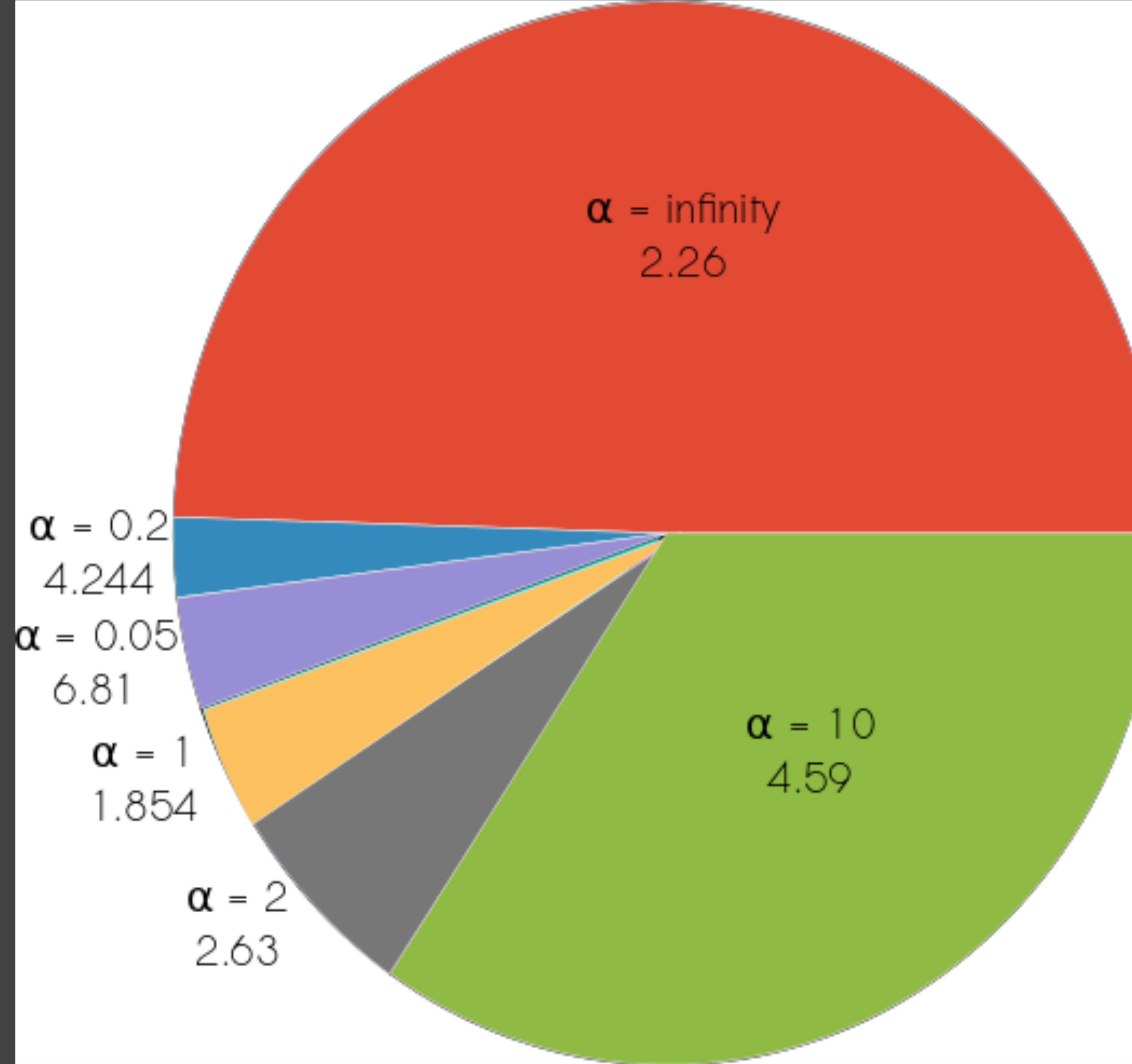
For model selection, SteppingStone
(Xie et al 2011) commonly used

Most datasets prefer an Mk-Like model



Most datasets prefer an Mk-Like model

Some like a Dollo-like Model



How do we model morphological data

a $Q = \begin{pmatrix} -\mu_0 & \mu_1 \\ \mu_0 & -\mu_1 \end{pmatrix},$ $0 \longleftrightarrow 1$

b $Q = \begin{pmatrix} -\mu_0 & \mu_1 & \mu_2 & \mu_3 \\ \mu_0 & -\mu_1 & \mu_2 & \mu_3 \\ \mu_0 & \mu_1 & -\mu_2 & \mu_3 \\ \mu_0 & \mu_1 & \mu_2 & -\mu_3 \end{pmatrix},$

c $Q = \begin{pmatrix} -\mu_0\pi_0 & \mu_1\pi_1 \\ \mu_0\pi_0 & -\mu_1\pi_1 \end{pmatrix},$ $0 \longleftrightarrow 1$

d

	0	1
0	-	3
1	1	-

 $0 \longleftrightarrow 1$

- On the fly assessment of state spaces
 - Khakurel et al 2024
 - Mulvey et al 2025
- No common meaning to characters
 - Ronquist et al 2001 introduced a state frequency prior
 - Explored more in Klopstein and Ronquist 2015; Wright et al 2016
 - Recent recoding method introduced by Smith 2026

How do we model morphological data

a

$$Q = \begin{pmatrix} -\mu_0 & \mu_1 \\ \mu_0 & -\mu_1 \end{pmatrix},$$

0 $\longleftrightarrow$ 1

b

$$Q = \begin{pmatrix} -\mu_0 & \mu_1 & \mu_2 & \mu_3 \\ \mu_0 & -\mu_1 & \mu_2 & \mu_3 \\ \mu_0 & \mu_1 & -\mu_2 & \mu_3 \\ \mu_0 & \mu_1 & \mu_2 & -\mu_3 \end{pmatrix},$$

0 $\longleftrightarrow$ 1
2 $\longleftrightarrow$ 3
0 $\longleftrightarrow$ 2
1 $\longleftrightarrow$ 3

c

$$Q = \begin{pmatrix} -\mu_0\pi_0 & \mu_1\pi_1 \\ \mu_0\pi_0 & -\mu_1\pi_1 \end{pmatrix},$$

0 $\longleftrightarrow$ 1

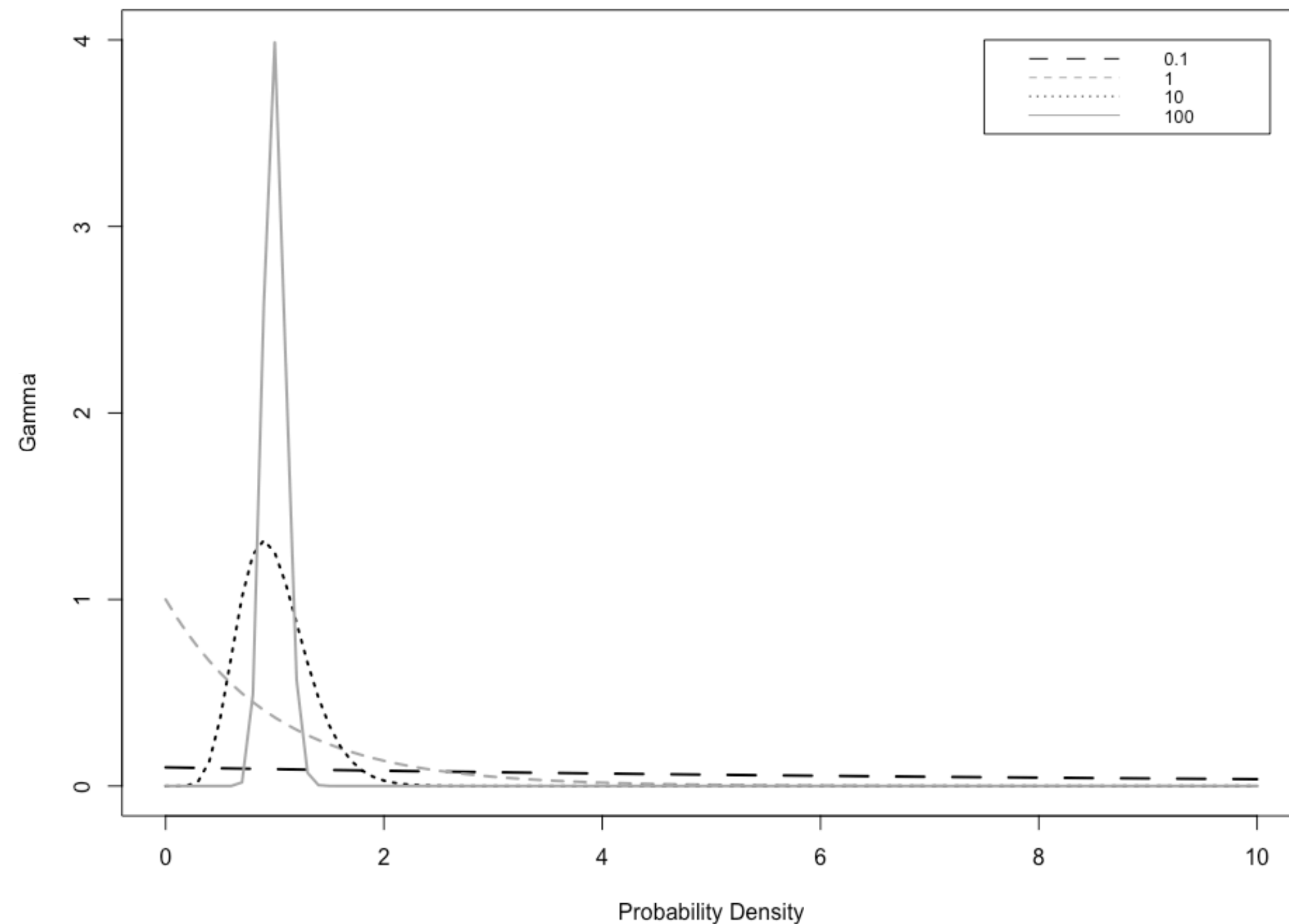
d

	0	1
0	-	3
1	1	-

0 $\longleftrightarrow$ 1

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How do we model morphological data



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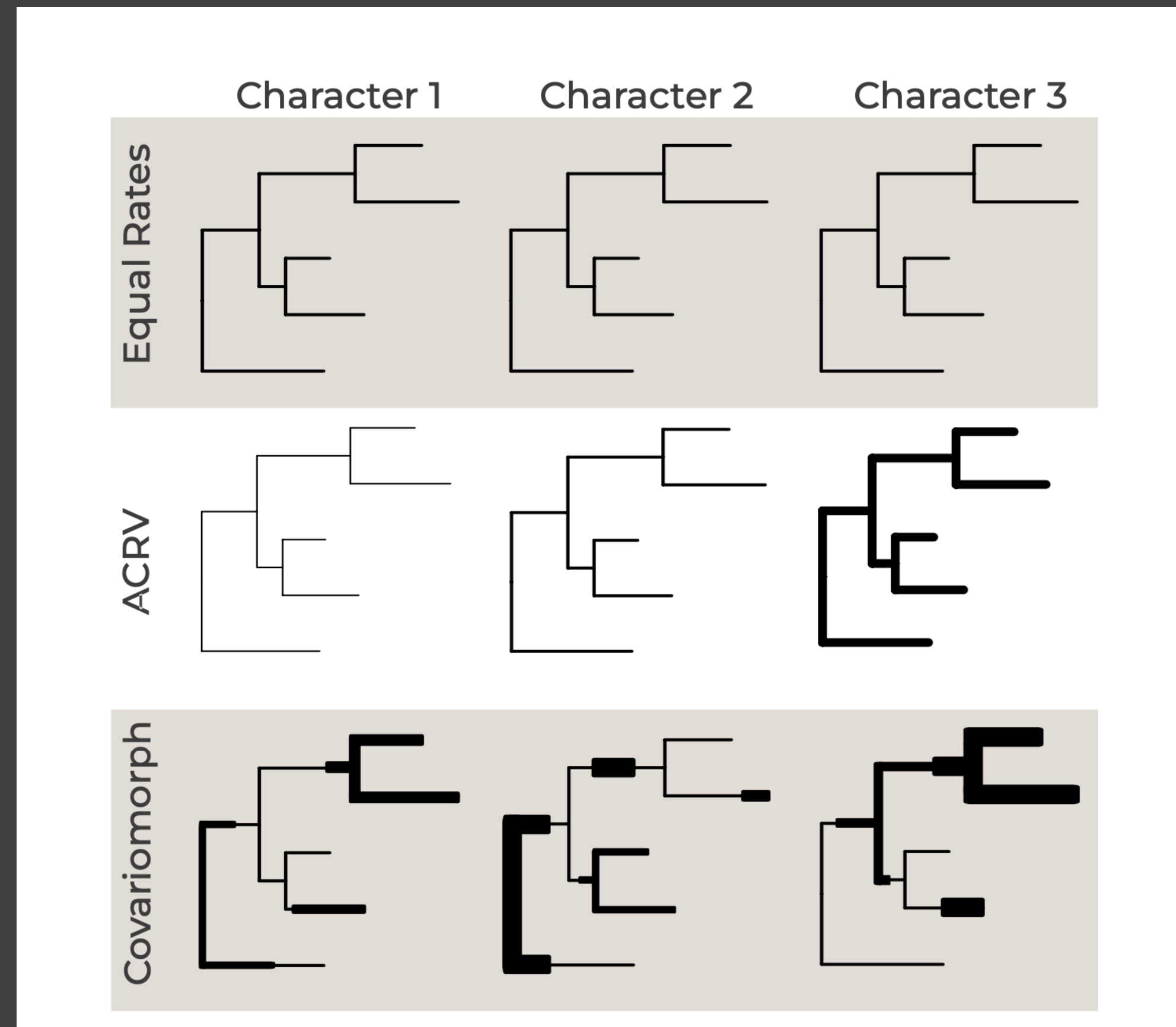
Yang 1994

Wagner 2012

Harrisson and Larson 2015

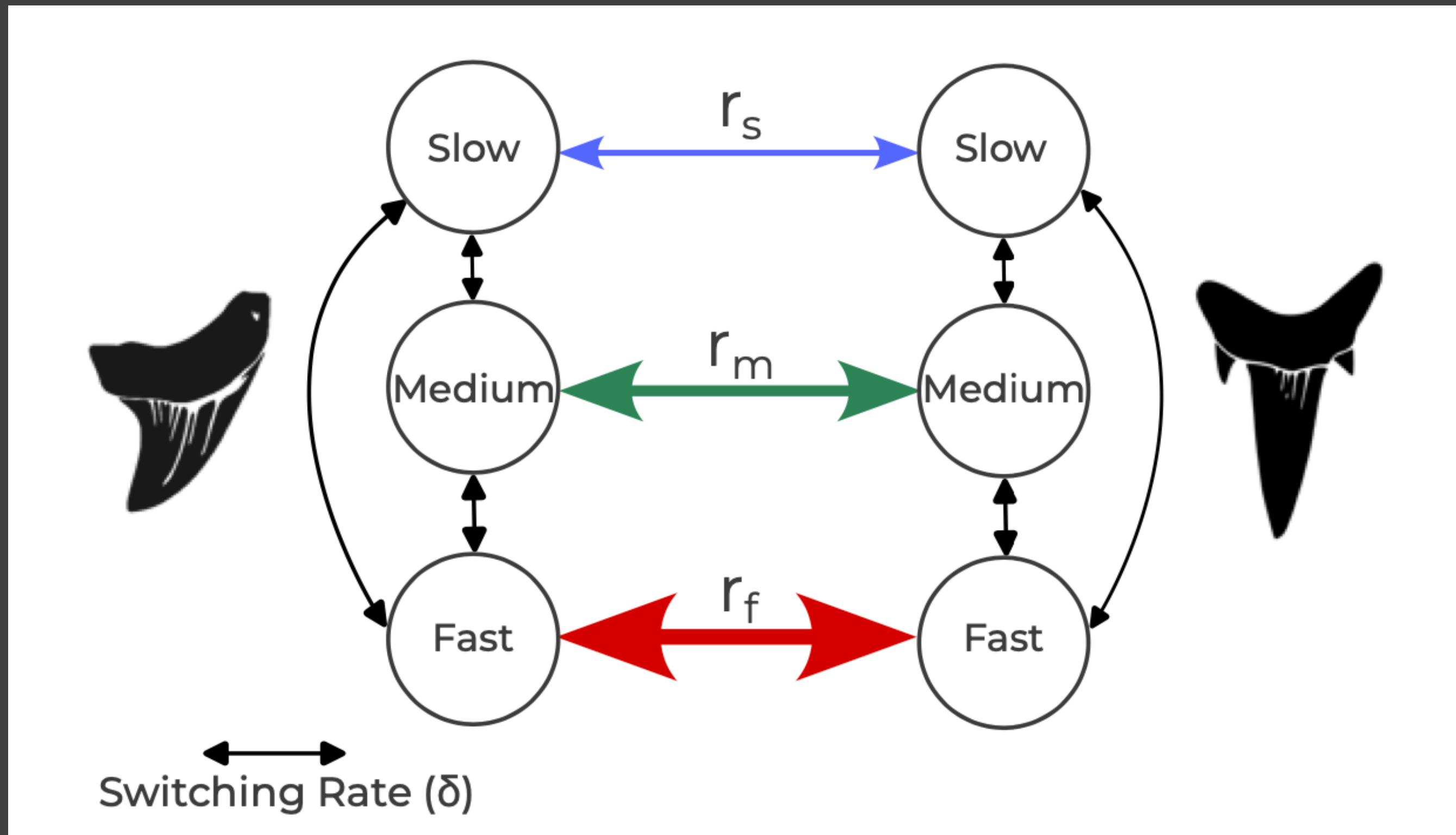
Wright and Wynd 2024

How do we model morphological data



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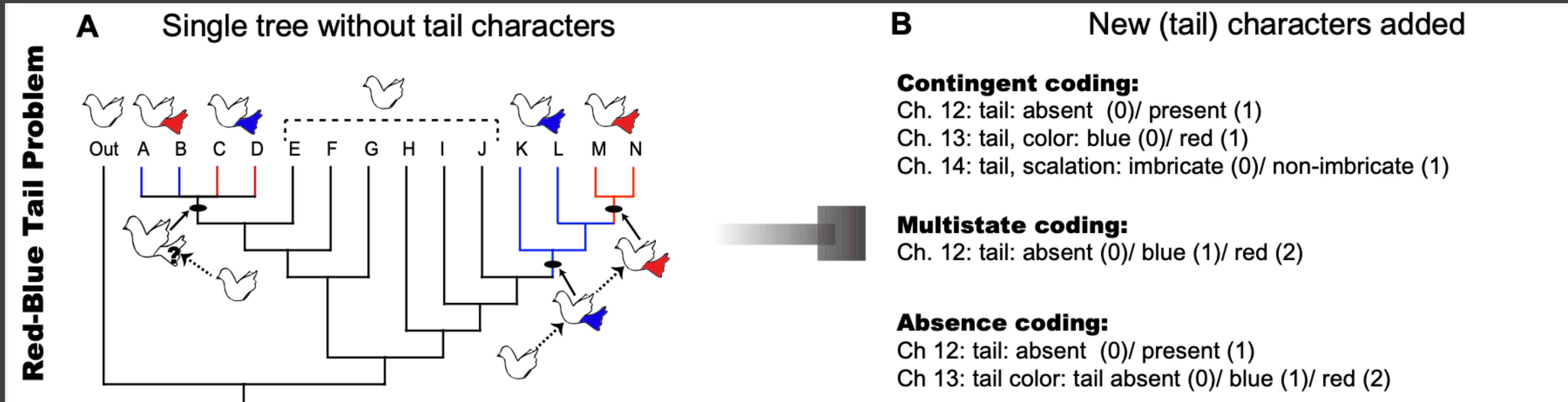
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How do we model morphological data

- Inapplicable characters
- More or less comes out in the bath (Simoes et al 2023)



How do we model morphological data



Morphological phylogenetics has been impeded by sluggish adoption of modern methods

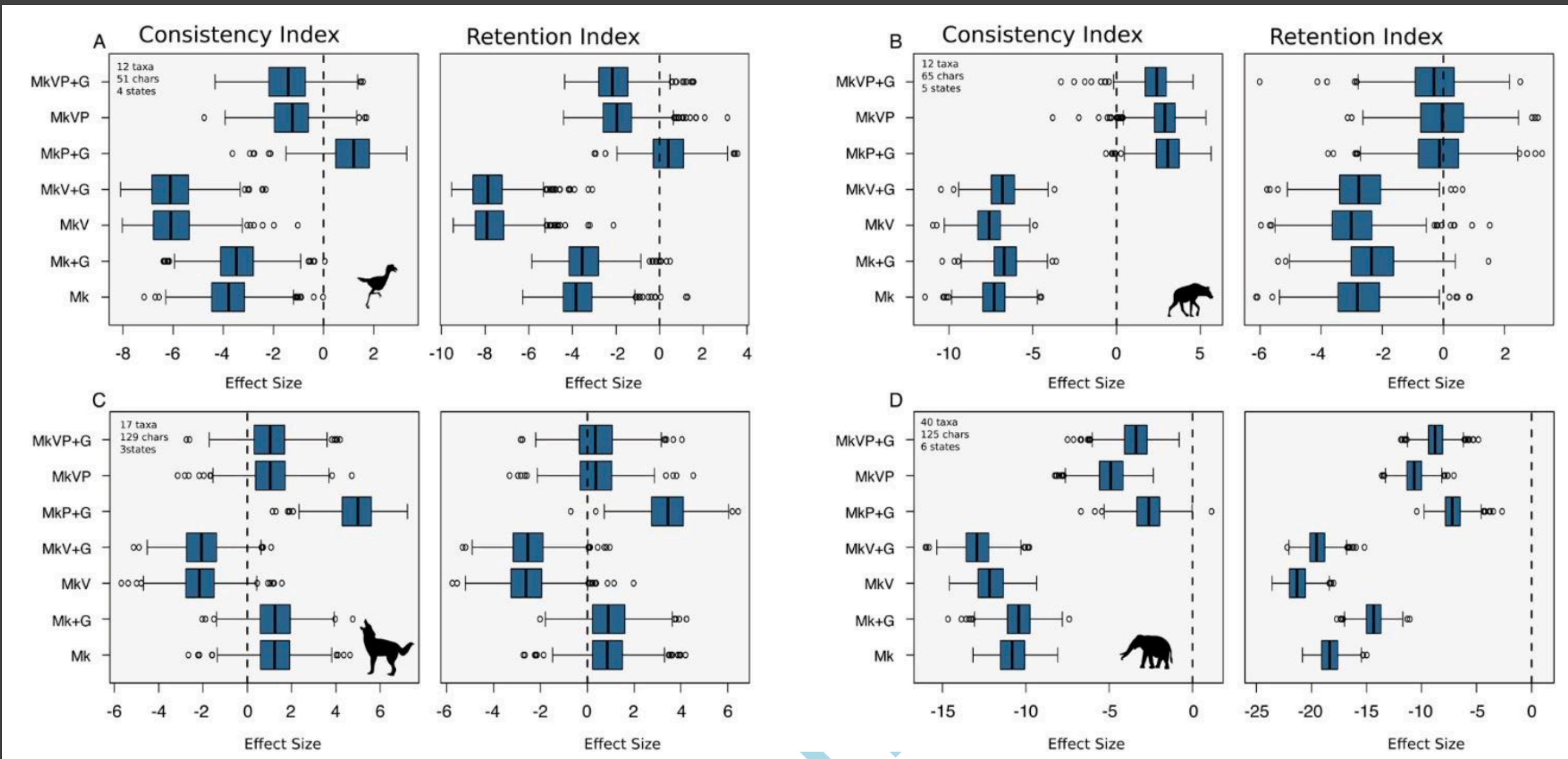
Still asking a lot of fundamental and basic questions

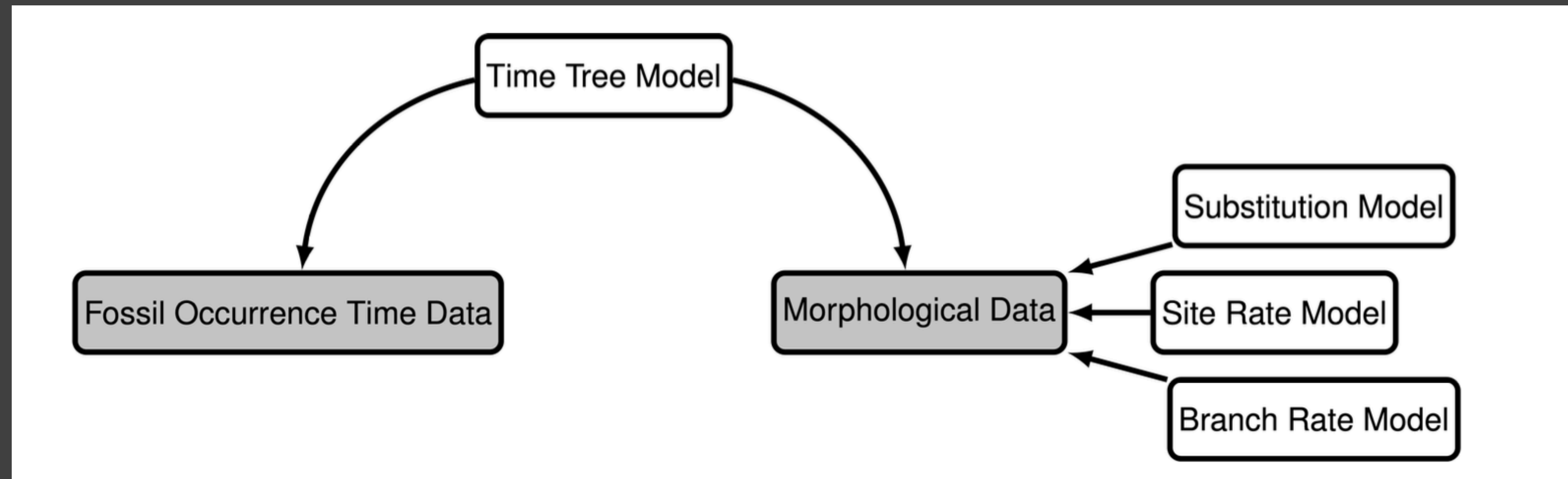
Wright and Hillis 2014

How do we model morphological data

Morphological phylogenetics has been impeded by sluggish adoption of modern methods

Still asking a lot of fundamental and basic questions



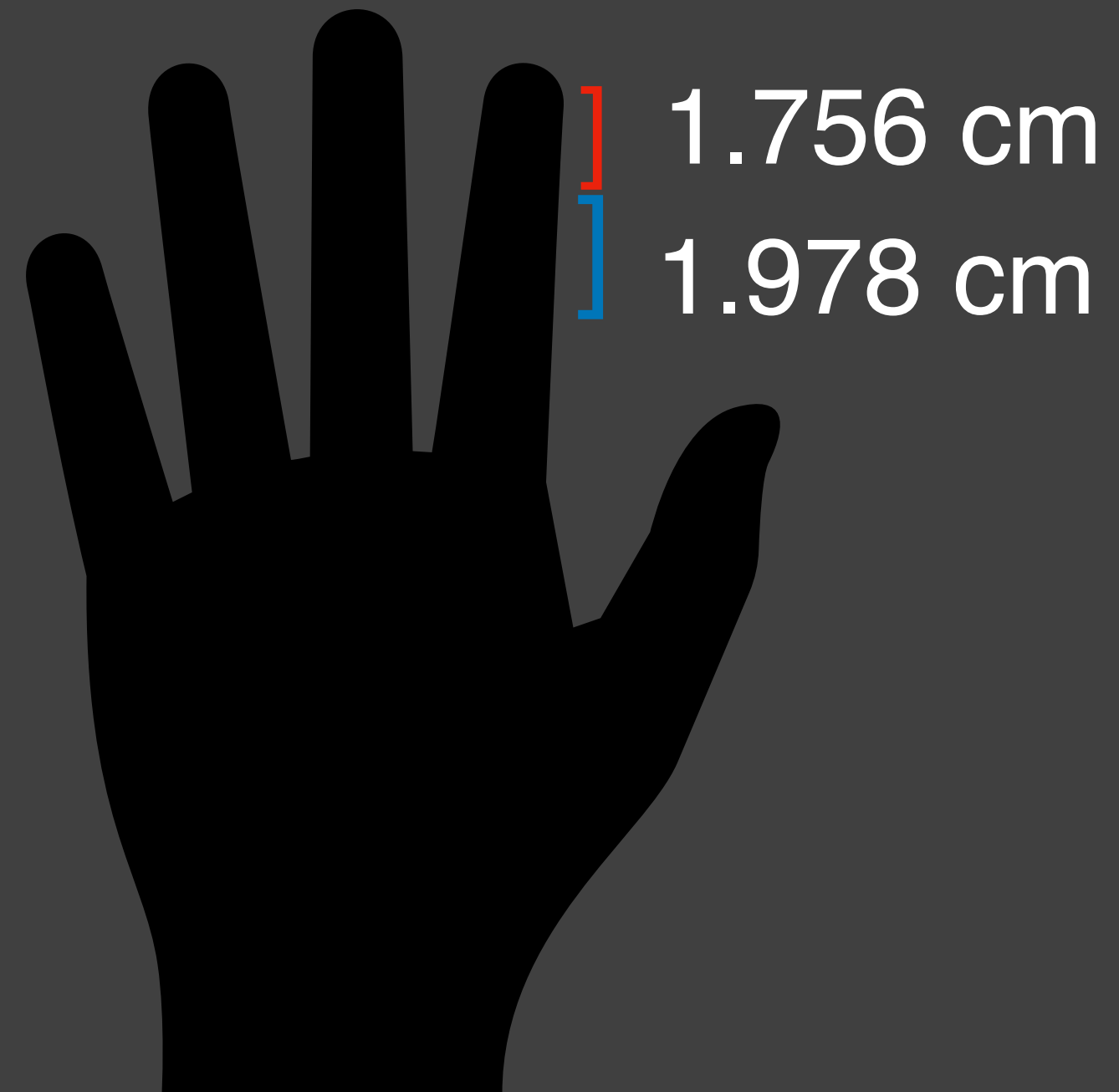


Continuous Trait Data

Measurements

Can't be broken down easily into discrete sets

Heights, weights, lengths and depths, ratios of features to one another



Continuous Trait Data

Objective

But still - need to agree on what to measure and how
(Houle - Meaning and Measurement in Biology is required reading)



But continuous data break all our ways of thinking about models

Continuous Trait Data

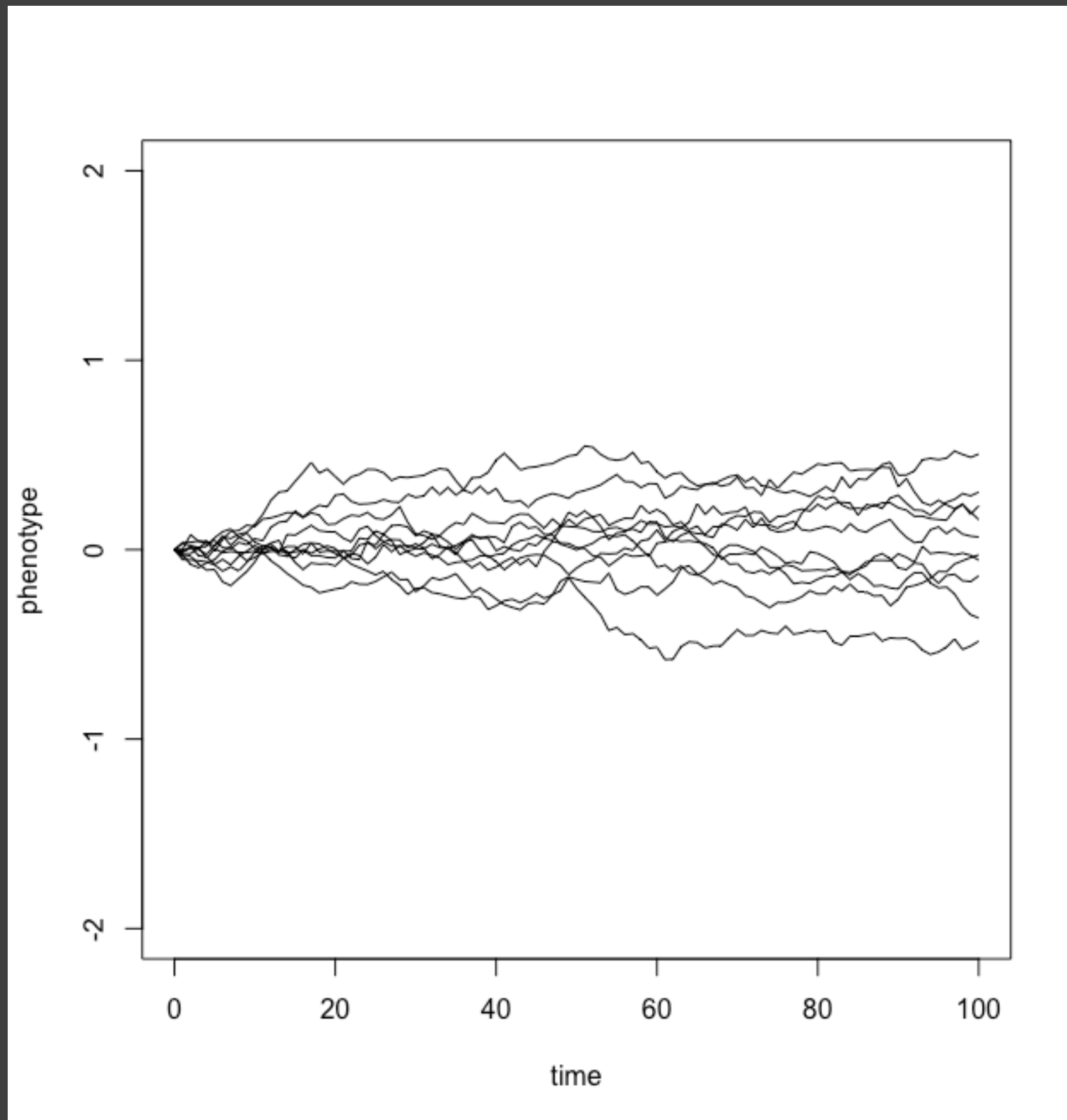
But continuous data break all our ways of thinking about models,
which rely on data being discrete

People have tried to clobber the data into that form: discretizing traits

E.g.,
> 1 cm is a 0,
1-2 cm is a 1,
2-3 cm is a 2

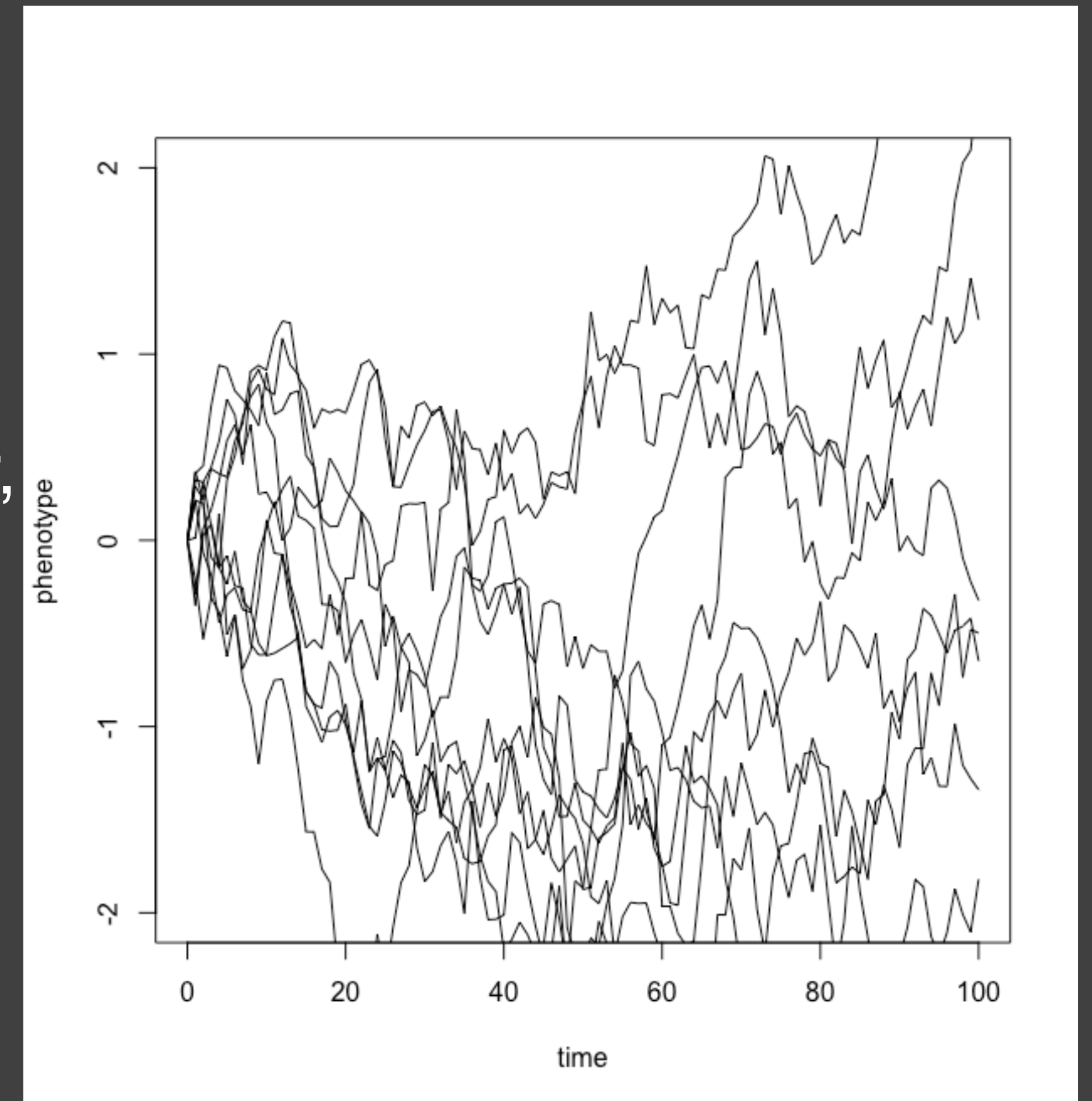
Continuous Trait Data

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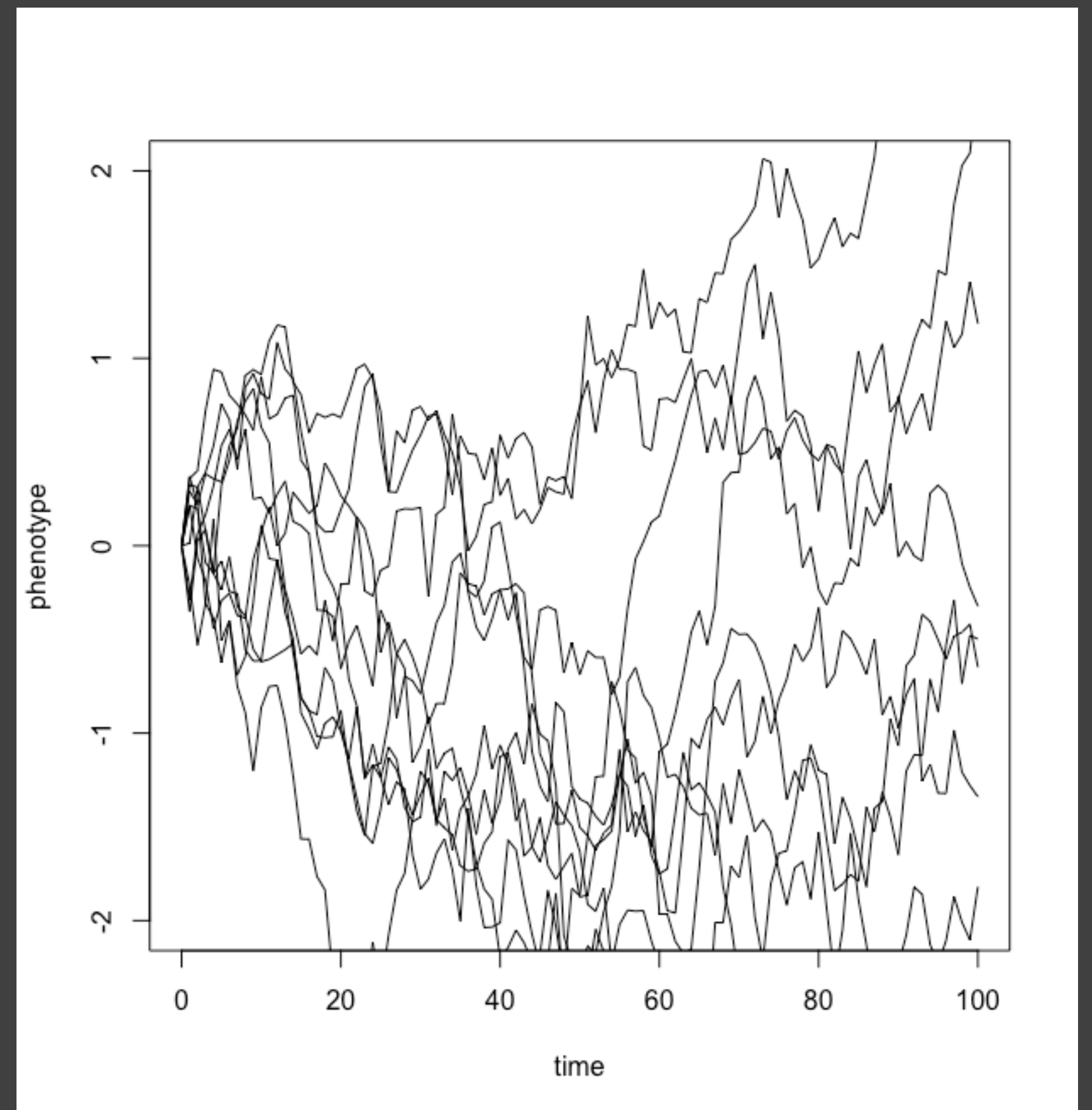
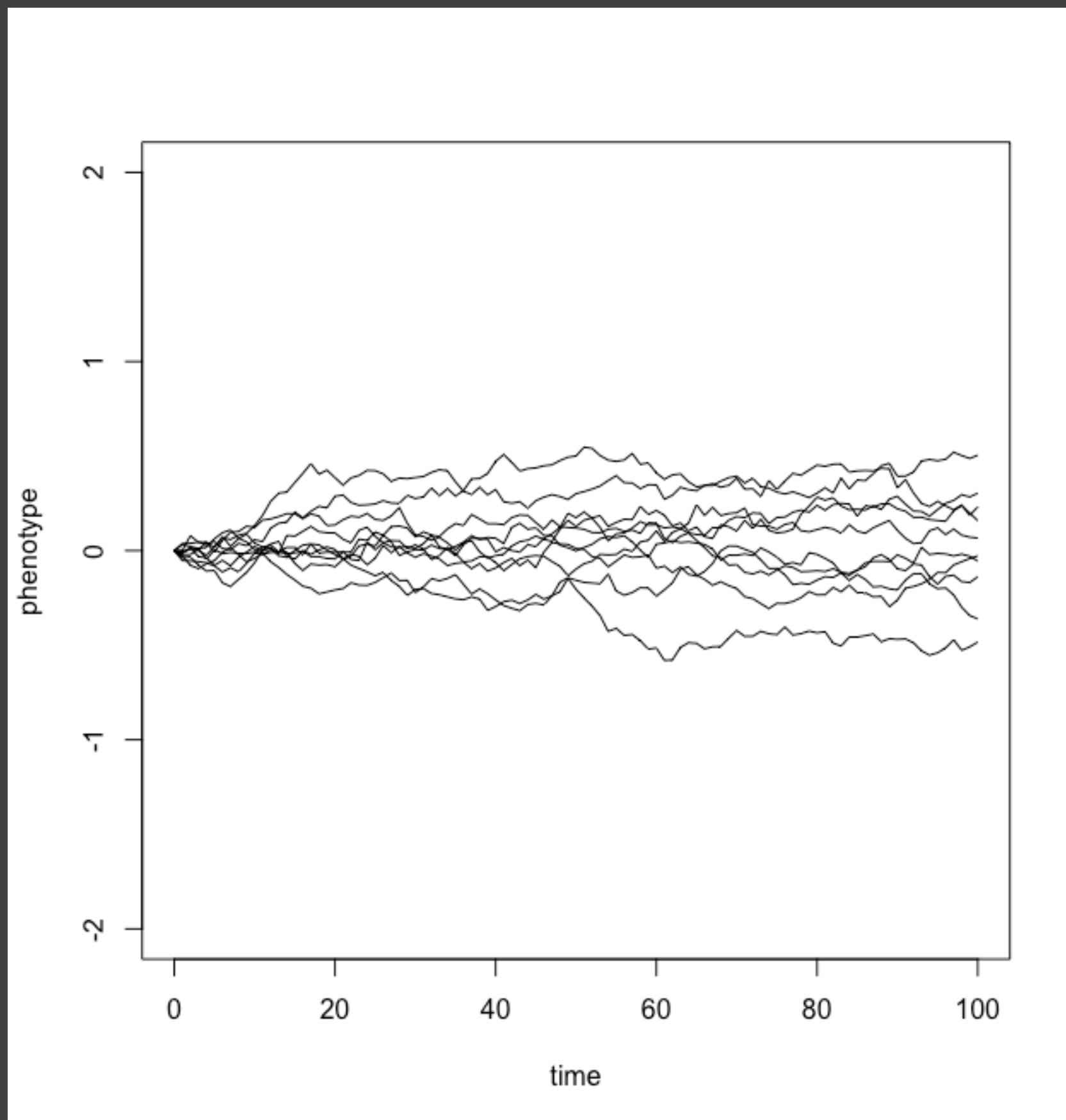
Brownian motion: traits
move in their space of
possible values
according
to a variance parameter,
 σ^2

Similar to the idea of
Farris (1970),
except movement is
bidirectional



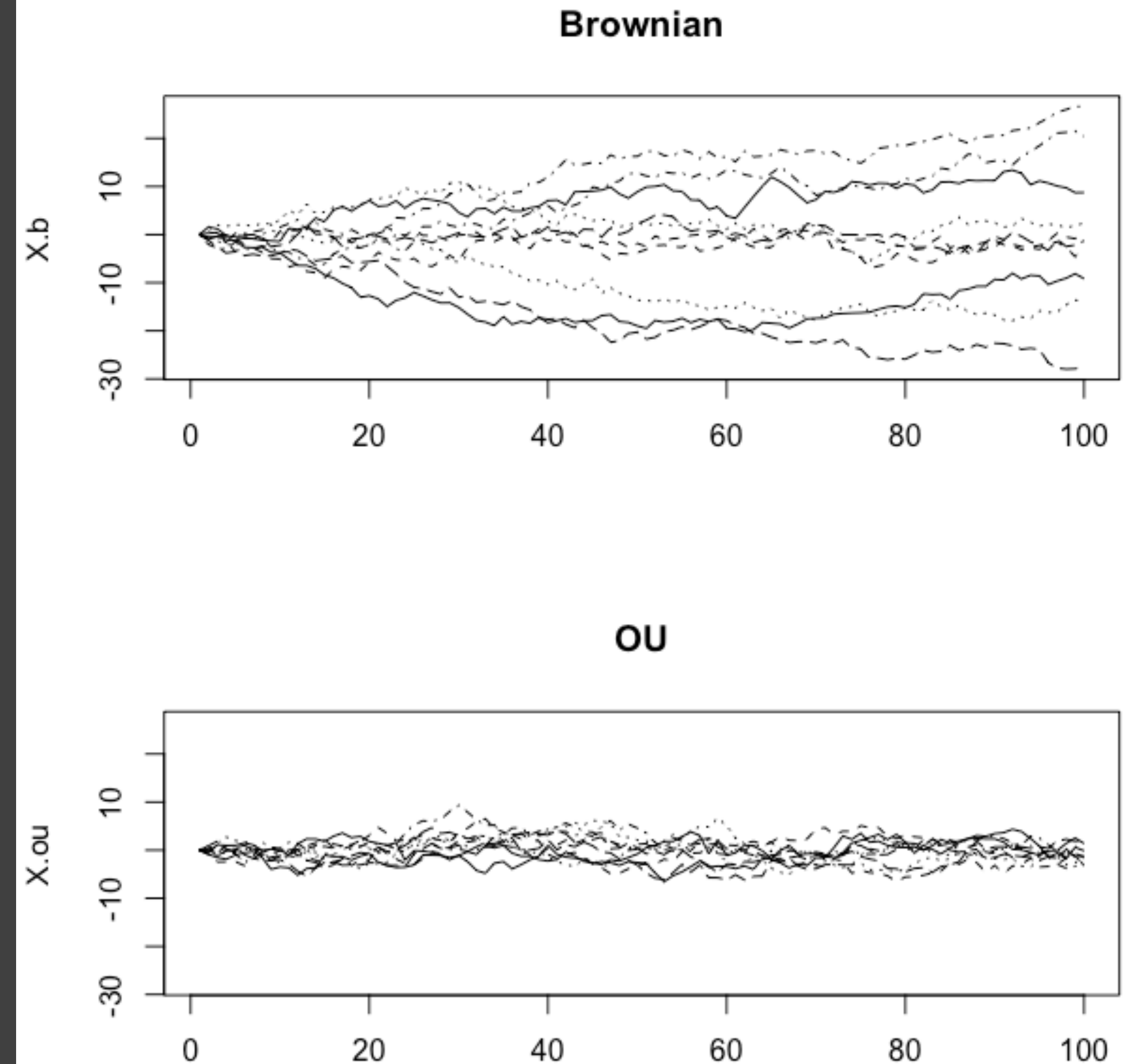
Continuous Trait Data

Widely used in
phylogenetic
comparative methods



Continuous Trait Data

Can also use Ornstein-Uhlenbeck:
a bounded selection model implying
there is a trait optimum and the trait will
come back if it gets too far



The data

AND/OR

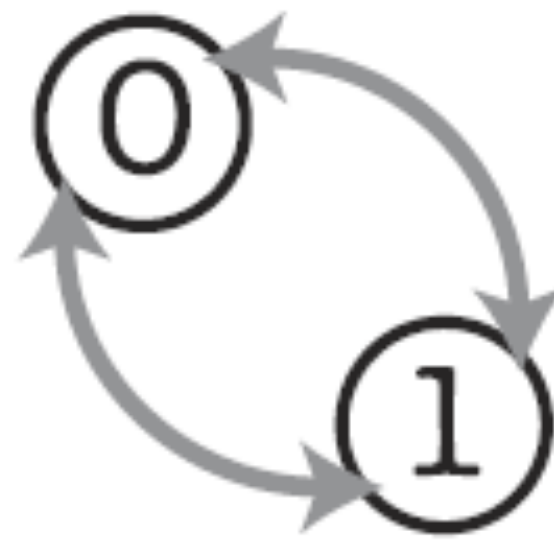
0101... ATTG...
1101... TTGC...
0100... ATTC...



Phylogenetic
characters

Fossil
ages

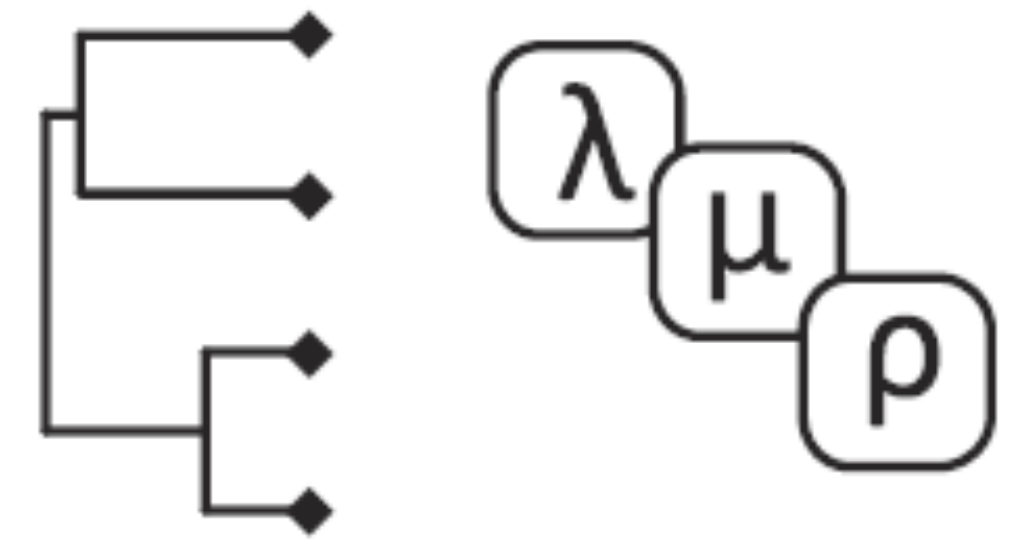
Tripartite model components



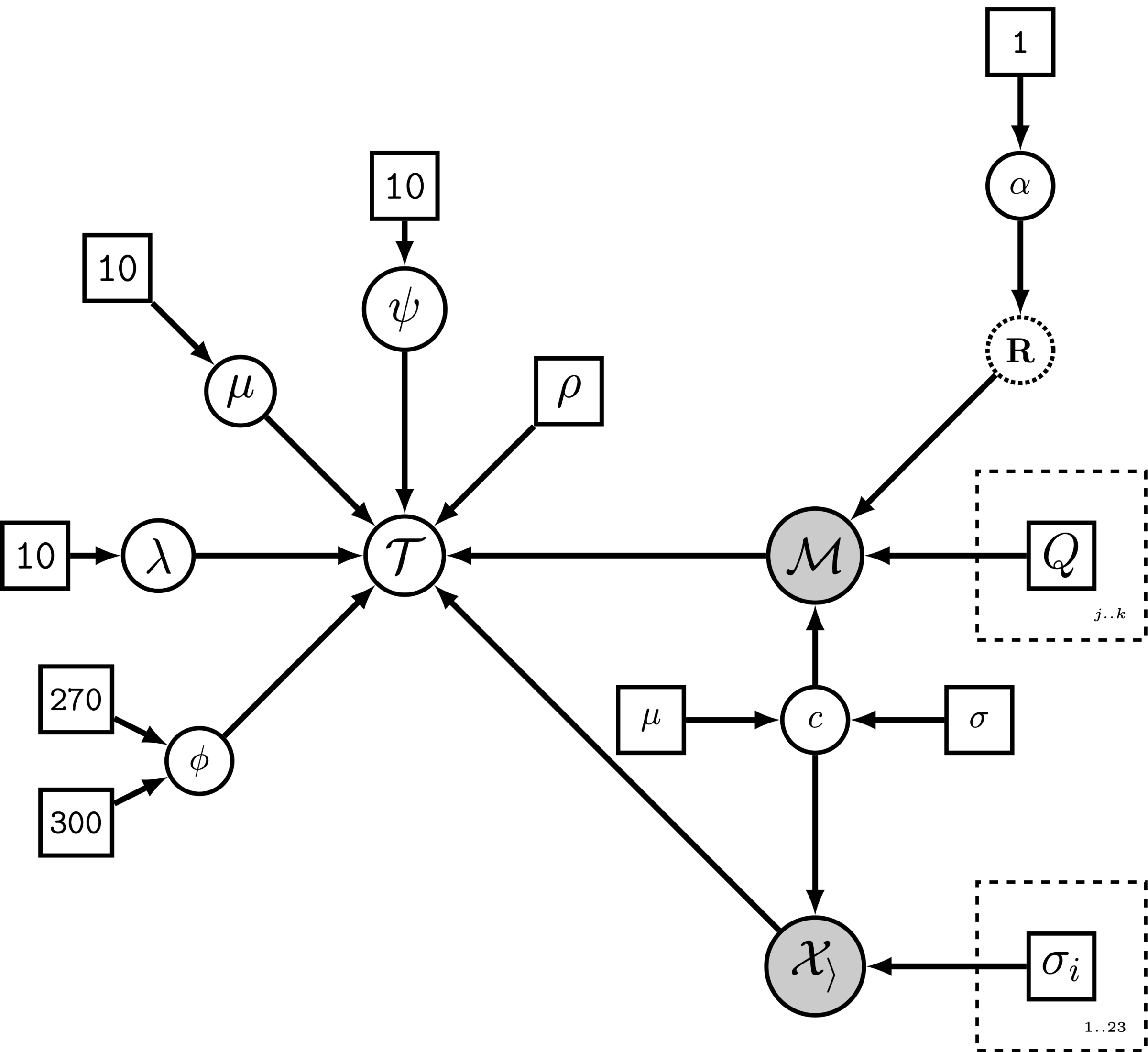
Substitution
model



Clock
model



Tree and tree
model



Time Tree Model

origin time	$\phi \sim \text{Uniform}(270, 300)$
speciation rate	$\lambda \sim \text{Exponential}(10)$
extinction rate	$\mu \sim \text{Exponential}(10)$
fossil sampling rate	$\psi \sim \text{Exponential}(10)$
sampling probability	$\rho = 1$
tree topology & divergence times	$\mathcal{T} \sim \text{FBD}(\phi, \lambda, \mu, \psi, \rho)$

Branch Rate Model

Uncorrelated Lognormal clock rate	$c \sim \text{Lognormal}(\sigma, \mu)$
-----------------------------------	--

Discrete Substitution Model

gamma shape parameter	$\alpha \sim \text{Exponential}(1)$
4 site-rate categories	$\mathbf{R} := \text{Discretized Gamma}(\alpha, \alpha, 4)$

Multi-state Mk rate matrix	$Q = \begin{bmatrix} * & \frac{1}{k-1} & \cdots & \frac{1}{k-1} \\ \frac{1}{k-1} & * & \cdots & \frac{1}{k-1} \\ \cdots & \cdots & \ddots & \cdots \\ \frac{1}{k-1} & \frac{1}{k-1} & \cdots & * \end{bmatrix}$
----------------------------	---

Data Likelihoods

likelihood of continuous character data $\mathcal{X}_>$	$f(\mathcal{X} \mid \mathcal{T}, c, BM_\sigma)$
likelihood of discrete character data $\mathcal{M}$	$f(\mathcal{M} \mid \mathcal{T}, c, \mathbf{R}, Q)$

